Starting your single cell analysis on Biowulf

NIA BDMS (October 19-20, 2026)
Dominic J. Acri, PhD
Laboratory of Neurogenetics, NIA/IRP

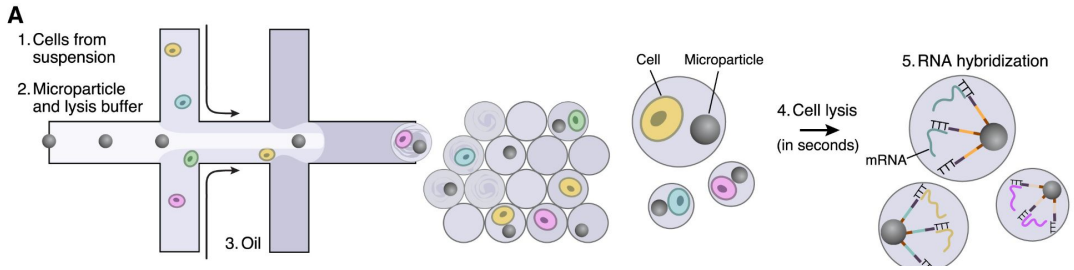
Course Schedule

Monday 1-3PM

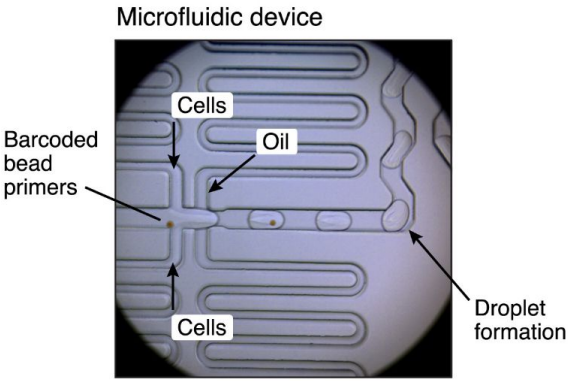
Tuesday 1-3PM

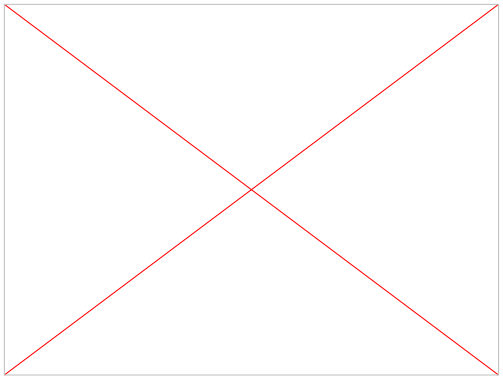
4 modules:

- Cellranger
- Integration/Annotation
- Differential Expression
- Advanced Analyses



World before 10x Genomics - PMID26000488





Big Learning Objectives (2 day)

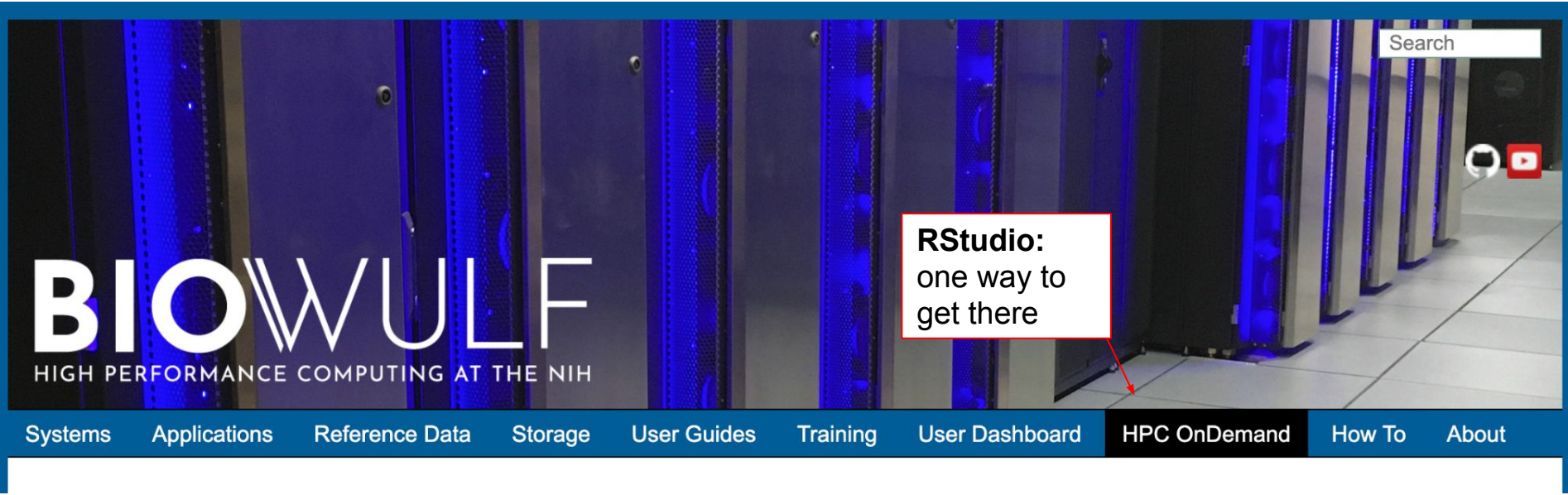
What You Can Expect

1. How to start from “true raw data” from sc/sn RNA seq
2. Decide how to perform the minimum pre-processing
3. *Audience:* anyone wanting to understand the **bridge between** the wet-lab and dry-lab analyses with NIH specific resources (biowulf)

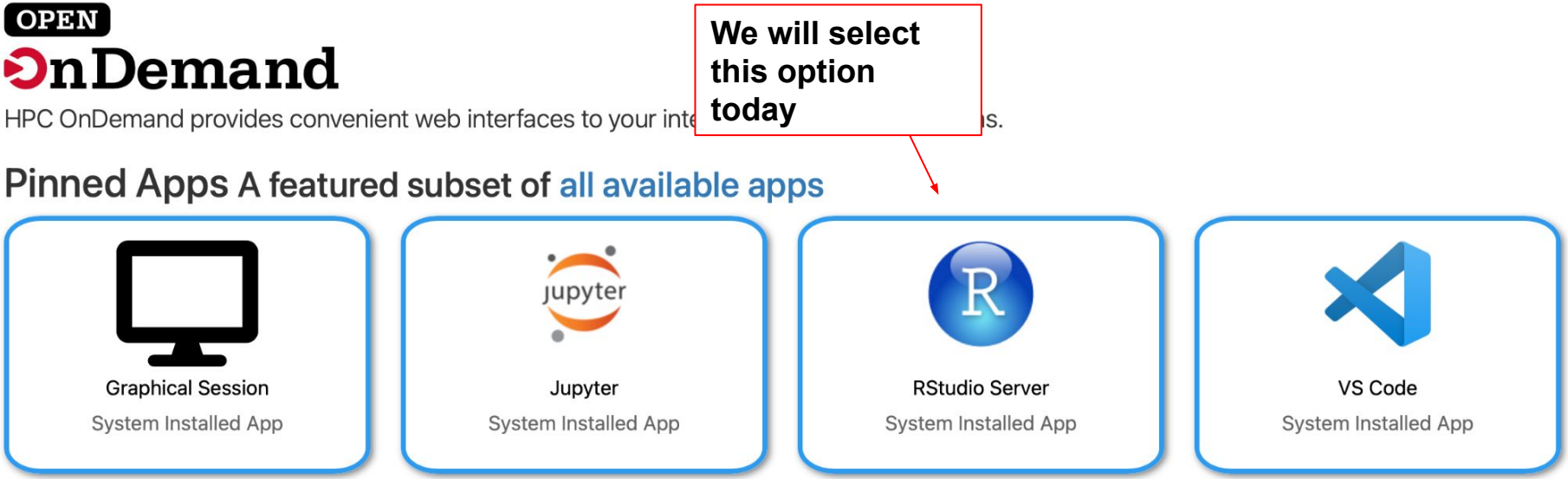
NIH Biowulf

Course pre-requiste: NIH biowulf account

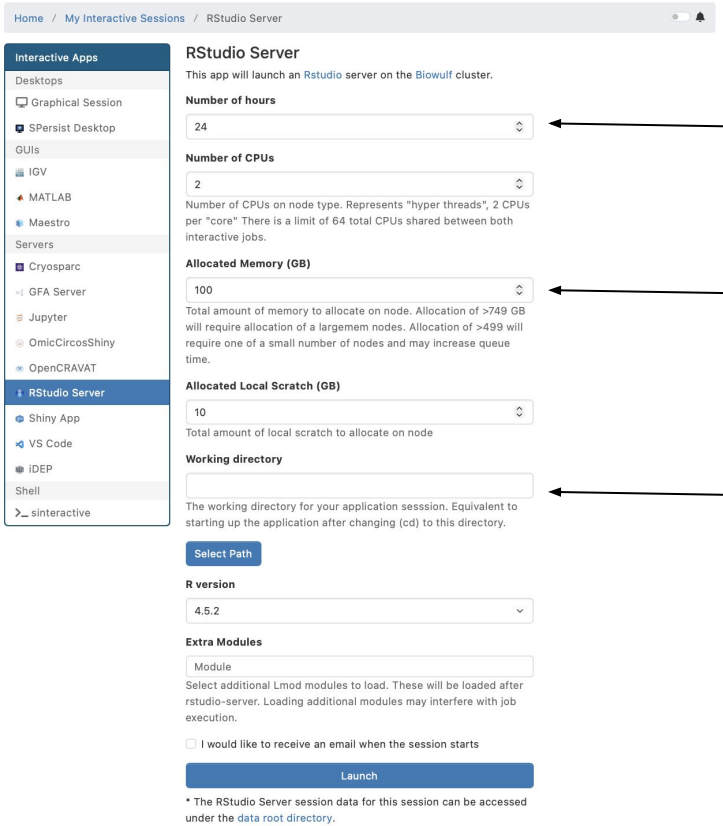
Limited course-specific accounts organized by Deb Croteau



(Optional) HPC OnDemand



(Optional) HPC OnDemand



24 hours is ok we will open a new one each day though

100 Gb RAM for this workshop

Up to you, if you don't regularly work in biowulf I'd recommend:

'/data/{YOUR_USERNAME}'

Starting your single cell analysis on Biowulf

Day 1: Cellranger & Integration to Annotation

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NIH Intramural Research Program
Our Research Changes Lives
ONE PROGRAM, MANY PEOPLE, INFINITE POSSIBILITIES

Today's workshop

Learning objectives

1. Perform cellranger from sample fastq to get to Seurat-ready input (1 human PBMC sample)
2. Investigate the need for integration (cross 5 mouse brain samples from 3 different sources)
3. Define approaches to annotate clusters

Where is the code

Biowulf

- *For today*
- All things downloaded and ready for us to use today and tomorrow
- You will create **a copy** of all materials wherever you'd like to work

GitHub

- *For posterity*
- No access to data/results
- All transparent steps to recreate and redo (if/when you'd like)



Part 1: *Concepts & Cellranger*

Start by exploring RStudio

Download today's materials:

1) Move into your preferred directory
2) Bottom left → move to terminal
3) Copy from location on biowulf:

```
cp /data/CARD_datasets/acridj/BDMS_download.tar ./
```

4) Open to compressed file

```
tar -xf ./BDMS_download.tar
```

Just in case…

0_setup.R

Source on Save

```r
# Run once in the R console before the workshop, using the course RStudio session.
# Open BDMS_sc_2026_student.Rproj first; all paths start in this folder.

install.packages(c(
  "Seurat", "ggplot2", "dplyr", "tibble", "qs2", "harmony", "clustree",
  "patchwork", "eulerr", "remotes", "BiocManager", "rmarkdown", "knitr"
))
BiocManager::install(c("MAST", "edgeR", "ComplexHeatmap"), update = FALSE, ask = FALSE)
remotes::install_github("immunogenomics/presto", upgrade = "never")
remotes::install_github("neurogenetics/SAHA", upgrade = "never")
remotes::install_github("neurogenetics/SAHAdata", upgrade = "never")

```

Good practices in data stewardship

```
(r-reticulate) [acridj@cn0215 BDMS_sc_2026]$ pwd
/data/acridj/project/teaching/BDMS_sc_2026
(r-reticulate) [acridj@cn0215 BDMS_sc_2026]$ tree -L 2 ./
./
├── data
│   ├── 10x_mouse
│   ├── fastq_files
│   ├── Karahan_2025
│   ├── Kersey_2026
│   └── unpacked
├── plots
│   ├── day1
│   └── day2
├── README.md
├── results
│   ├── day1
│   └── day2
└── scripts
```

To learn more, I'd recommend NCI BTEP online classes: https://bioinformatics.ccr.cancer.gov/btep/classes/

Cellranger

10x GENOMICS Products Resources Support Hub Company Store Order status

Support home > Cell Ranger

Cell Ranger

- Overview
- Getting Started
- Supported Libraries
- Download Center
- Cloud Analysis
- Cell Ranger Commands
- Analysis
  - Inputs
  - Running Pipelines
  - Outputs
- Tutorials

Cell Ranger



What is Cell Ranger?

Cell Ranger is a set of analysis pipelines for processing Chromium single cell data. It performs barcode processing and UMI counting to quantify gene expression (from 3', 5', and Flex assays), assembles V(D)J immune receptor sequences, and analyzes Feature Barcode data for applications such as cell surface protein analysis and sample multiplexing.

Download Cell Ranger | Use Cloud Analysis | Get Started

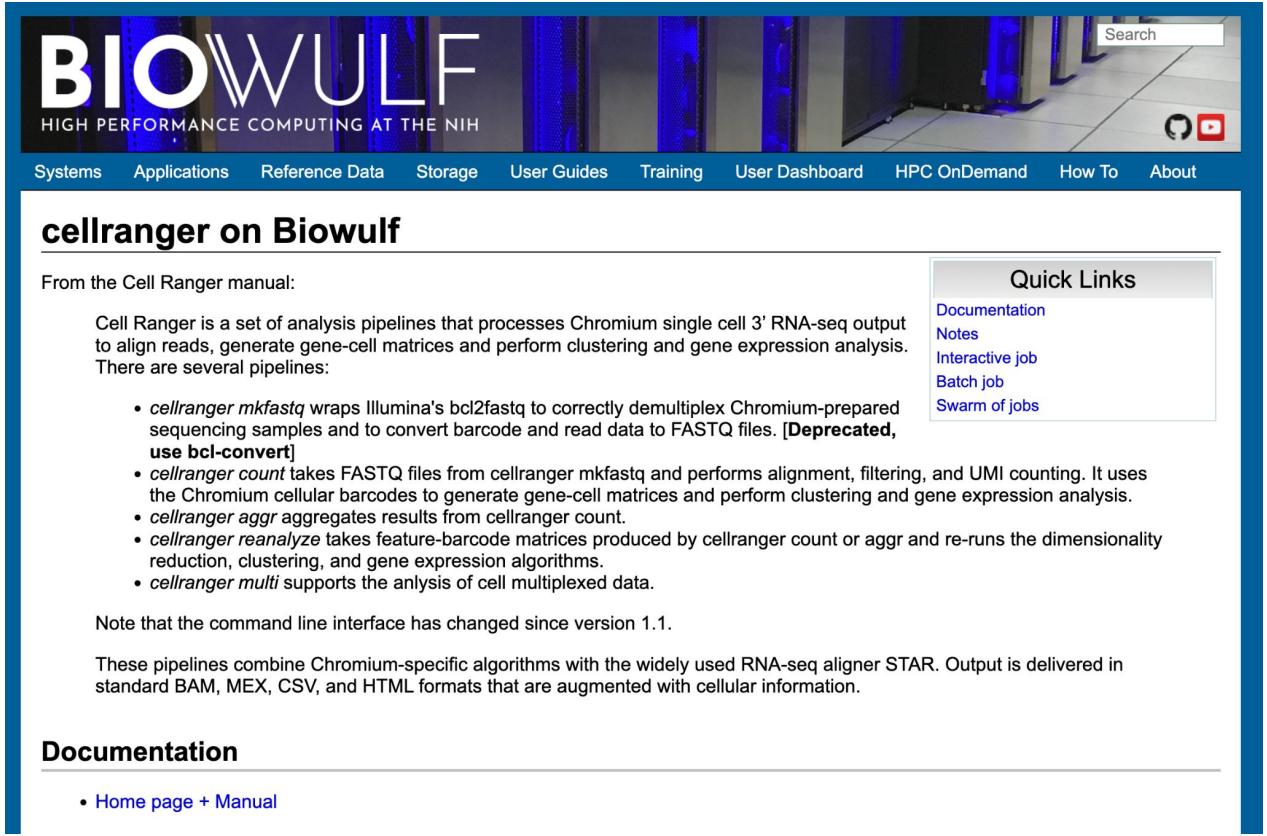
The `cellranger mkfastq` pipeline is deprecated and will be removed in a future release. Use Illumina's BCL Convert to generate Cell Ranger-compatible FASTQ files.

Cellranger

I would recommend starting with Biowulf help pages every time you are trying a new package/pipeline

1) Well-annotated

2) Specific to our resources with real advice to make life easier



Running Cellranger is very easy

1_cellranger_pbmc.sh

```
# you downloaded this from: /data/CARD_datasets/acridj/BDMS_download.tar
# set your working directory to /WHATEVER_YOU_WANT/BDMS_sc_2026/
# this file is in ./scripts/0_cellranger/

module load cellranger

cd ./results/day1/0_cellranger_pbmc

cellranger count \
  --id="demo_run" \
  --fastqs="../../../data/fastq_files/pbmc_1k_v3_fastqs" \
  --sample="pbmc_1k_v3" \
  --transcriptome="${CELLRANGER_REF}/refdata-gex-GRCh38-2020-A" \
  --localcores=4 \
  --localmem=16 \
  --create-bam=false
```

`count`
is to go from
fastq to “outs”

This is where
fastqs are

Biowulf helping
you out!!

Running Cellranger for real has many options

Swarm of Jobs

A swarm of jobs is an easy way to submit a set of independent commands requiring identical resources.

Please do **NOT** set `--jobmode=slurm` when submitting a swarm of cellranger jobs. Running several cellranger jobs in slurm mode can overwhelm the SLURM controller with each job in the swarm polling the controller for the status of all sub-jobs started by cellranger.

Create a swarmfile (e.g. cellranger.swarm). For example:

```
cellranger mkfastq --run=./run1 --localcores=$SLURM_CPUS_PER_TASK --localmem=34
cellranger mkfastq --run=./run2 --localcores=$SLURM_CPUS_PER_TASK --localmem=34
cellranger mkfastq --run=./run3 --localcores=$SLURM_CPUS_PER_TASK --localmem=34
```

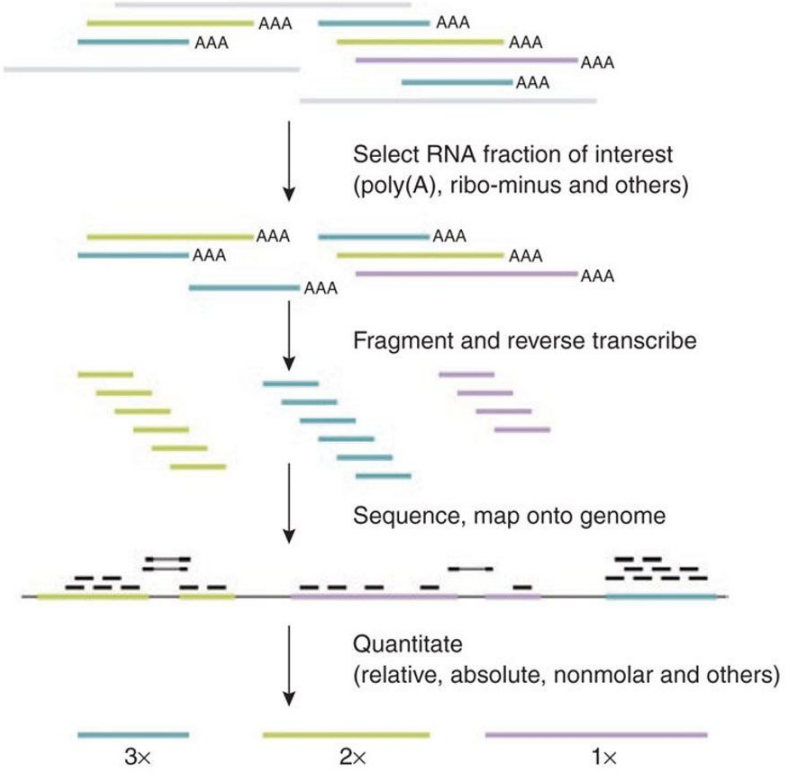
Submit this job using the swarm command.

```
swarm -f cellranger.swarm -g 35 -t 12 --module cellranger
```

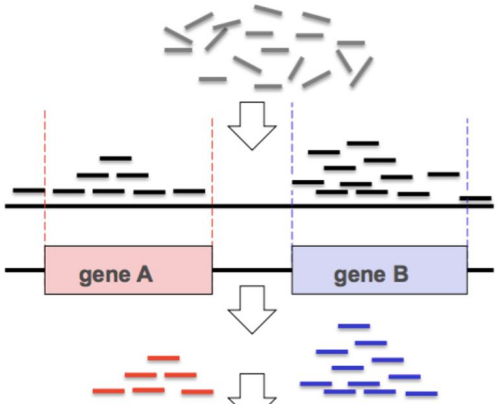
where

-g #	Number of Gigabytes of memory required for each process (1 line in the swarm command file)
-t #	Number of threads/CPUs required for each process (1 line in the swarm command file).
--module cellranger	Loads the cellranger module for each subjob in the swarm

Back to basics: Bulk RNA-seq



Back to basics: Bulk RNA-seq → differential expression



Fastq

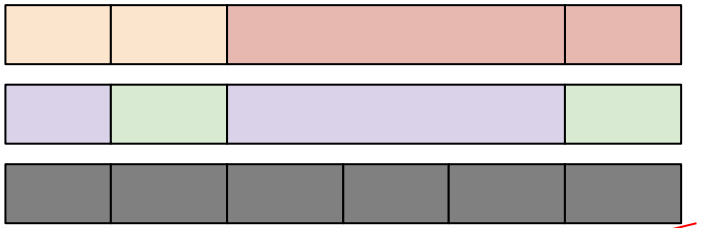
Bam (from HISAT2)

	Control 1	Control 2	Control 3	Sample 1	Sample 2	Sample 3
Gene A	6	5	7	17	10	11
Gene B	11	11	10	3	4	2
Gene C	200	150	355	50	1	3
Gene D	0	1	0	2	0	1

Counts (from featureCounts)

You might do:
Control v Sample
in DESeq2

Back to basics: Bulk RNA-seq → differential expression



Samples (index)

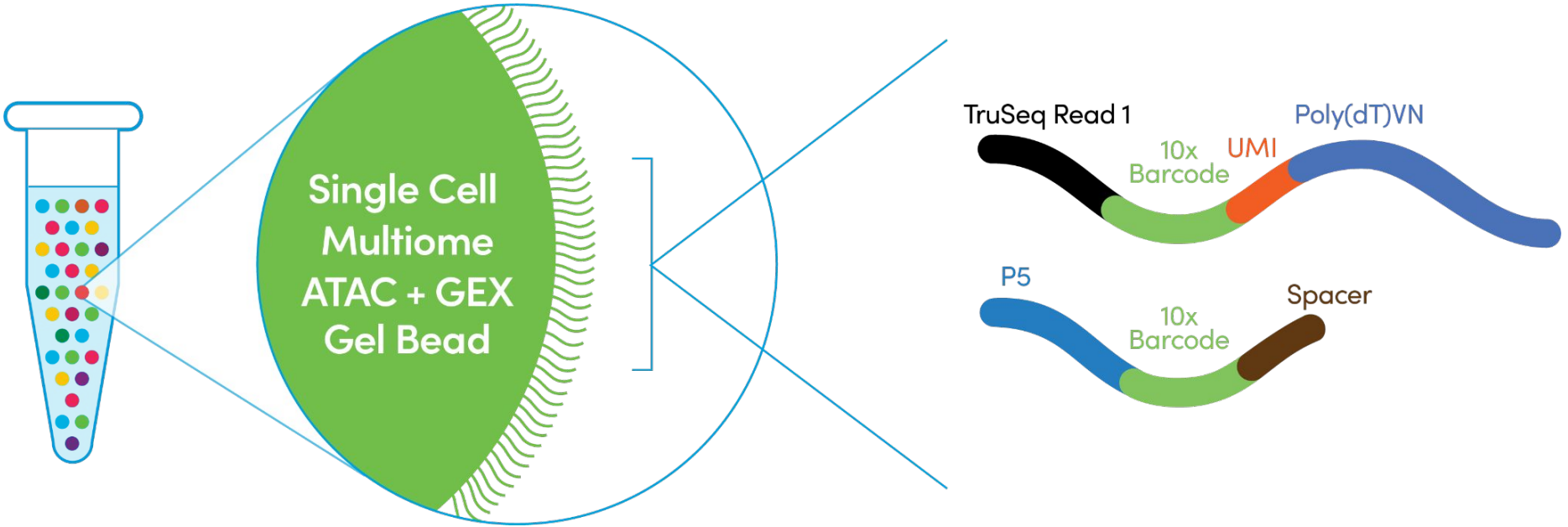
Cell type (need to define)

Cells (barcode)

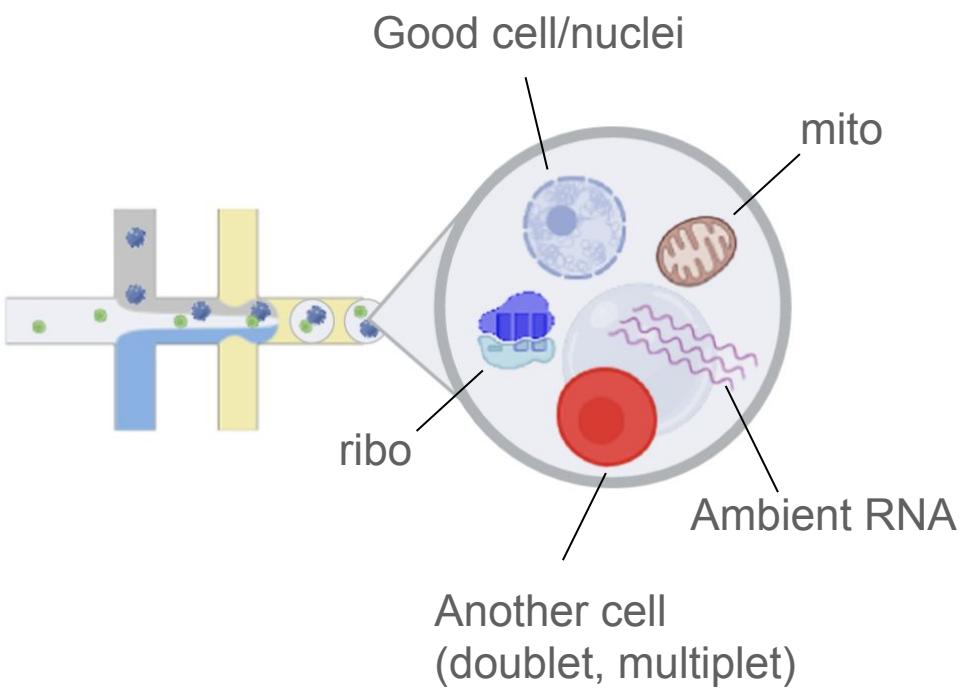
	Control 1	Control 2	Control 3	Sample 1	Sample 2	Sample 3
Gene A	6	5	7	17	10	11
Gene B	11	11	10	3	4	2
Gene C	200	150	355	50	1	3
Gene D	0	1	0	2	0	1

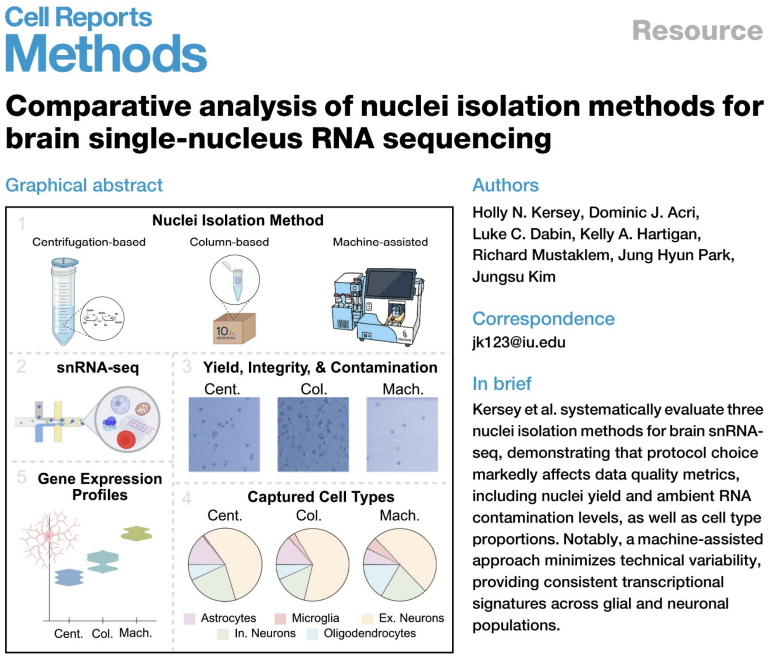
Added dimensionalities (and this is even only RNAseq)

Single Cell Additions: Barcodes



Single Cell Additions: Barcodes, Contamination




Cell Reports
Methods

Resource

Comparative analysis of nuclei isolation methods for brain single-nucleus RNA sequencing

Graphical abstract

Authors

Holly N. Kersey, Dominic J. Acri, Luke C. Dabin, Kelly A. Hartigan, Richard Mustaklem, Jung Hyun Park, Jungsu Kim

Correspondence

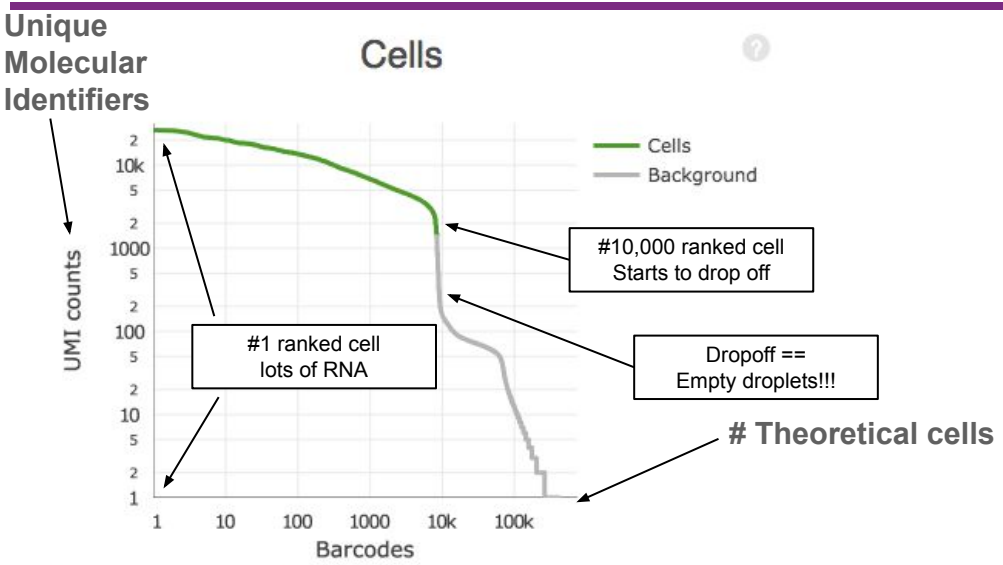
jk123@iu.edu

In brief

Kersey et al. systematically evaluate three nuclei isolation methods for brain snRNA-seq, demonstrating that protocol choice markedly affects data quality metrics, including nuclei yield and ambient RNA contamination levels, as well as cell type proportions. Notably, a machine-assisted approach minimizes technical variability, providing consistent transcriptional signatures across glial and neuronal populations.

What you put in determines what you get out, esp. for nuclei… PMID41875869

Single Cell Additions: Barcodes, Contamination, Sparsity

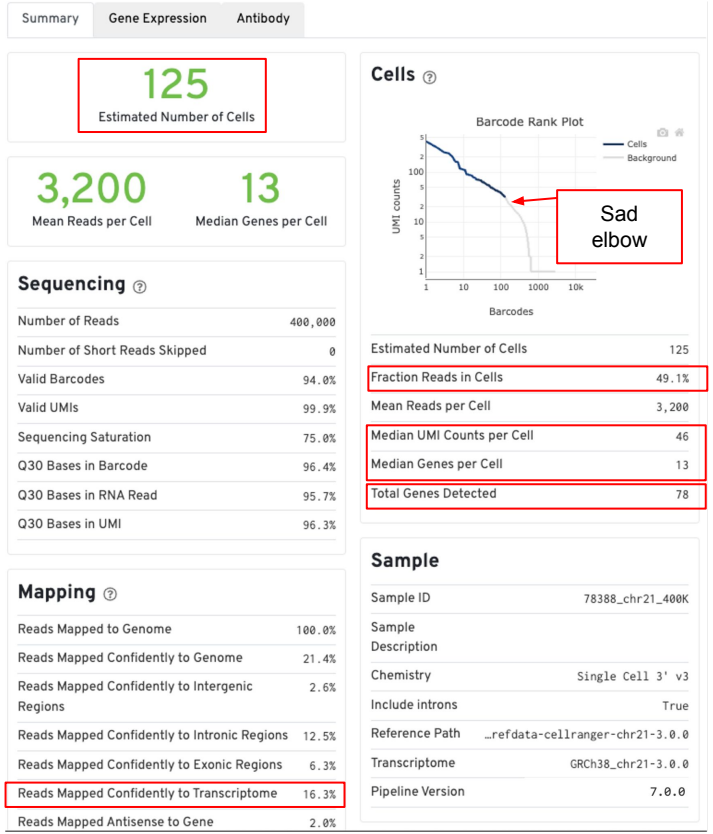


Estimated Number of Cells	8,381
Fraction Reads in Cells	93.1%
Mean Reads per Cell	93,552
Median Genes per Cell	1,297
Total Genes Detected	21,425
Median UMI Counts per Cell	4,084

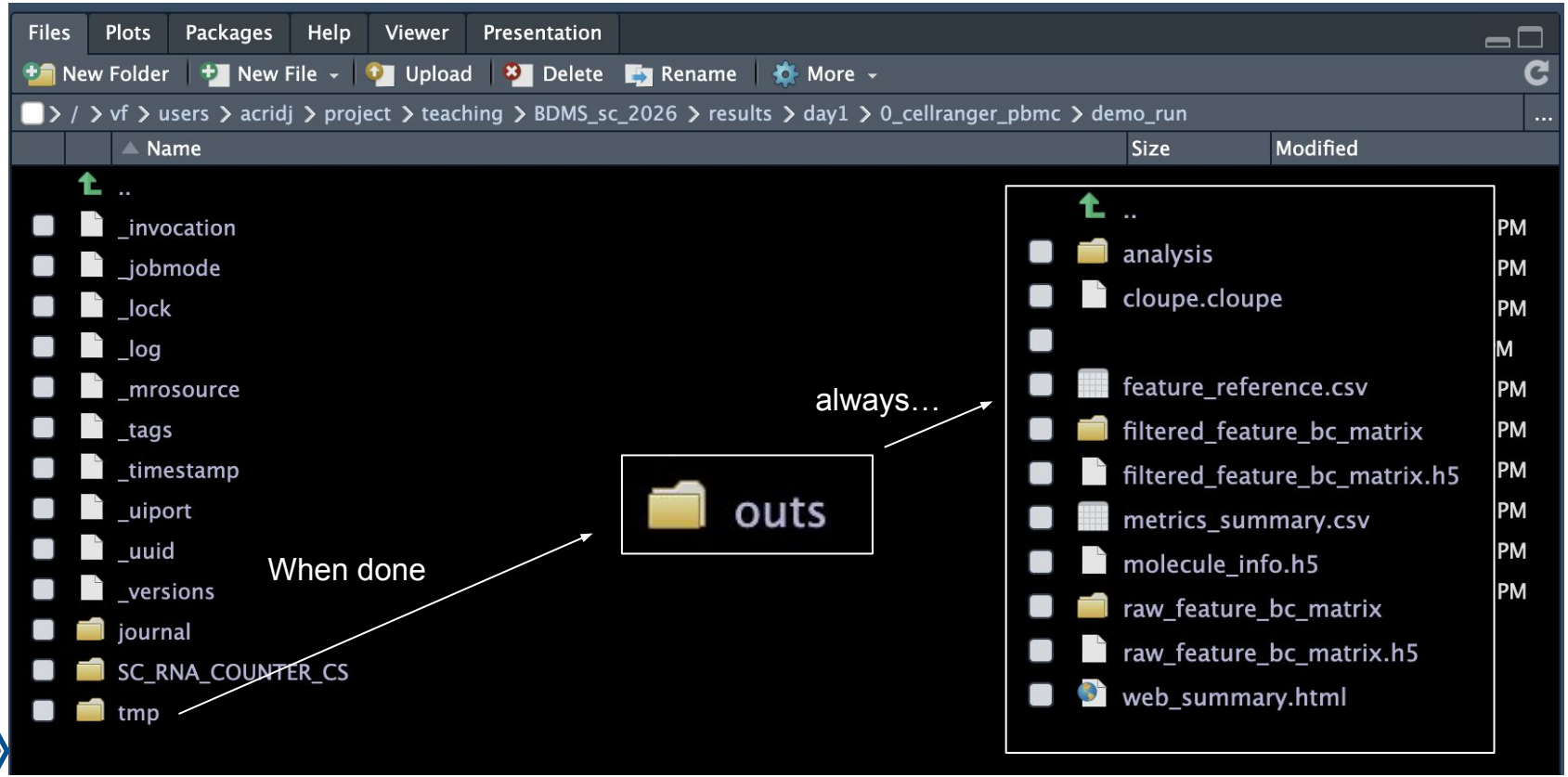
Nice!! RNA in cells

Ok… not whole genome, but…

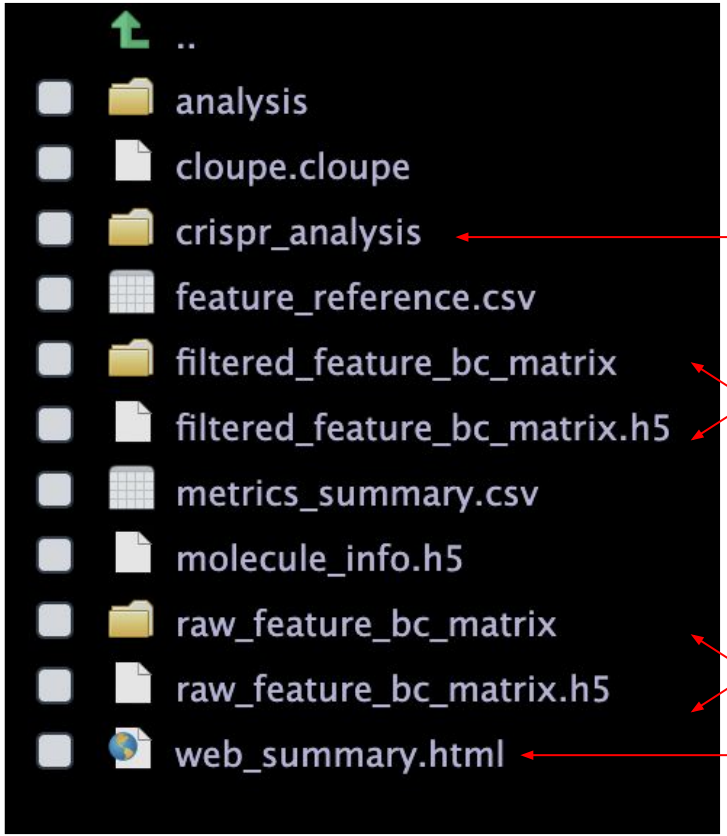
10,000 x 4,000 = 40M UMI Sequencing $$$



Let’s look at cellranger outputs…



Let's look at cellranger outputs…



There are different versions of analysis in cellranger… may have additional

Two legacy formats (folder of matrices or single .h5) can use either

Raw (unfiltered) for specific analyses NOT basic

The QC html (with elbow and stats)

Part 2: *Cellranger outputs → clusters*

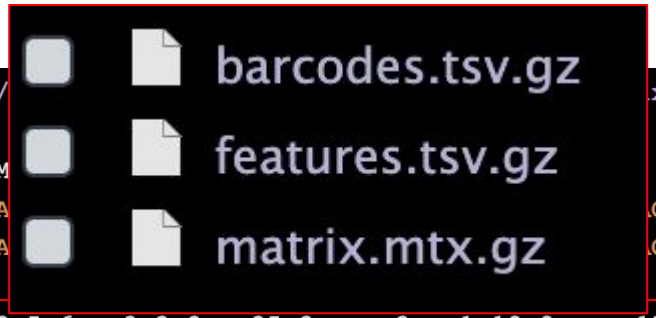
Take the cellranger (1 sample) and turn into seurat obj

```
####################
#the simplest load in possible
####################
test <- Seurat::Read10X(data.dir  = "./data/10x_mouse/filtered_feature_bc_matrix/")
test
obj_test <- CreateSeuratObject(test,assay = "RNA",project = "test_demo",min.cells = 3,min.features = 0)
obj_test
obj_test[["percent.mt"]] <- Seurat::PercentageFeatureSet(
  obj_test, pattern = "^mt-"
)
colnames(obj_test@meta.data)
VlnPlot(obj_test,features = c("nCount_RNA","nFeature_RNA","percent.mt"))
```

Take the cellranger (1 sample) and turn into seurat obj

genes

barcodes



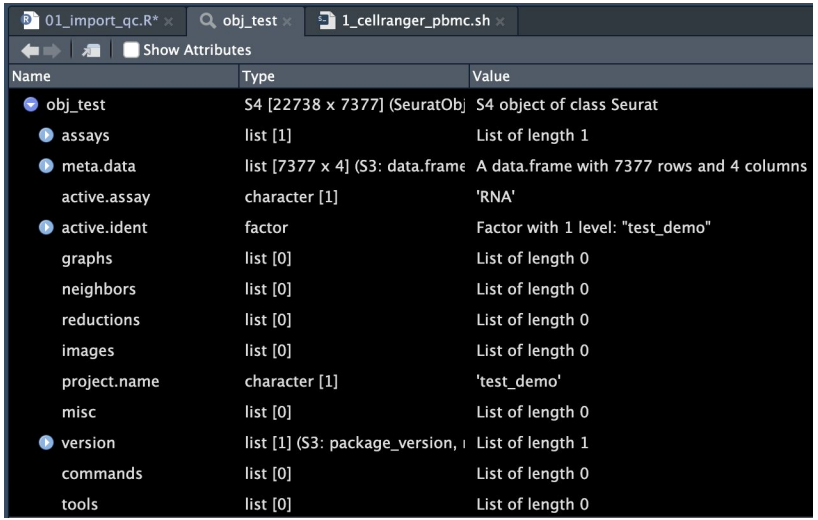

```
> test <- Seurat::Read10X(data.dir  = "./                                    x/")
> test
32285 x 7377 sparse Matrix of class "dgCM
  [[ suppressing 59 column names 'AAACCCA                                  CCCACAAAGGCTG-1' ... ]]
  [[ suppressing 59 column names 'AAACCCA                                  CCCACAAAGGCTG-1' ... ]]

Xkr4    3 3 9 12 19 8 2 1 . 5 . . 3 4 4 8 5 6 . 3 2 2 . 25 2 . . 2 . 1 18 8 . . 15 . 8 4 4 1 21 4 1 . 11 3 . 2 . .
Gm1992  . . 1  .  . . . . . . . . . 1 . 2 1 . . . . . .  3 . . . . 4  2 . . .  3 . . . . .  . 1 . .  2 . . . . .
Gm19938 . . 3  1  . . . . . 1 . . 2 1 1 . 2 . . 1 . . .  1 . . . 1 . .  1 1 . .  . . . . . 1  1 . . .  1 . . . . 1
Gm37381 . . .  .  . . . . . . . . . . . . . . . . . . .  . . . . . . .  . . . .  . . . . . . .  . . . .  . . . . . .
Rp1     . . .  .  . . . . . . . . . . . . . . . . 1 . . .  . . . . . . .  . . . .  . . . . . . .  . . . .  . . . . . .
Sox17   . . .  .  . . . . . . . . . . . . . . . . . . . .  . . . . . . .  . . . .  . . . . . . .  . . . .  . . . . . .
Gm37587 . . .  .  . . . . . . . . . . . . . . . . . . . .  . . . . . . .  . . . .  . . . . . . .  . . . .  . . . . . .
Gm37323 . . .  .  . . . . . . . . . . . . . . . . . . . .  . . . . . . .  . . . .  . . . . . . .  . . . .  . . . . . .
```

matrix

Seurat object structure

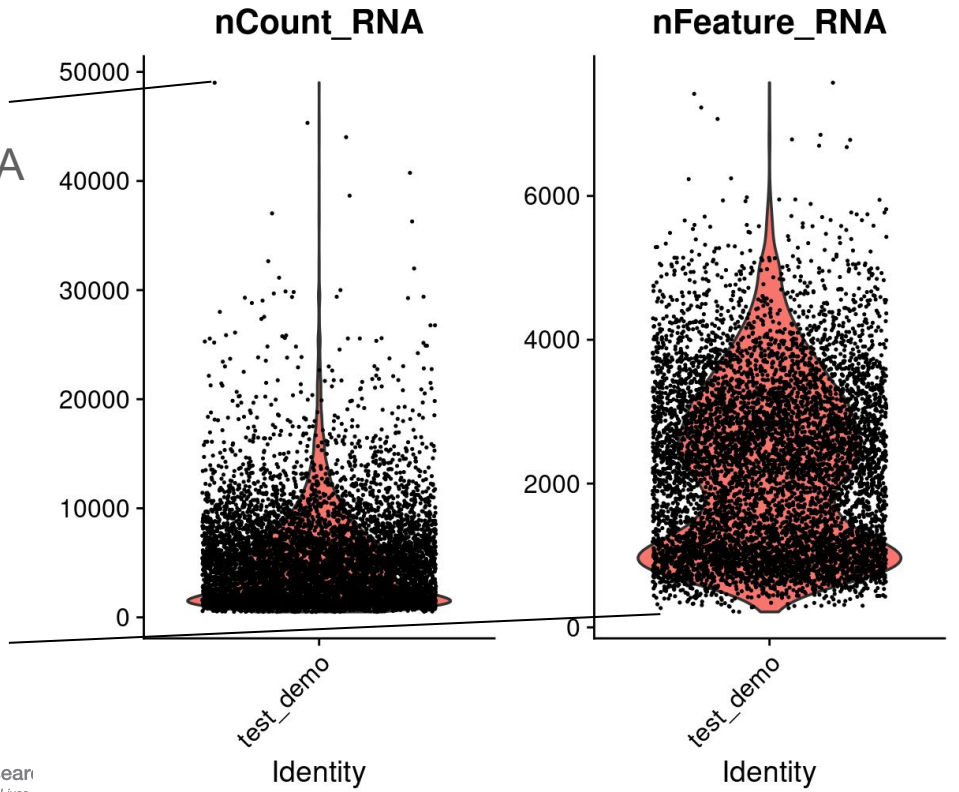
```
> obj_test
An object of class Seurat
22738 features across 7377 samples within 1 assay
Active assay: RNA (22738 features, 0 variable features)
 1 layer present: counts
```



QC per sample (in vignette, but wouldn't recommend)

This cell has a lot more RNA

This cell has a few genes



Most counts are mt-gene (bad if nuclei)

Keep cells..
< 10%
< 5%?
< 2%

Use LLMs as a tool to develop more reproducible code…

```
> seurat_samples
$`10x_mouse`
An object of class Seurat
22738 features across 7377 samples within 1 assay
Active assay: RNA (22738 features, 0 variable features)
 1 layer present: counts

$GSM8824189_S_1
An object of class Seurat
24648 features across 15241 samples within 1 assay
Active assay: RNA (24648 features, 0 variable features)
 1 layer present: counts

$GSM8824190_S_2
An object of class Seurat
24269 features across 19053 samples within 1 assay
Active assay: RNA (24269 features, 0 variable features)
 1 layer present: counts

$`A1331_Idol-Camk2a-WT-5xFAD`
An object of class Seurat
22513 features across 11518 samples within 1 assay
Active assay: RNA (22513 features, 0 variable features)
 1 layer present: counts

$`A2022_Idol-Camk2a-WT-5xFAD`
An object of class Seurat
23506 features across 13413 samples within 1 assay
Active assay: RNA (23506 features, 0 variable features)
 1 layer present: counts
```

```
obj <- base::merge(seurat_samples[[1]], y = seurat_samples[-1],
                   add.cell.ids = samples$sample_id, project = "BDMS_sc_2026")
DefaultAssay(obj) <- "RNA"
saveRDS(obj, file.path(output_dir, "01_merged_qc.rds"))
```

Simply put together or “merged”...

```
> obj
An object of class Seurat
25752 features across 64917 samples within 1 assay
Active assay: RNA (25752 features, 0 variable features)
 5 layers present: counts.10x_mouse, counts.GSM8824189_S_1, count
s.GSM8824190_S_2, counts.A1331_Idol-Camk2a-WT-5xFAD, counts.A2022_
Idol-Camk2a-WT-5xFAD
```

What happens if you take a Seurat object and just run…

Seurat 5.6.0 | Install | Get started | Vignettes | Extensions | FAQ | News | Reference | Archive

Seurat - Guided Clustering Tutorial

Compiled: April 28, 2026

Source: `vignettes/pbmc3k_tutorial.Rmd`

Set up the Seurat object

For this tutorial, we will be analyzing a dataset of Peripheral Blood Mononuclear Cells (PBMC) freely available from 10X Genomics. There are 2,700 single cells that were sequenced on the Illumina NextSeq 500. The raw data can be found here.

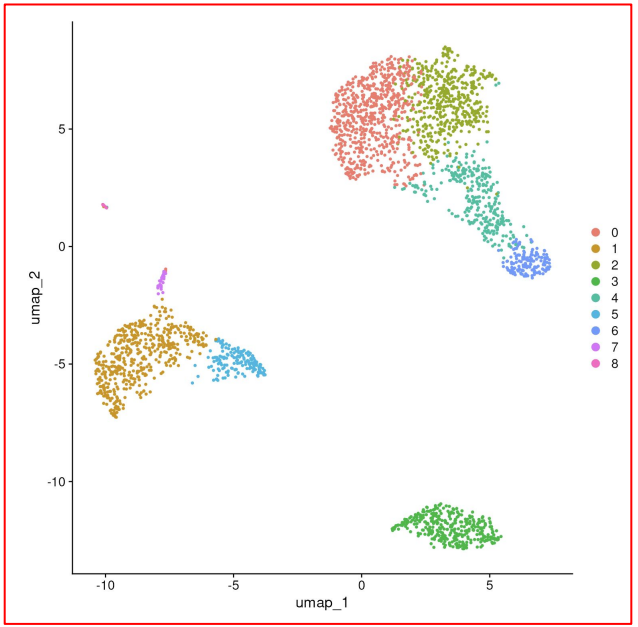
We start by reading in the data. The `Read10X()` function reads in the output of the cellranger pipeline from 10X, returning a unique molecular identified (UMI) count matrix. The values in this matrix represent the number of molecules for each feature (i.e. gene; row) that are detected in each cell (column). Note that more recent versions of cellranger now also output using the h5 file format, which can be read in using the `Read10X_h5()` function in Seurat.

We next use the count matrix to create a `Seurat` object. The object serves as a container that contains both data (like the count matrix) and analysis (like PCA, or clustering results) for a single-cell dataset. For more information, check out the docs for SeuratObject or the section on object interaction in our list of essential commands. For example, in Seurat v5, the count matrix is stored in `pbmc[["RNA"]]$counts`.

```
library(dplyr)
library(Seurat)
library(patchwork)

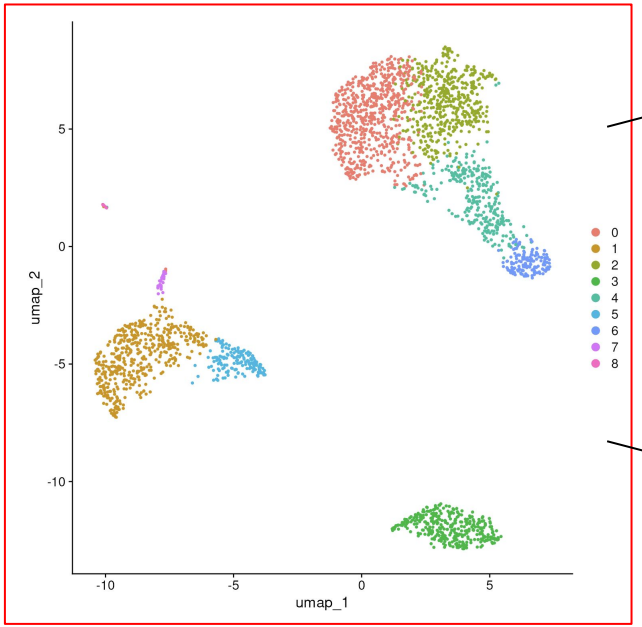
# Load the PBMC dataset
pbmc.data <- Read10X(data.dir = "/brahms/mollag/practice/filtered_gene_bc_matrices/hg19/")
# Initialize the Seurat object with the raw (non-normalized data).
pbmc <- CreateSeuratObject(counts = pbmc.data, project = "pbmc3k", min.cells = 3, min.features =
200)
pbmc
```

Famous “UMAP” but only 1 sample



What happens if you take a Seurat object and just run…

Famous “UMAP” but only 1 sample



Annotate: how to name

Assigning cell type identity to clusters

Fortunately in the case of this dataset, we can use canonical markers to easily match the unbiased clustering to known cell types:

Cluster ID	Markers	Cell Type
0	IL7R, CCR7	Naive CD4+ T
1	CD14, LYZ	CD14+ Mono
2	IL7R, S100A4	Memory CD4+
3	MS4A1	B
4	CD8A	CD8+ T
5	FCGR3A, MS4A7	FCGR3A+ Mono
6	GNLY, NKG7	NK
7	FCER1A, CST3	DC
8	PPBP	Platelet

Integrate: batches & technical artifacts

Seurat 5.6.0 | Install | Get started | Vignettes | Extensions | FAQ | News | Reference | Archive

Introduction to scRNA-seq integration

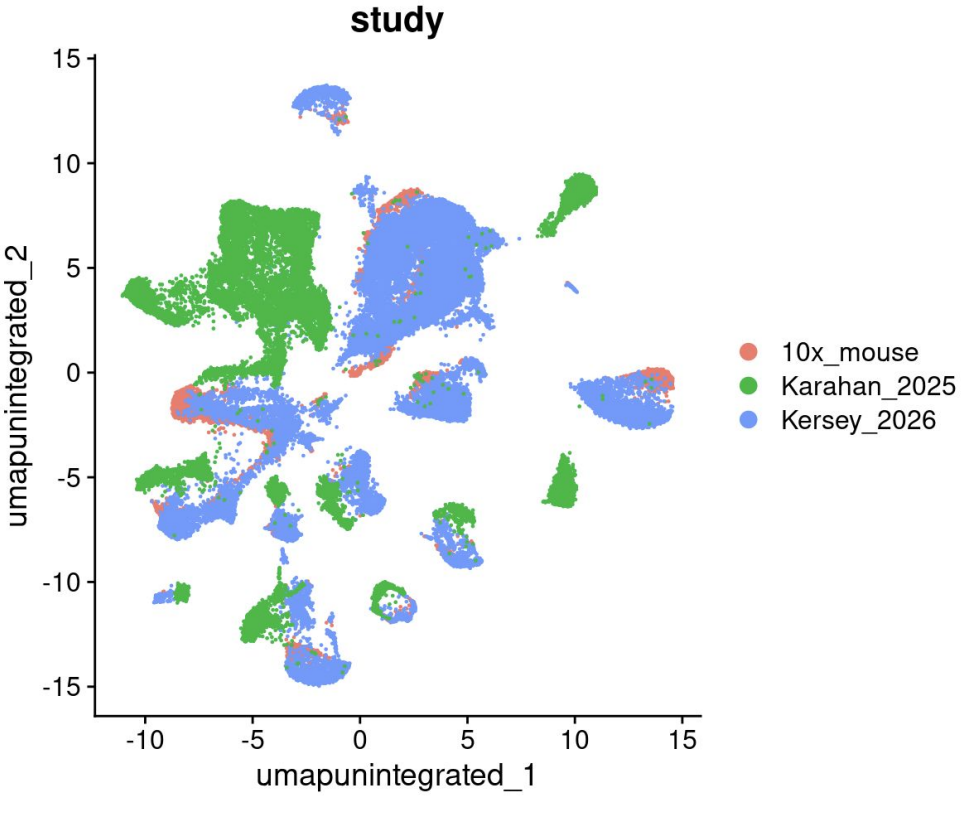
Compiled: May 22, 2026

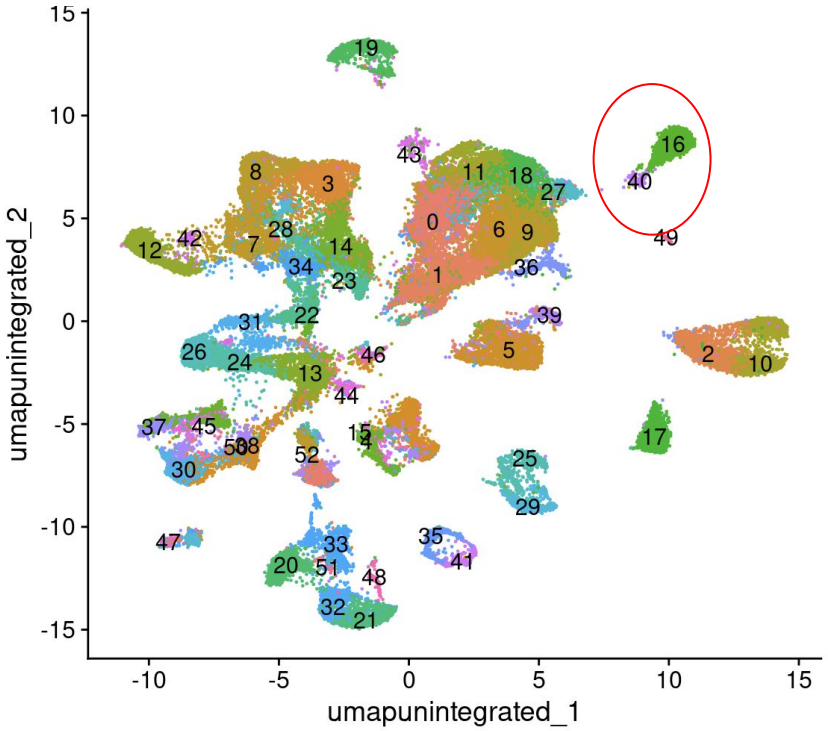
Source: `vignettes/integration_introduction.Rmd`

Introduction to scRNA-seq integration

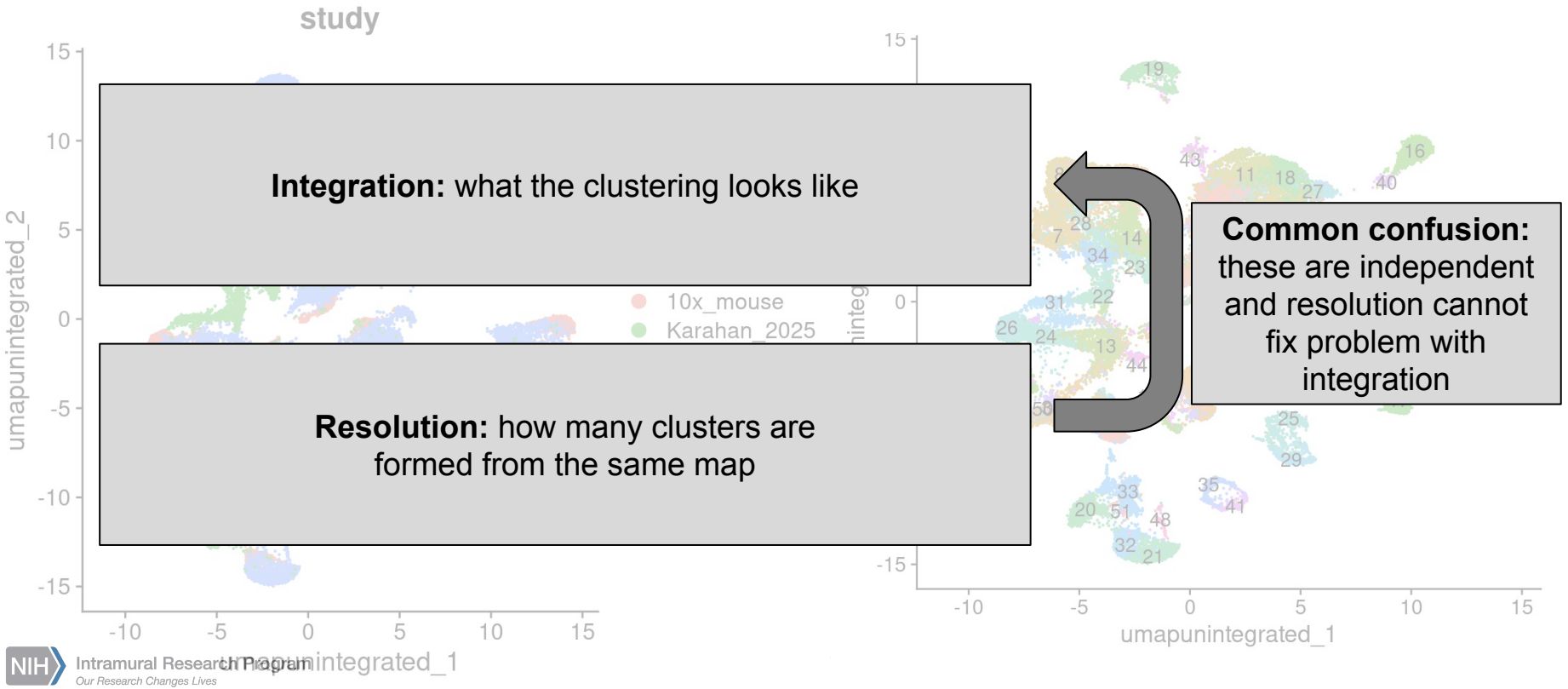
Integration of single-cell sequencing datasets, for example across experimental batches, donors, or conditions, is often an important step in scRNA-seq workflows. Integrative analysis can help to match shared cell types and states across datasets, which can boost statistical power, and most importantly, facilitate accurate comparative analysis across datasets. In previous versions of Seurat we introduced methods for integrative analysis, including our ‘anchor-based’ integration workflow. Many labs have also published powerful and pioneering methods, including Harmony and scVI, for integrative analysis. Please see our Integrating scRNA-seq data with multiple tools vignette.

If you don’t integrate…





If you don’t integrate…



Things to think about

Resolution

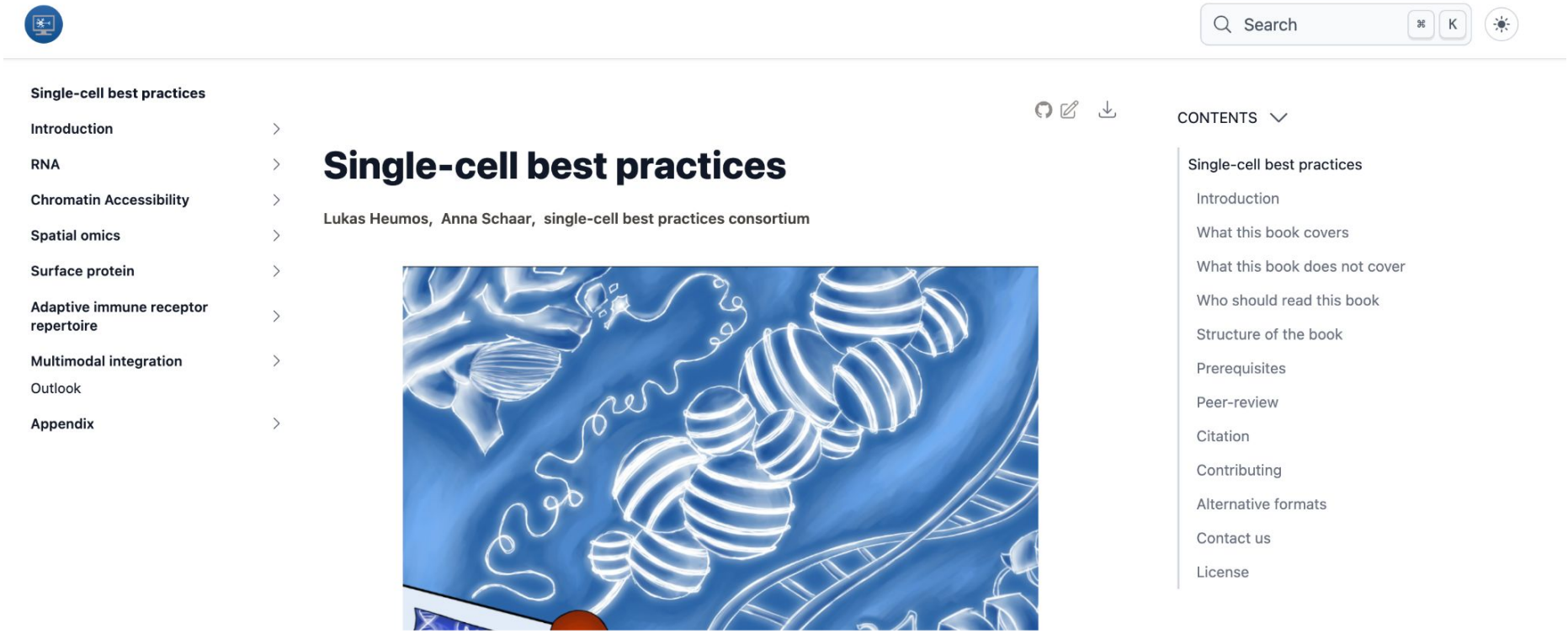
1. Higher resolution, more clusters from the same map → relative # (often 0.1-1.0)
2. “Overclustering” is a risk, or calling the same cell group as “two clusters”

Integration

1. Requires explicit batch handling, some discovery, or a decision about how to put samples together
2. “Under integrating” is a risk, what would you do with a cell type that is “only in one sample”?

Types of integration

https://www.sc-best-practices.org/



Types of integration

https://www.sc-best-practices.org/

Single-cell best practices
Introduction
RNA
Preprocessing and visualization
Identifying cellular structure
Clustering
Annotation
Data integration
Inferring trajectories
Dealing with conditions
Modeling mechanisms
Deconvolution
Chromatin Accessibility
Spatial omics
Surface protein
Adaptive immune receptor repertoire
Multimodal integration
Outlook
Appendix

Search ⌘ K

the human lung, where the variance attributable to different technical covariates was used to make this choice [Sikkema *et al.*, 2022].

Types of integration models

Methods that remove batch effects in scRNA-seq are typically composed of (up to) three steps:

1. Dimensionality reduction
2. Modeling and removing the batch effect
3. Projection back into a high-dimensional space

While modeling and removing the batch effect (Step 2) is the central part of any batch removal method, many methods first project the data to a lower dimensional space (Step 1) to improve the signal-to-noise ratio (see the dimensionality reduction chapter) and perform batch correction in that space to achieve better performance (see [Luecken *et al.*, 2021]). In the third step, a method may project the data back into the original high-dimensional feature space after removing the fitted batch effect, thereby outputting a batch-corrected gene expression matrix.

Batch-effect removal methods can vary in each of these three steps. They may use various linear or non-linear dimensionality reduction approaches, linear or non-linear batch effect models, and output different formats of batch-corrected data. Overall, we can divide methods for batch effect removal into 4 categories. In their order of development, these are global models, linear embedding models, graph-based methods, and deep learning approaches (Fig. I1).

CONTENTS

Motivation
Types of integration models
Batch removal complexity
Comparisons of data integration methods
Choosing an integration method
Dataset
Unintegrated data
Batch-aware feature selection
Variational autoencoder (VAE) based integration
Data preparation
Building the model
Training the model
Extracting the embedding
Calculate a batch-corrected UMAP
VAE integration using cell labels
Graph-based integration

Harmony integration

nature methods

ARTICLES

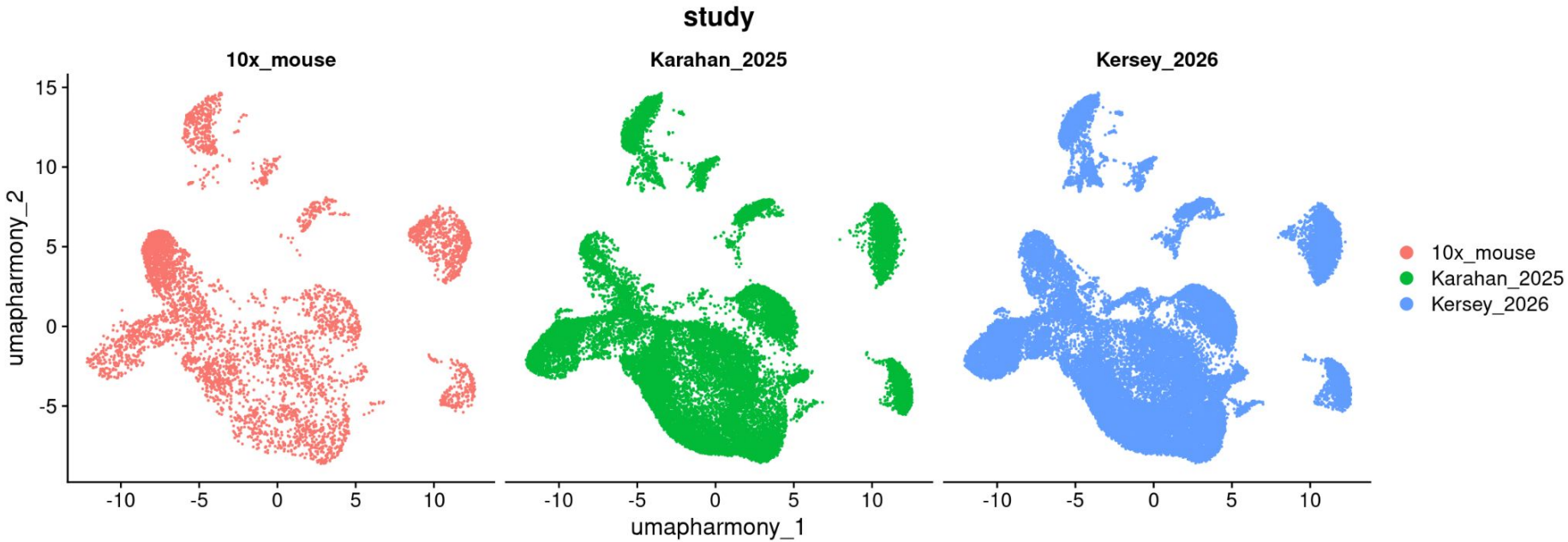
https://doi.org/10.1038/s41592-019-0619-0

Fast, sensitive and accurate integration of single-cell data with Harmony

Ilya Korsunsky[1,2,3,4], Nghia Millard[1,2,3,4], Jean Fan[5], Kamil Slowikowski[1,2,3,4], Fan Zhang[1,2,3,4], Kevin Wei[2], Yuriy Baglaenko[1,2,3,4], Michael Brenner[2], Po-ru Loh[1,3,4] and Soumya Raychaudhuri[1,2,3,4,6*]

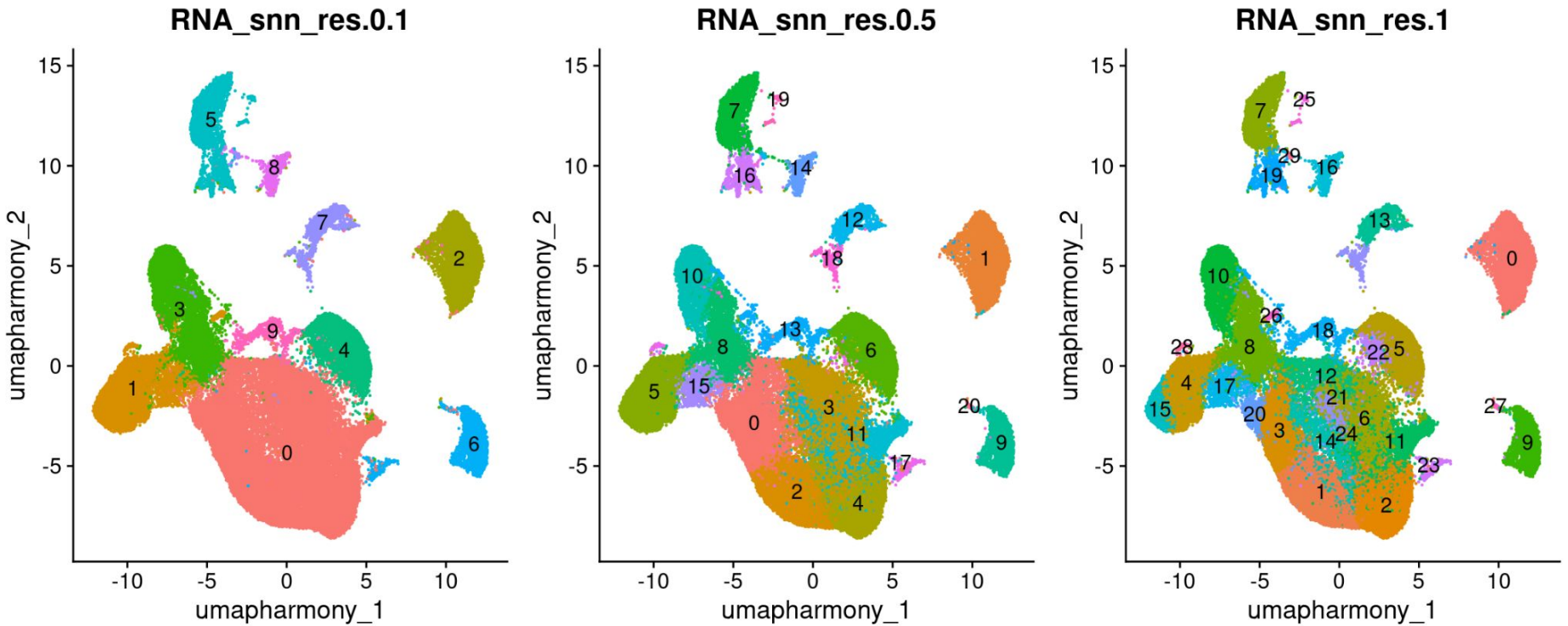
The emerging diversity of single-cell RNA-seq datasets allows for the full transcriptional characterization of cell types across a wide variety of biological and clinical conditions. However, it is challenging to analyze them together, particularly when datasets are assayed with different technologies, because biological and technical differences are interspersed. We present Harmony (https://github.com/immunogenomics/harmony), an algorithm that projects cells into a shared embedding in which cells group by cell type rather than dataset-specific conditions. Harmony simultaneously accounts for multiple experimental and biological factors. In six analyses, we demonstrate the superior performance of Harmony to previously published algorithms while requiring fewer computational resources. Harmony enables the integration of ~10^6 cells on a personal computer. We apply Harmony to peripheral blood mononuclear cells from datasets with large experimental differences, five studies of pancreatic islet cells, mouse embryogenesis datasets and the integration of scRNA-seq with spatial transcriptomics data.

Harmony integration

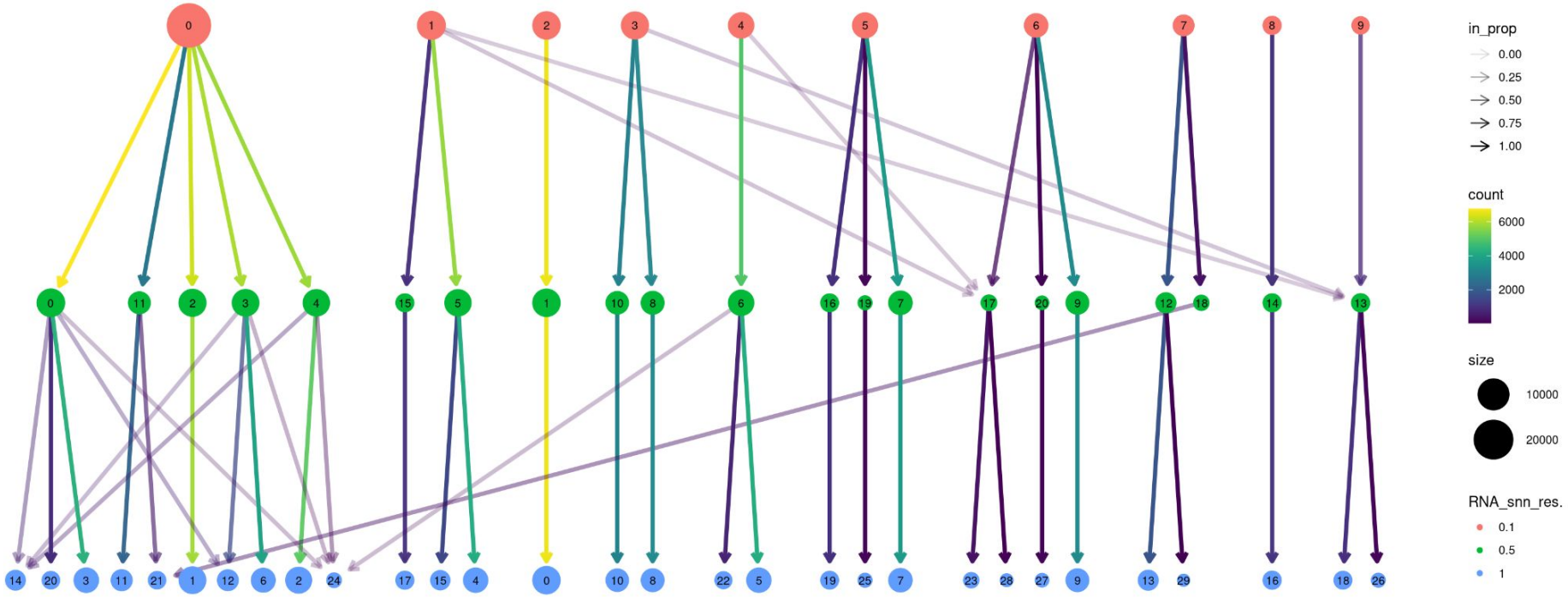


Once integrated…

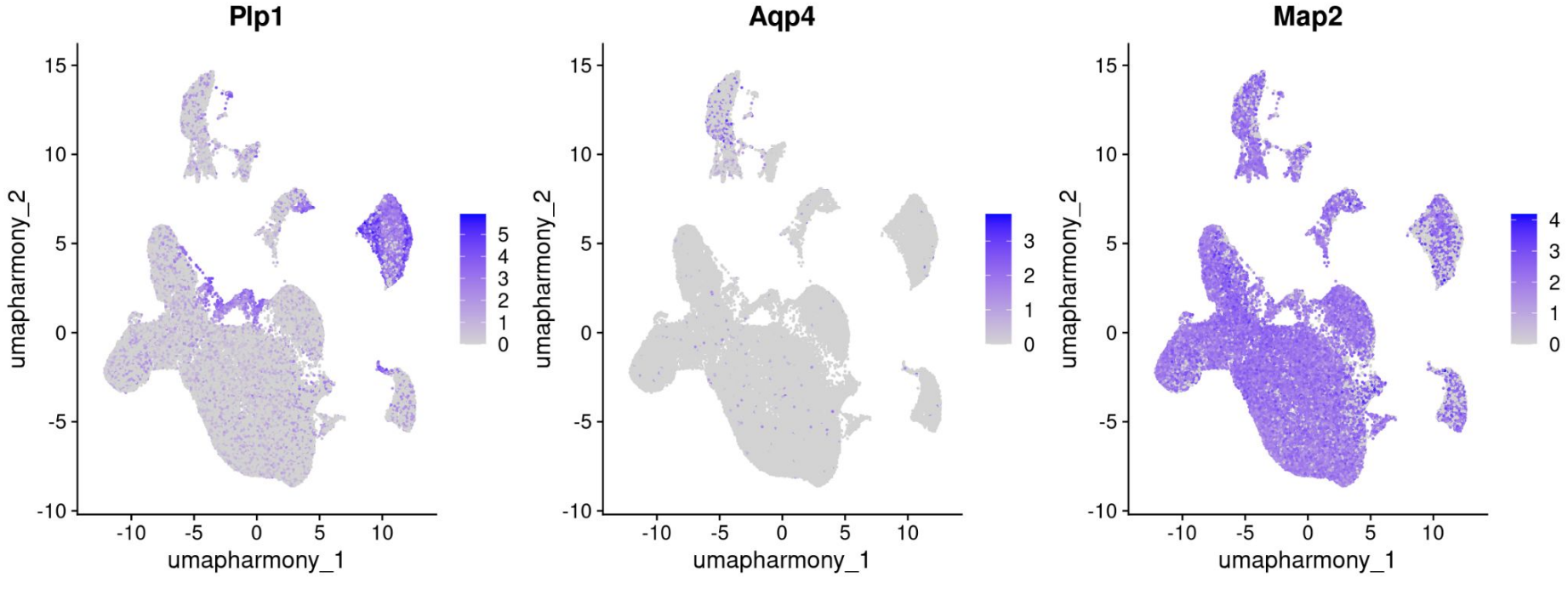
Increasing resolution… more clusters from the same map



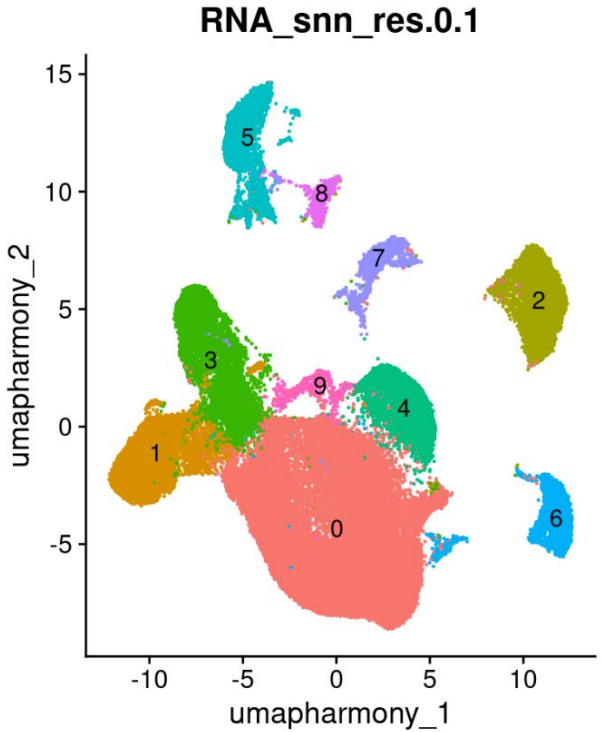
The tough thing about “resolutions”

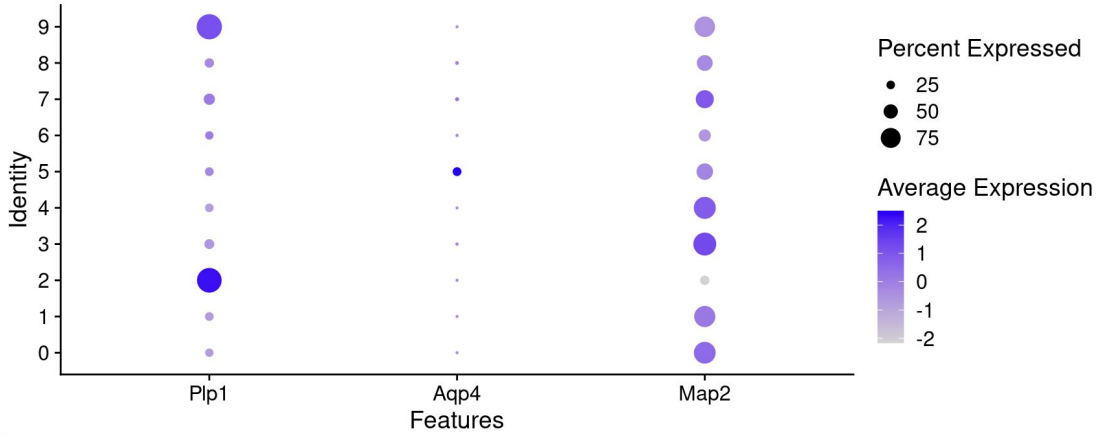


The tough thing about “annotations”

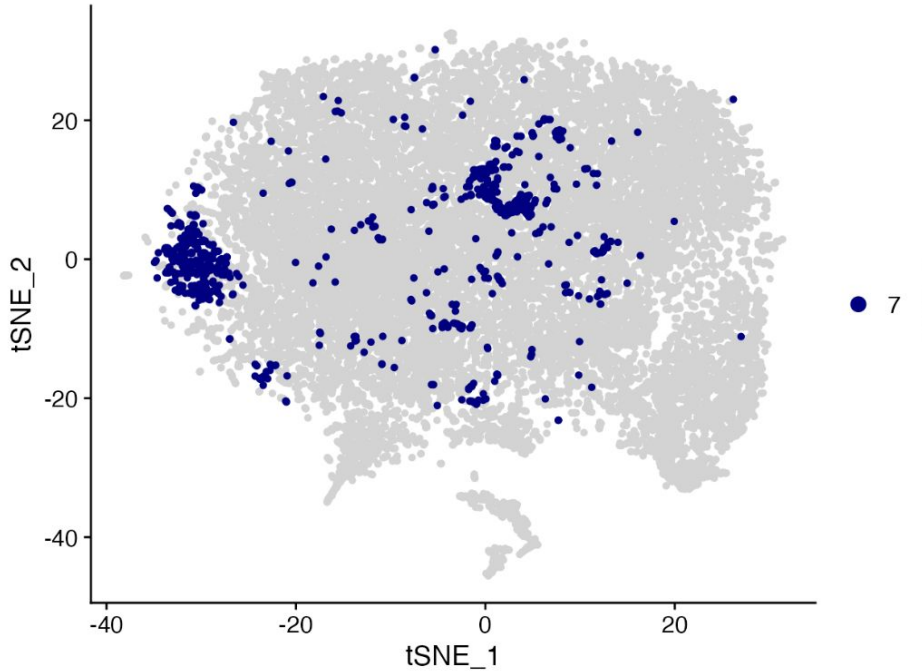


The tough thing about “annotations”

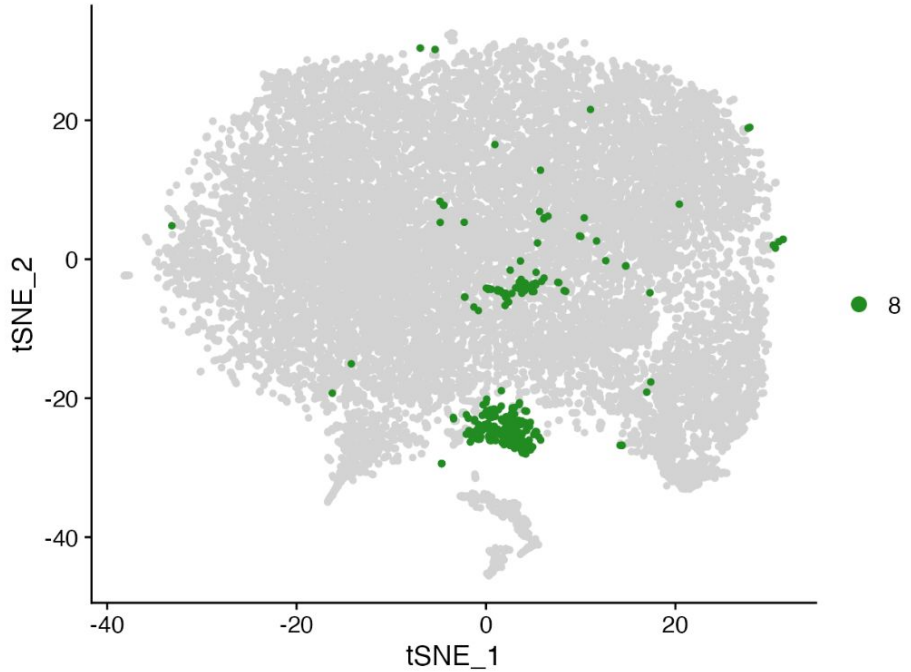




Calculate markers



Markers: **7** vs all



Markers: **8** vs all

Marker lists

Pct.1: percent cluster expressing

Pct.2: percent non-cluster expressing

```
> head(markers)
        p_val avg_log2FC pct.1 pct.2 p_val_adj cluster    gene
Pdzrn3      0   1.984200 0.733 0.259         0       0 Pdzrn3
Fstl4       0   1.526483 0.708 0.297         0       0  Fstl4
Cdh12       0   1.627394 0.820 0.416         0       0  Cdh12
Tafa1       0   1.894564 0.826 0.425         0       0  Tafa1
Slit3       0   1.358482 0.833 0.434         0       0  Slit3
Sv2b        0   1.089754 0.769 0.375         0       0   Sv2b
```

Cluster it is marker for…

Part 2.5: *Annotation Strategies*

Why do this at all?

Wait, how do I pick these names?



Richie Mustaklem, MS
IUSM Kim Lab (previously)

Seurat 5.5.1 | Install | Get started | Vignettes | Extensions | FAQ | News | Reference | Archive

Seurat - Guided Clustering Tutorial

Compiled: April 28, 2026

Source: `vignettes/pbmc3k_tutorial.Rmd`

Set up the Seurat object

For this tutorial, we will be analyzing a dataset of Peripheral Blood Mononuclear Cells (PBMC) freely available from 10X Genomics. There are 2,700 single cells that were sequenced on the Illumina NextSeq 500. The raw data can be found here.

We start by reading in the data. The `Read10X()` function reads in the output of the cellranger pipeline from 10X, returning a unique molecular identified (UMI) count matrix. The values in this matrix represent the number of molecules for each feature (i.e. gene; row) that are detected in each cell (column). Note that more recent versions of cellranger now also output using the h5 file format, which can be read in using the `Read10X_h5()` function in Seurat.

We next use the count matrix to create a `Seurat` object. The object serves as a container that contains both data (like the count matrix) and analysis (like PCA, or clustering results) for a single-cell dataset. For more information, check out the docs for SeuratObject or the section on object interaction in our list of essential commands. For example, in Seurat v5, the count matrix is stored in `pbmc[["RNA"]]$counts`.

```
library(dplyr)
library(Seurat)
library(patchwork)

# Load the PBMC dataset
pbmc.data <- Read10X(data.dir = "/brahms/mollag/practice/filtered_gene_bc_matrices/hg19/")
# Initialize the Seurat object with the raw (non-normalized data).
pbmc <- CreateSeuratObject(counts = pbmc.data, project = "pbmc3k", min.cells = 3, min.features =
200)
pbmc
```

```
## An object of class Seurat
## 13714 features across 2700 samples within 1 assay
## Active assay: RNA (13714 features, 0 variable features)
##  1 layer present: counts
```

▶ What does data in a count matrix look like?

Assigning cell type identity to clusters

Fortunately in the case of this dataset, we can use canonical markers to easily match the unbiased clustering to known cell types:

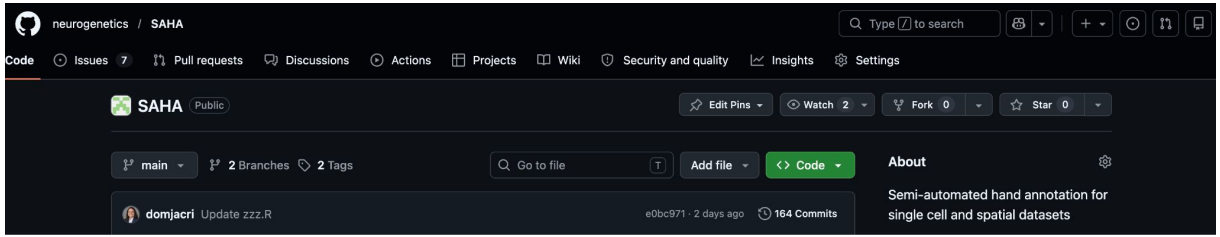
Cluster ID	Markers	Cell Type
0	IL7R, CCR7	Naive CD4+ T
1	CD14, LYZ	CD14+ Mono
2	IL7R, S100A4	Memory CD4+
3	MS4A1	B
4	CD8A	CD8+ T
5	FCGR3A, MS4A7	FCGR3A+ Mono
6	GNLY, NKG7	NK
7	FCER1A, CST3	DC
8	PPBP	Platelet

```
new.cluster.ids <- c("Naive CD4 T", "CD14+ Mono", "Memory CD4 T", "B", "CD8 T", "FCGR3A+ Mono",
    "NK", "DC", "Platelet")
names(new.cluster.ids) <- levels(pbmc)
pbmc <- RenameIdents(pbmc, new.cluster.ids)
DimPlot(pbmc, reduction = "umap", label = TRUE, pt.size = 0.5) + NoLegend()
```

SAHA pipeline (annotation for any sc/spatial)

Semi-automated annotation refinement accelerates cell-type identification in brain spatial and single-cell studies

Dominic J. Acri[1,§], Richard Mustaklem[2,§], Liam Horan-Portelance[1,3], Luke C. Dabin[2], Jung Hyun Park[2], Kelly Hartigan[2], Holly Kersey[2], Monica E. Mesecar[1], J. Raphael Gibbs[1], Mark R. Cookson[1]*, Jungsu Kim[2]*

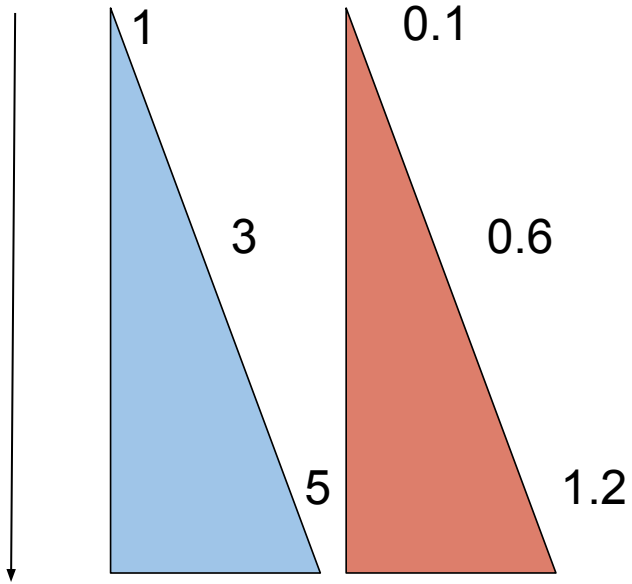


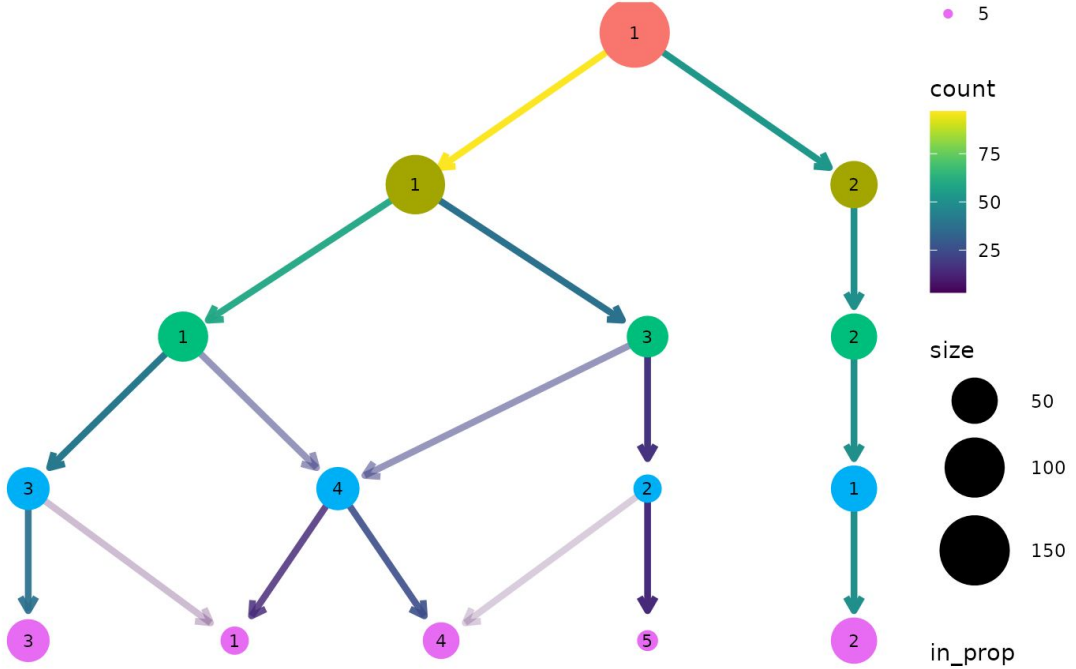
Interactive task:

Go to github/neurogenetics

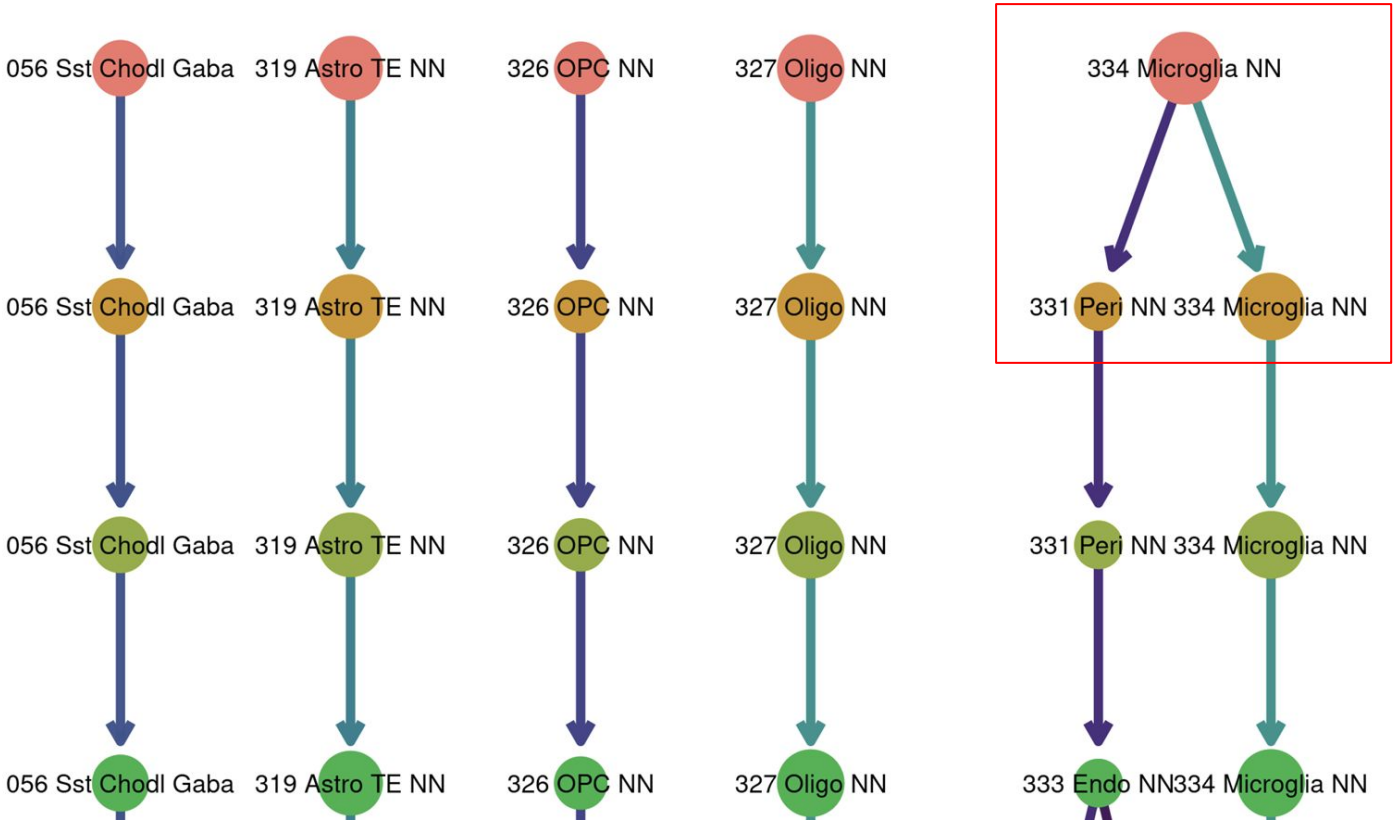
Find
1) SAHA
2) SAHAdata
3) download them

Can we name the clusters, quickly, and when?



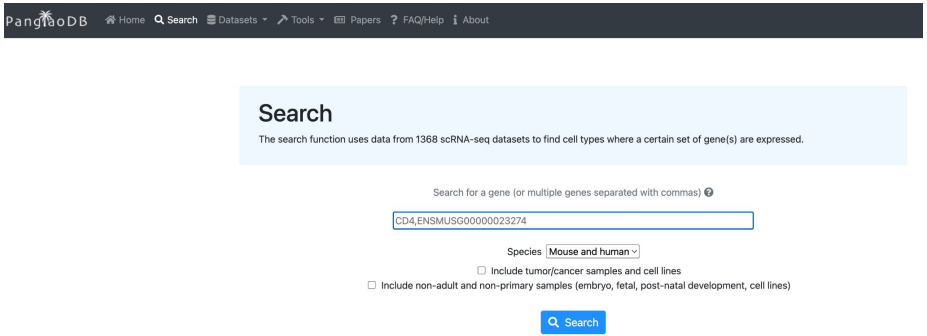


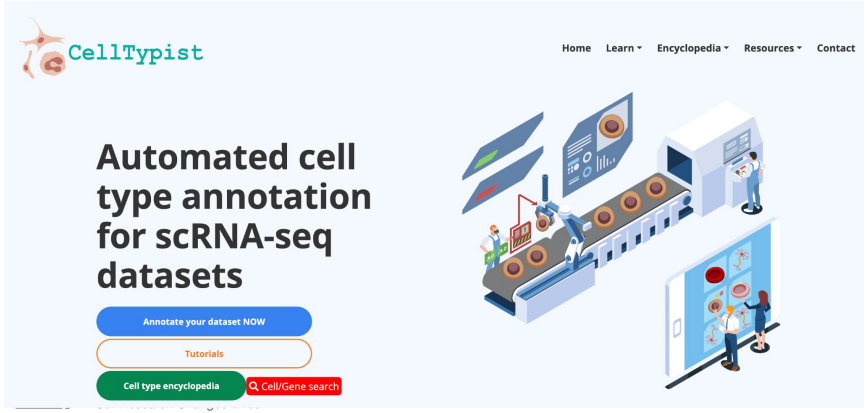
What we really want…

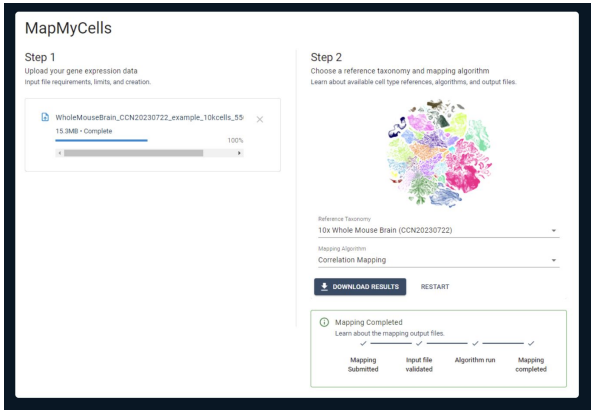


What exists out there…



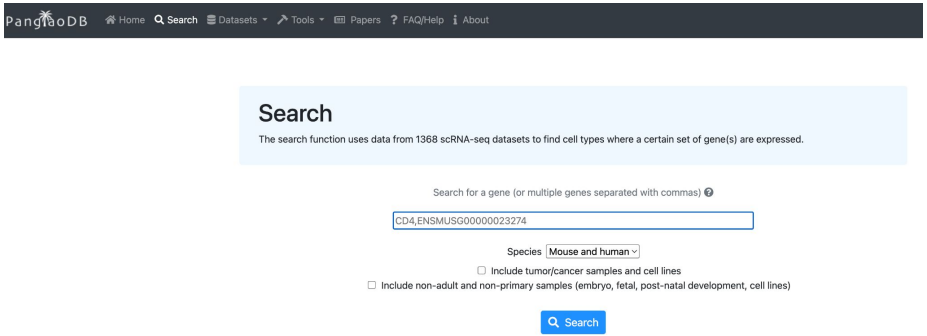


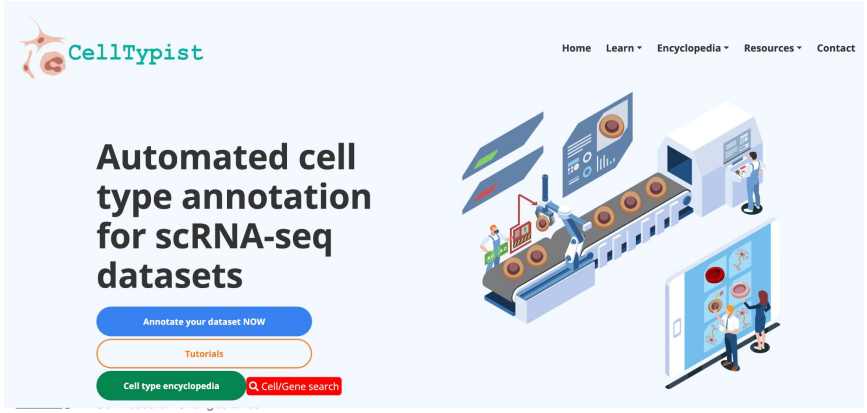


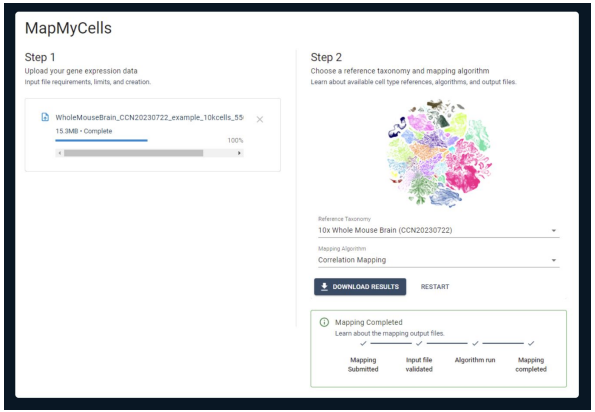


What exists out there…



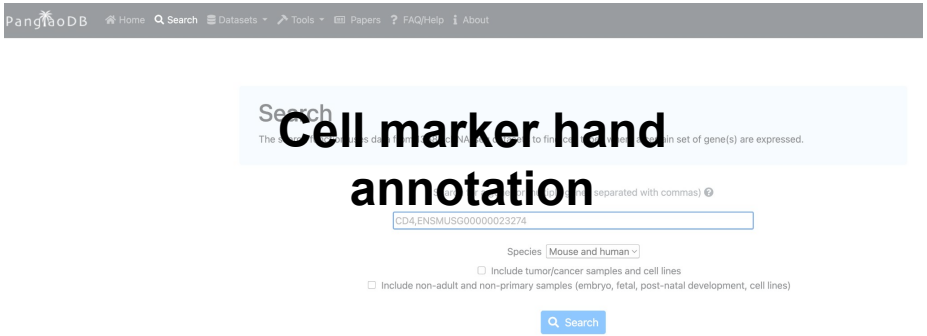


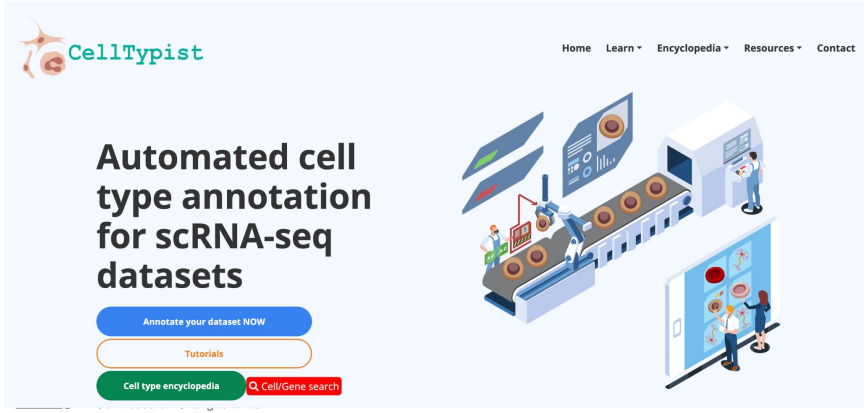


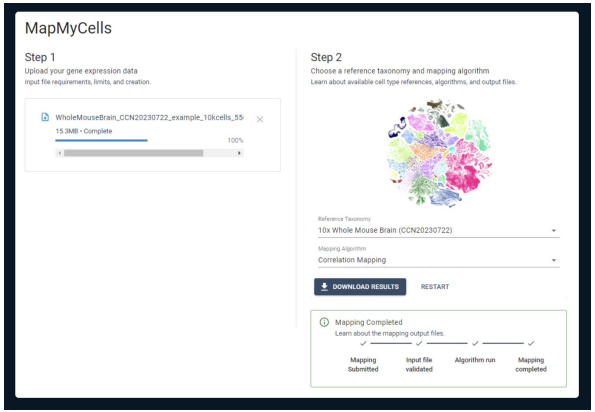


What exists out there…



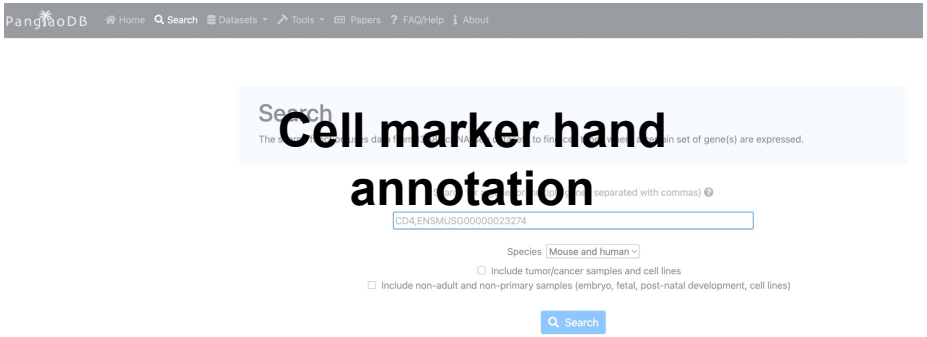


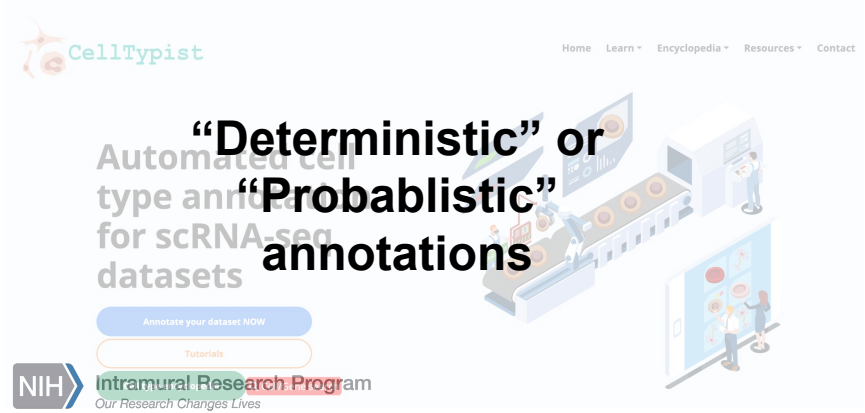


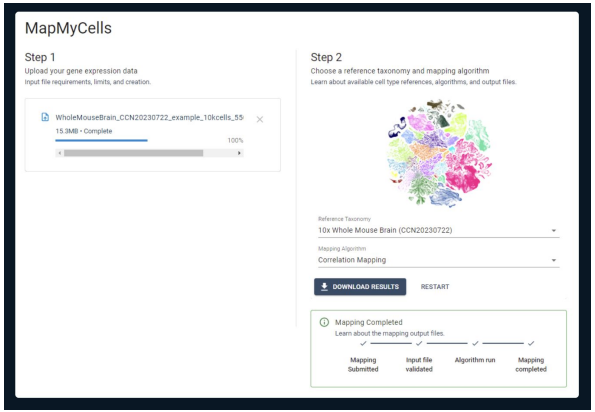


What exists out there…

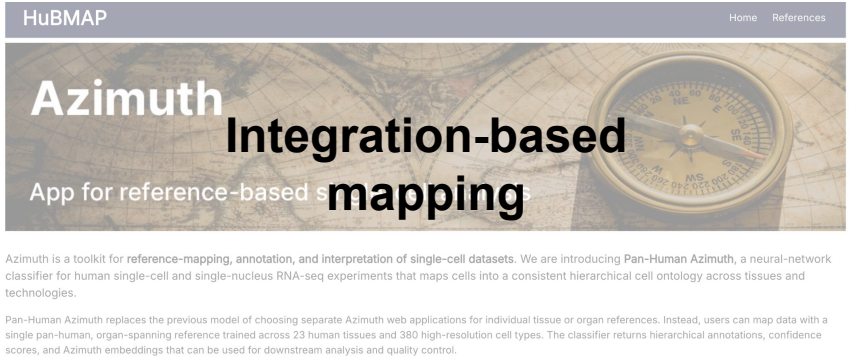




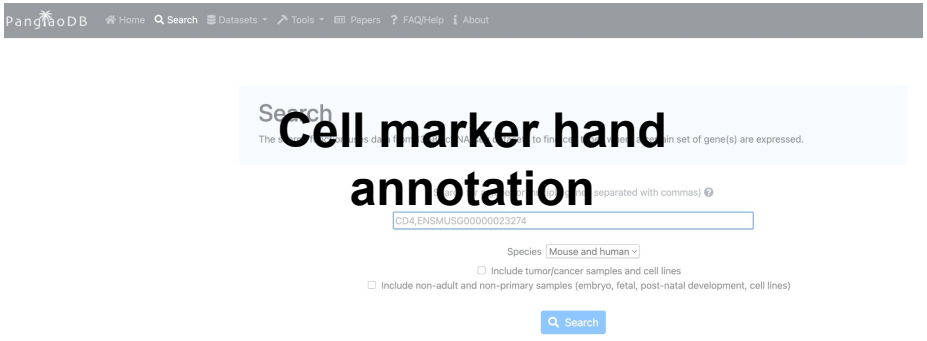




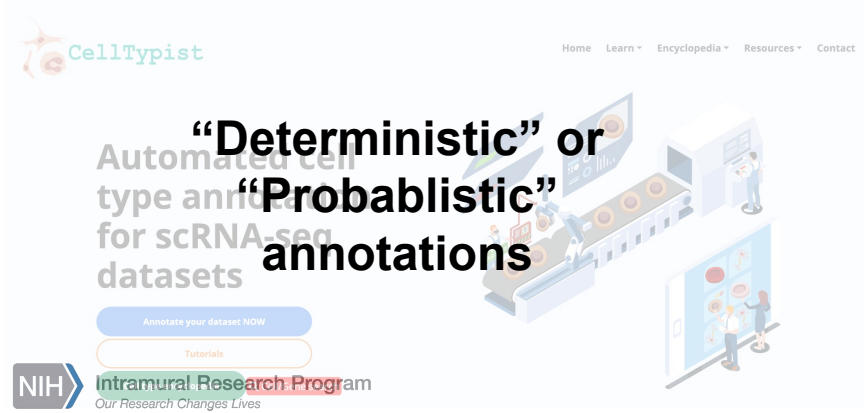
What exists out there…



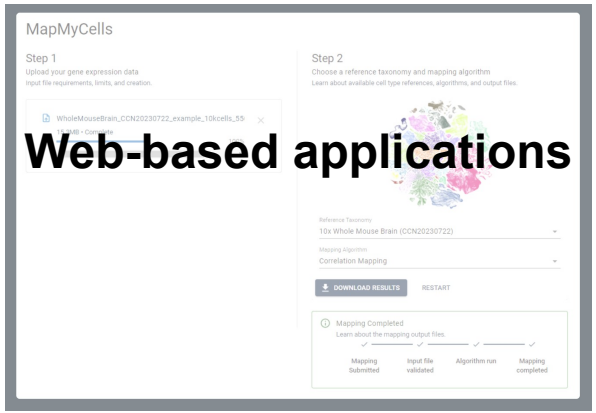
Integration-based mapping



Cell marker hand annotation



"Deterministic" or "Probablistic" annotations

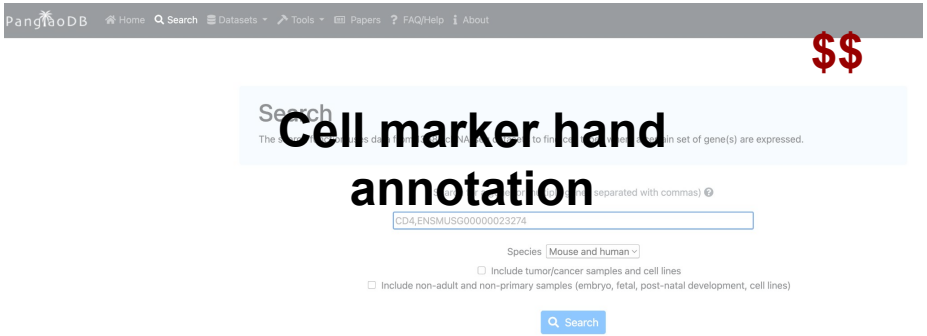


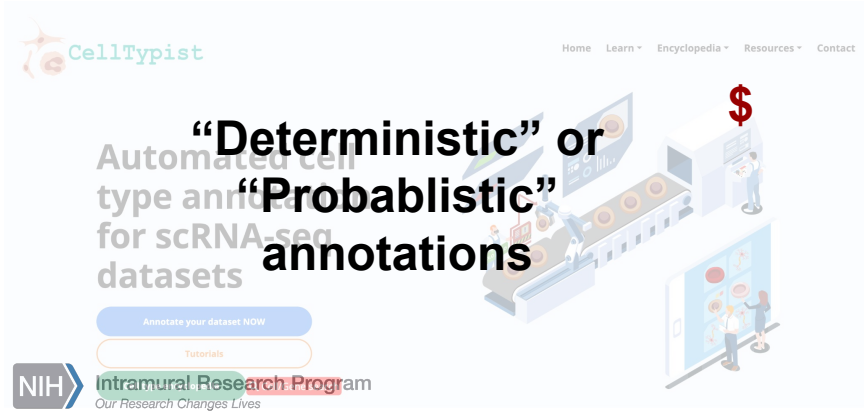
Web-based applications

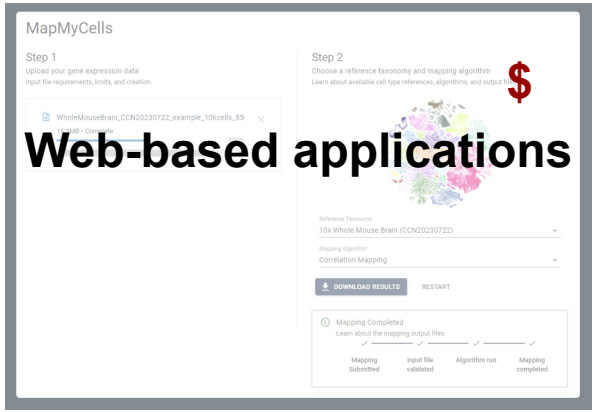
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Azimuth
App for reference-based
Integration-based mapping
Azimuth is a toolkit for reference-mapping, annotation, and interpretation of single-cell datasets. We are introducing Pan-Human Azimuth, a neural-network classifier for human single-cell and single-nucleus RNA-seq experiments that maps cells into a consistent hierarchical cell ontology across tissues and technologies.
Pan-Human Azimuth replaces the previous model of choosing separate Azimuth web applications for individual tissue or organ references. Instead, users can map data with a single pan-human, organ-spanning reference trained across 23 human tissues and 380 high-resolution cell types. The classifier returns hierarchical annotations, confidence scores, and Azimuth embeddings that can be used for downstream analysis and quality control.

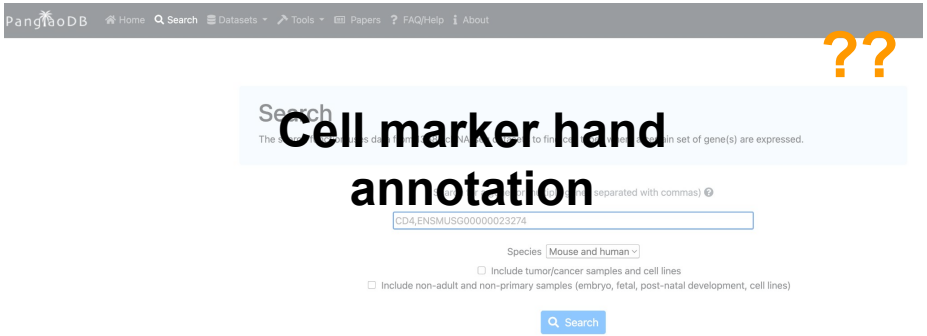

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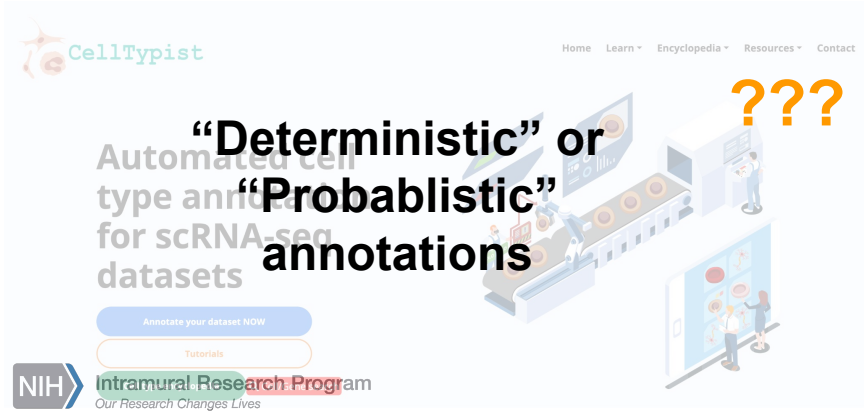

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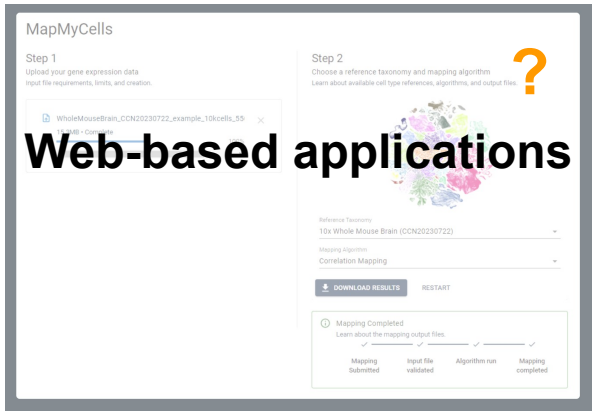

MapMyCells
Step 1
Step 2
$
Web-based applications
10x Whole Mouse Brain (CCN20230722)
Correlation Mapping
DOWNLOAD RESULTS
RESTART
Mapping Completed

Interpretation



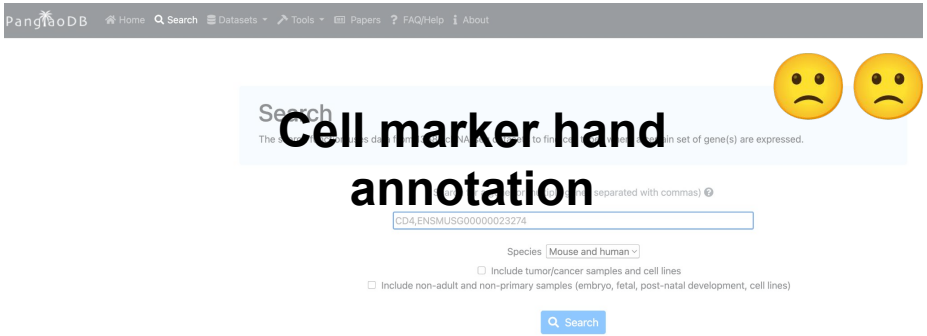


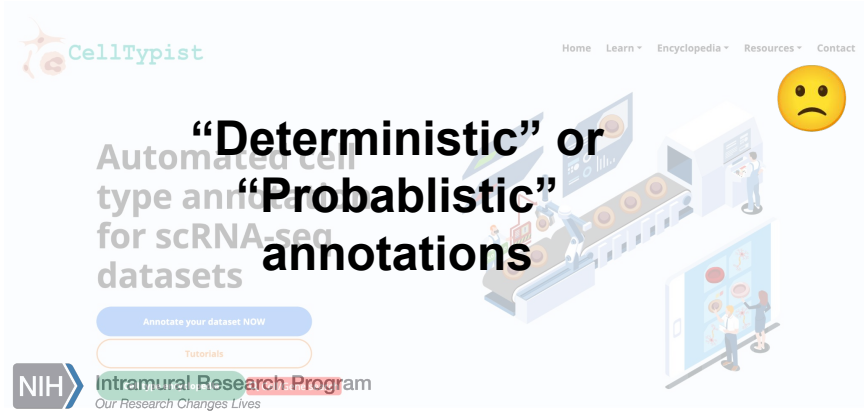


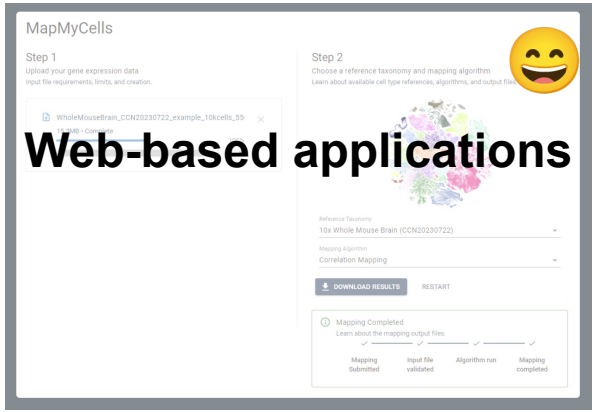


Reproducibility

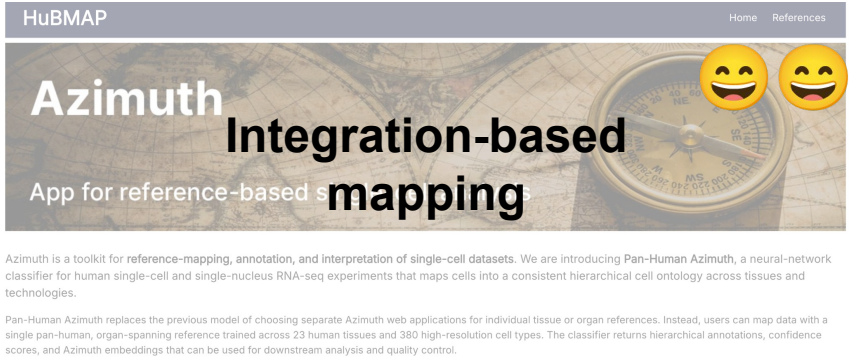


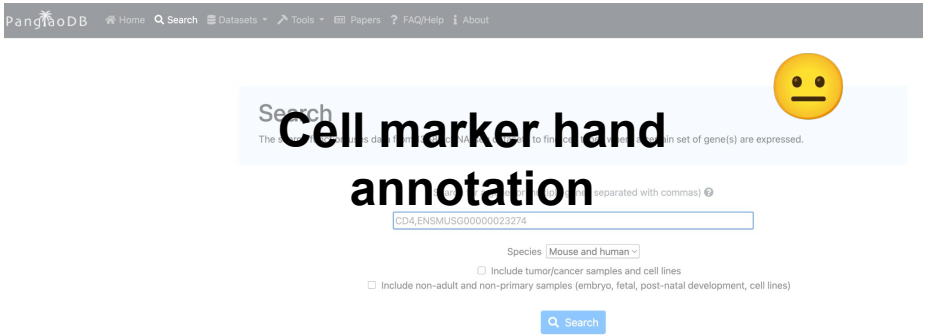


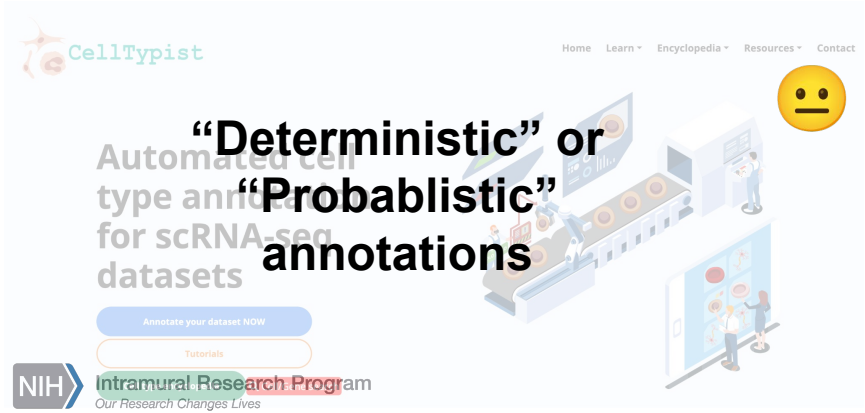


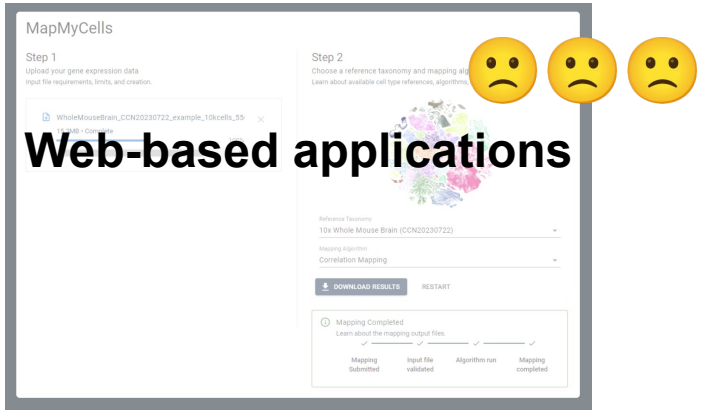


Safety!!!! (Human data)

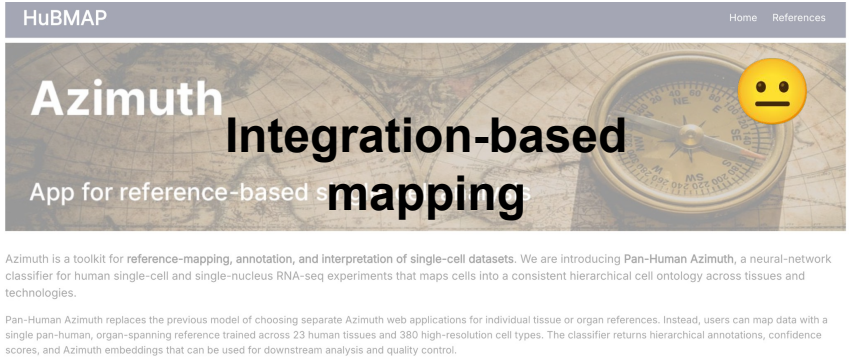


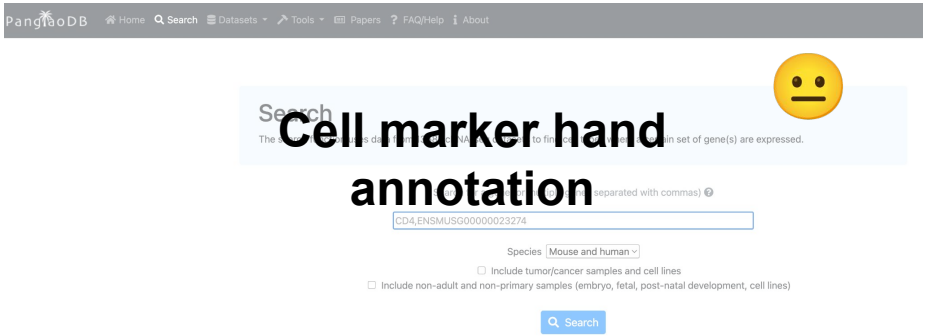


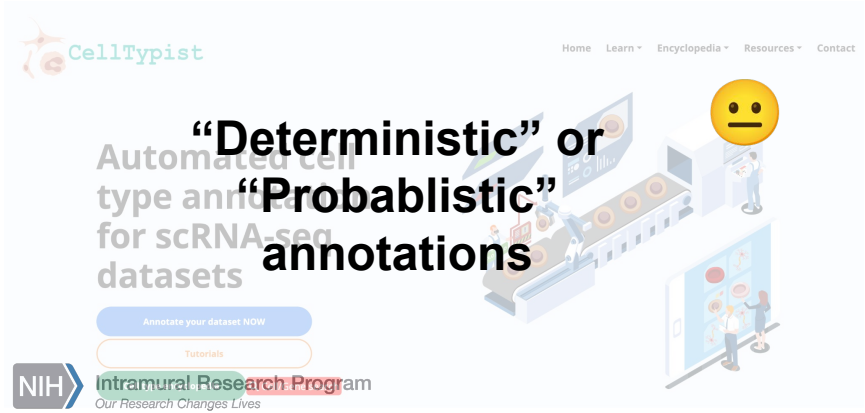


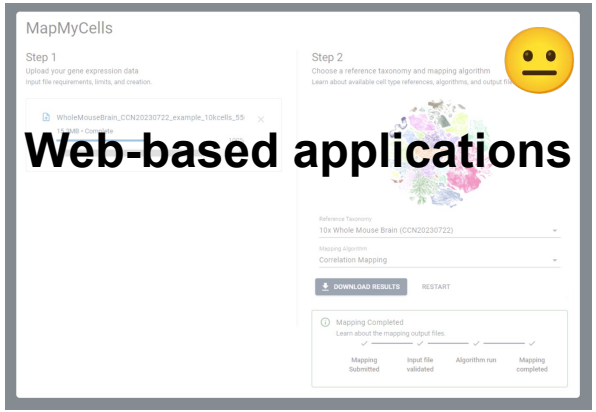


Truth?

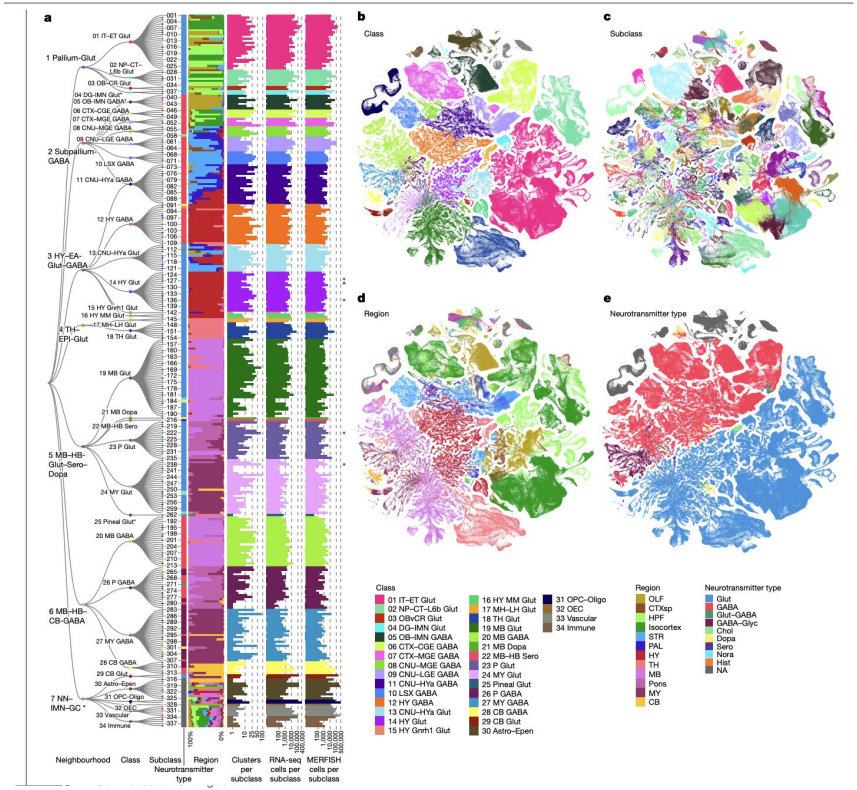

HuBMAP
Home
References
Azimuth
App for reference-based
Integration-based mapping
Azimuth is a toolkit for reference-mapping, annotation, and interpretation of single-cell datasets. We are introducing Pan-Human Azimuth, a neural-network classifier for human single-cell and single-nucleus RNA-seq experiments that maps cells into a consistent hierarchical cell ontology across tissues and technologies.
Pan-Human Azimuth replaces the previous model of choosing separate Azimuth web applications for individual tissue or organ references. Instead, users can map data with a single pan-human, organ-spanning reference trained across 23 human tissues and 380 high-resolution cell types. The classifier returns hierarchical annotations, confidence scores, and Azimuth embeddings that can be used for downstream analysis and quality control.


PangloaDB
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CD4,ENSMUSG00000023274
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Include tumor/cancer samples and cell lines
Include non-adult and non-primary samples (embryo, fetal, post-natal development, cell lines)
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Resources
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Intramural Research Program
Our Research Changes Lives


MapMyCells
Step 1
Step 2
Web-based applications
10x Whole Mouse Brain (CCN20230722)
Correlation Mapping
DOWNLOAD RESULTS
RESTART
Mapping Completed

One place to get the truth… a good reference

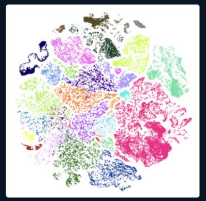


Available taxonomies

10x Whole mouse brain taxonomy (CCN20230722)

A high-resolution transcriptomic and spatial atlas of cell types in the whole mouse brain.

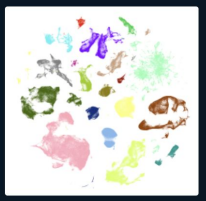
It contains 5,322 clusters that are organized in a hierarchical manner with nested groupings of 34 classes, 338 subclasses, 1,201 supertypes and 5,322 types/clusters.



10x Whole human brain taxonomy (CCN20240330)

Transcriptomic diversity of cell types in adult human brain.

It is clustered into hierarchical groups of 31 superclusters, 461 clusters, and 3313 subclusters.



10x Human MTG SEA-AD taxonomy (CCN20230505)

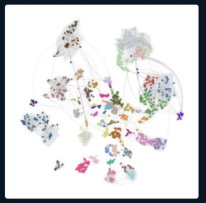
A high-resolution transcriptomic atlas of cell types from middle temporal gyrus from the SEA-AD aged human cohort that spans the spectrum of Alzheimer's disease.

It is hierarchically organized into nested levels of classification: 3 classes, 24 subclasses, and 139 supertypes.



Consensus Basal Ganglia taxonomy (CCN20250428)

A multi-species, multi-omic cell type atlas of the primate basal ganglia that integrates over two million single nuclei from eight major basal ganglia structures profiled using single-nucleus RNA-seq, ATAC-seq, spatial transcriptomics, morphology, and electrophysiology.



SEA-AD Multi-region taxonomy (CCN20260630)

A high-resolution transcriptomic atlas of cell types from ten neo- and allocortical brain regions from the SEA-AD aged human cohort that spans the spectrum of Alzheimer's disease. It is hierarchically organized, extending the MTG SEA-AD taxonomy to 3 classes, 28 subclasses, and 207 nested supertypes, while retaining the original types (e.g., SST 25 in both taxonomies represent the same cell type).

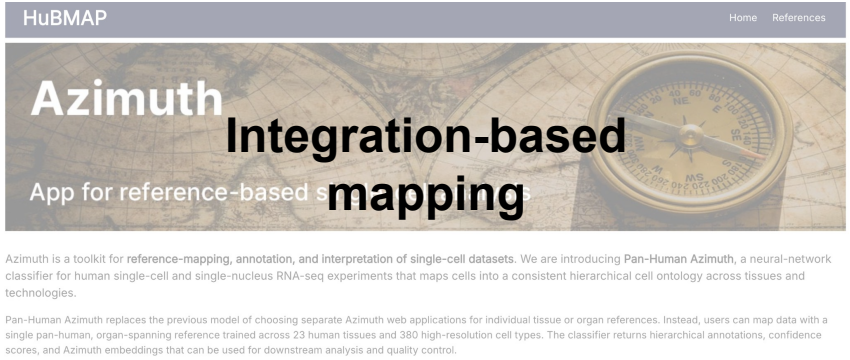


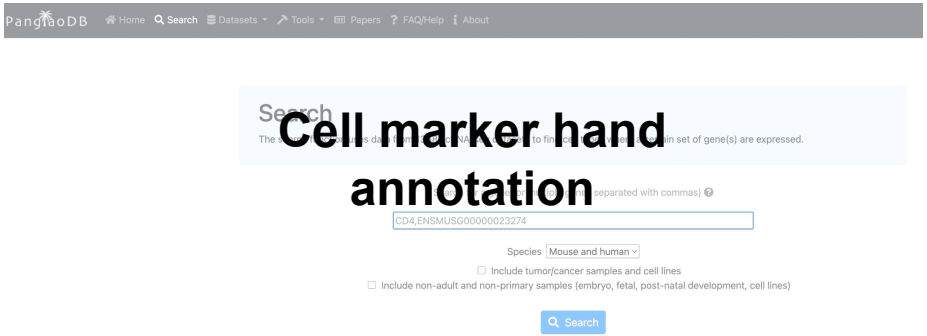
Developing mouse visual cortex (CCN20260131)

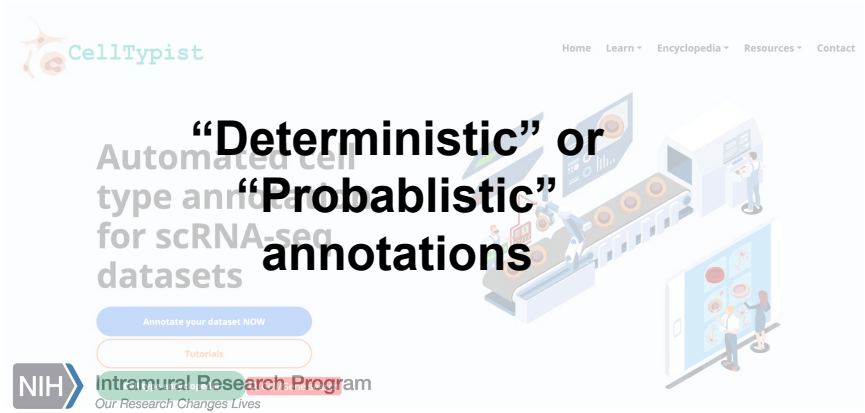
A high-resolution transcriptomic and epigenomic cell-type atlas of the developing mouse visual cortex, spanning embryonic to postnatal stages (E11.5–P56). It is organized in a hierarchical manner with nested groupings of 15 classes, 40 subclasses, 148 clusters, and 714 subclusters, and includes a reconstructed developmental trajectory map tracing the emergence of all excitatory, inhibitory, and non-neuronal cell types.

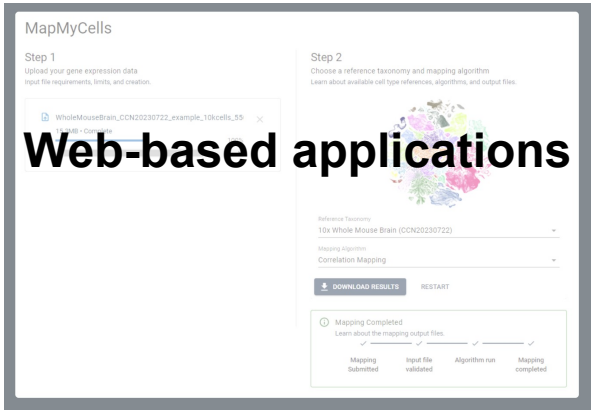


Up to you to use any combination of things….









What would we need…..

Cost: Fast, cheap (time and resources), easy to rerun QUICKLY

Interpretation: A lot of room for biologist intervention, iterative, quick

Reproducbility: Trace every decision, justify every name

Safety: No PHI shared, easily de-identified, prefer summary stats (not donor level)

Truth: One reference, two references, someone else's paper, all of them

Preference: Help brain researchers help themselves to get out of the sc-pit of despair (annotation, integration; i.e. everything before the fun stuff)

Summary

Today

1. cellranger
2. integration
3. annotate clusters

Tomorrow

1. Finalize annotations (SAHA)
2. Differential expression
3. Advanced

Starting your single cell analysis on Biowulf

Day 2: Differential Expression & Advanced Analyses

NIA BDMS (October 19-20, 2026)
Dominic J. Acri, PhD
Laboratory of Neurogenetics, NIA/IRP

Today's lecture

Learning objectives

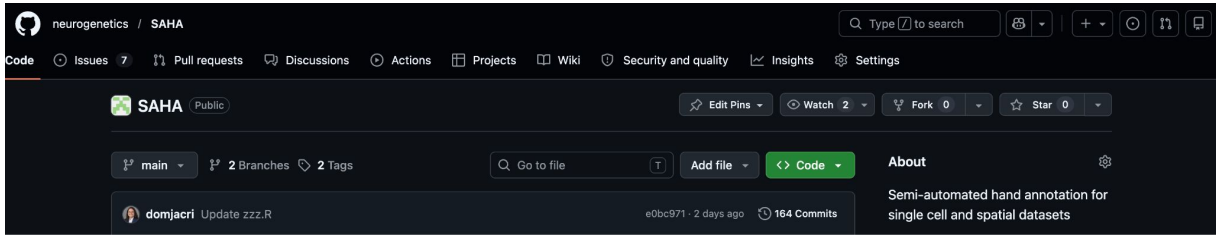
1. Perform SAHA pipeline to investigate annotation in a semi-automated manner
2. Understand differential expression approaches from sc data and how to apply them to Seurat objects
3. Appreciate what additional separate analyses can be done with single cell data once these steps are completed

Part 2.5: *Annotation Strategies (wrapping it up)*

SAHA pipeline (annotation for any sc/spatial)

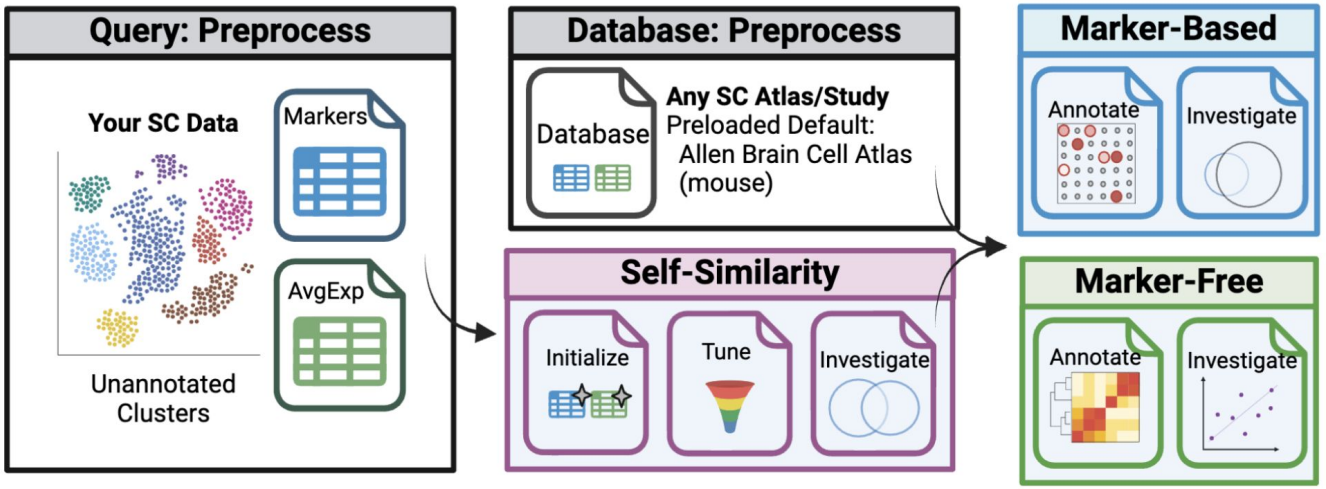
Semi-automated annotation refinement accelerates cell-type identification in brain spatial and single-cell studies

Dominic J. Acri[1,§], Richard Mustaklem[2,§], Liam Horan-Portelance[1,3], Luke C. Dabin[2], Jung Hyun Park[2], Kelly Hartigan[2], Holly Kersey[2], Monica E. Mesecar[1], J. Raphael Gibbs[1], Mark R. Cookson[1*], Jungsu Kim[2*]





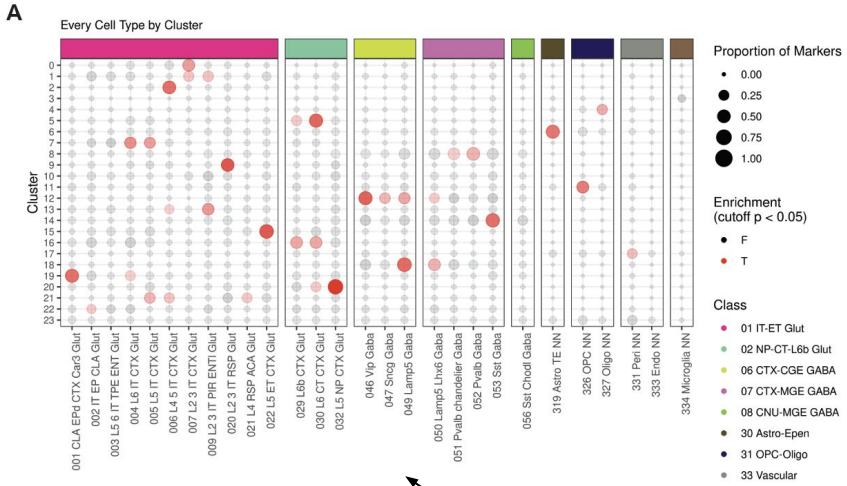
So what did we come up with…



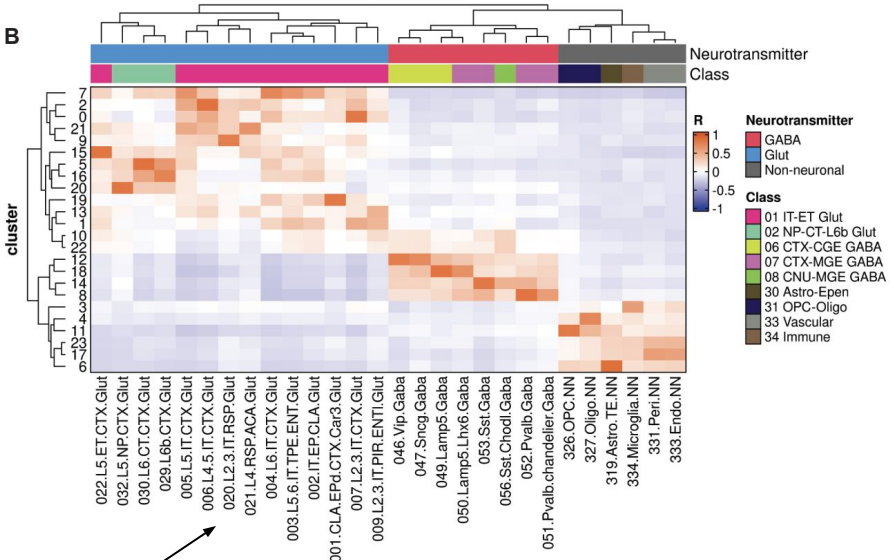
Acri et al., *Preprint*
bioRxiv 10.64898/2026.07.30.741795

So what did we come up with…

Marker-based (Fisher's exact test)



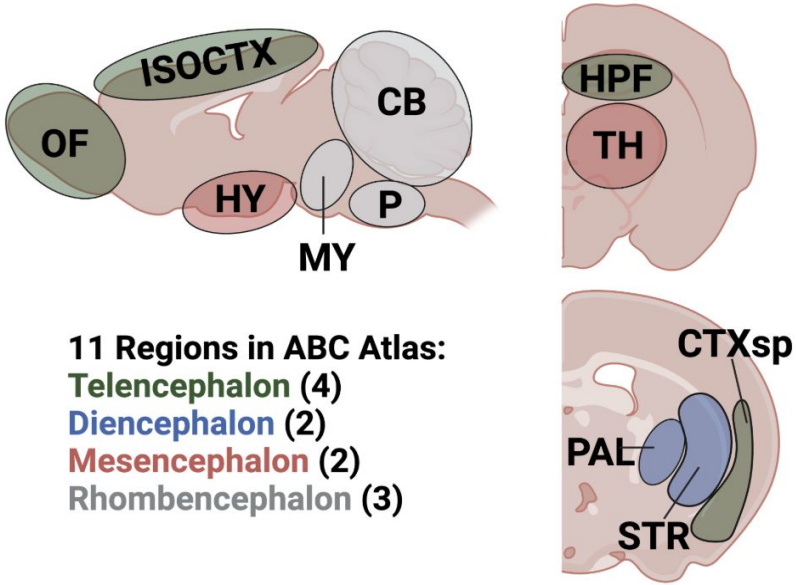
Marker-free (flexible correlation)



Comparison to a reference

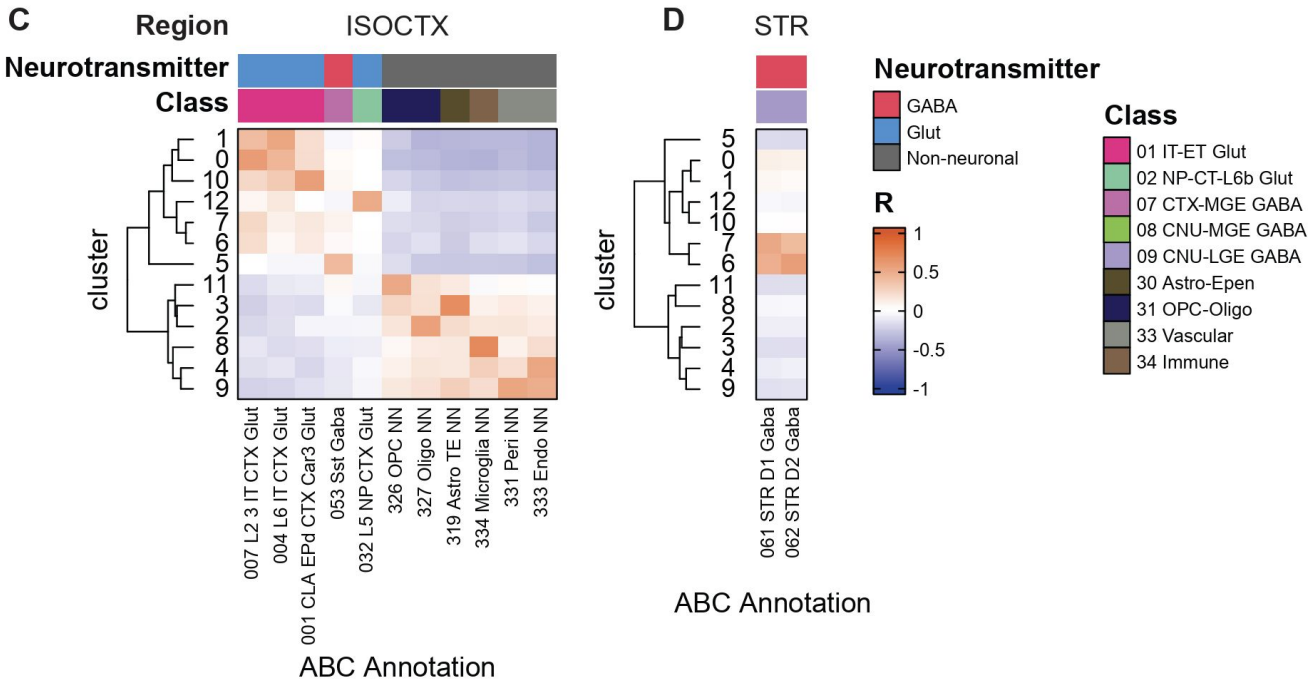
Acri et al., *Preprint*
bioRxiv 10.64898/2026.07.30.741795

Pre-loaded with the Allen Brain Cell Atlas (mouse)



The user is free to consider any “database” for which they have calculated markers or average expression profiles. However for ease-of-use, we provide the user wit hthe option to select inputs from the Allen Brain Cell Atlas (Yao et al., 2024 *Nature*: PMID-38948843).

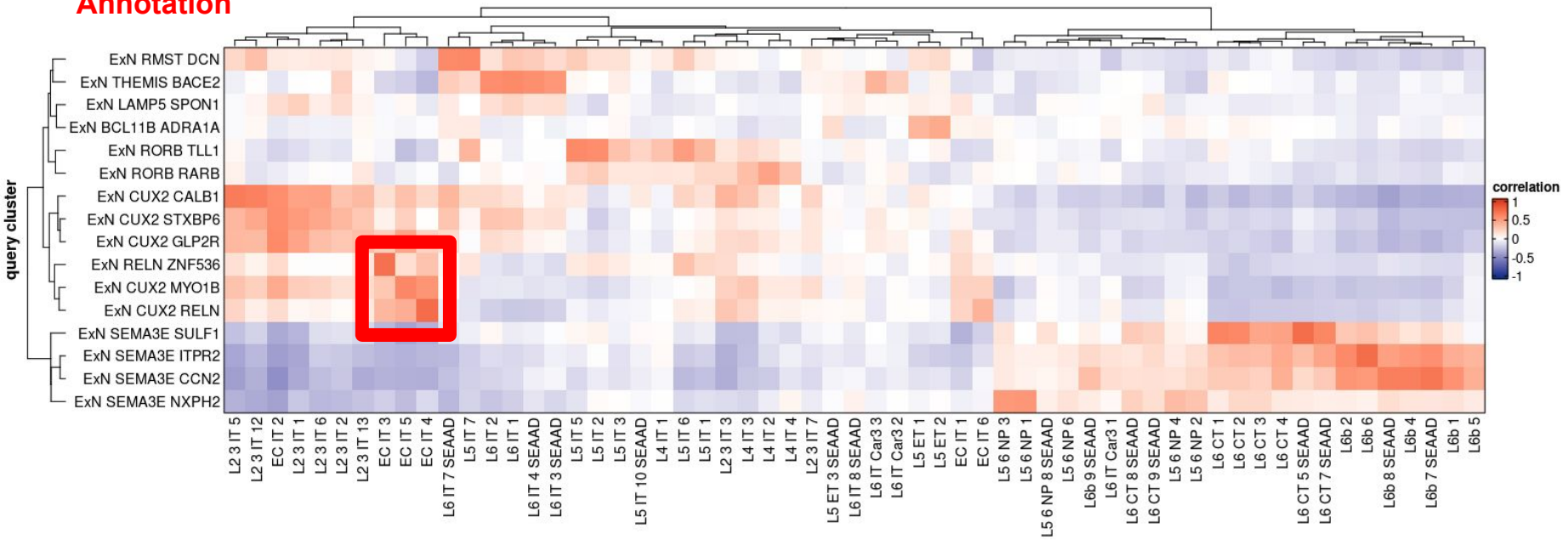
Pre-loaded with the Allen Brain Cell Atlas (mouse)



Acri et al., *Preprint*
bioRxiv 10.64898/2026.07.30.741795

With the freedom to change database to any study…

Cluster or Custom Annotation



Database: BRAIN Intiative (human)

Acri*, Mesecar* et al., *in prep.*
NIH/IRP/LNG unpublished

What is a marker?

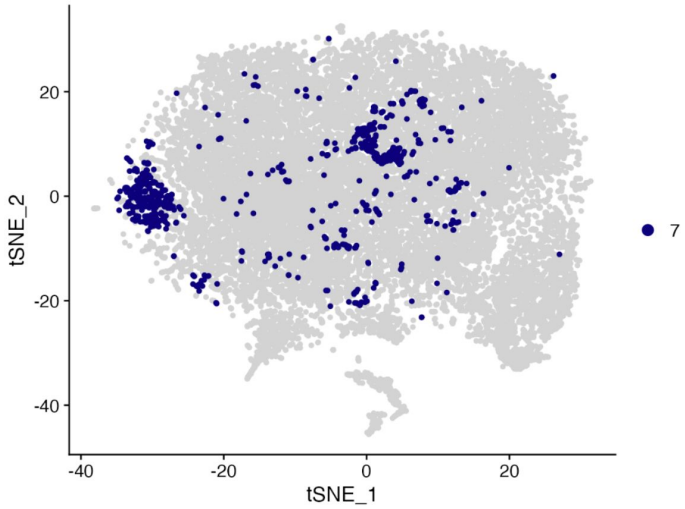
Query Input: Markers & Average Expression

Marker Genes:

A gene that represents the identity of a computationally-defined cluster through a one-versus-all statistical test, such that "Cluster 0" can have any number of markers up- or down-regulated compared to all others.

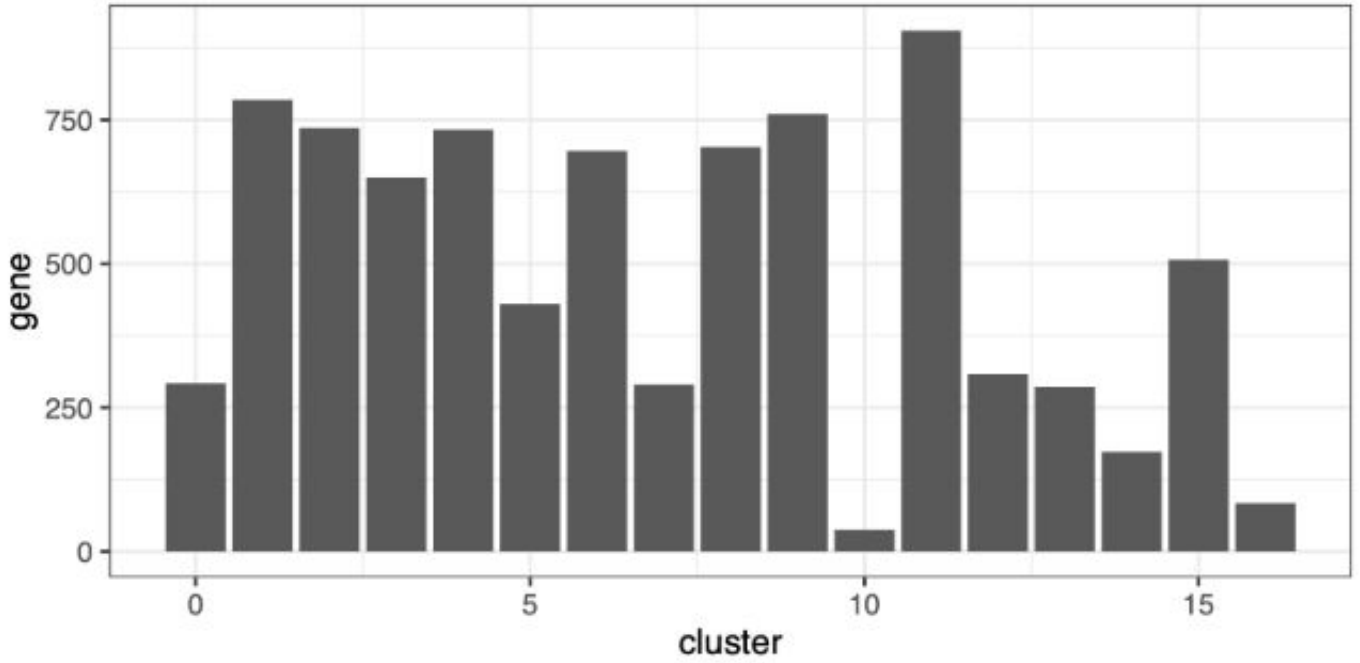
Average Expression:

Exponentiated, log-normalized counts averaged together across a computationally-defined cluster, normalized on a 0-to-1 scale so that "Cluster 0" has top expressed genes (1), median expressed genes (0.5) and so on.

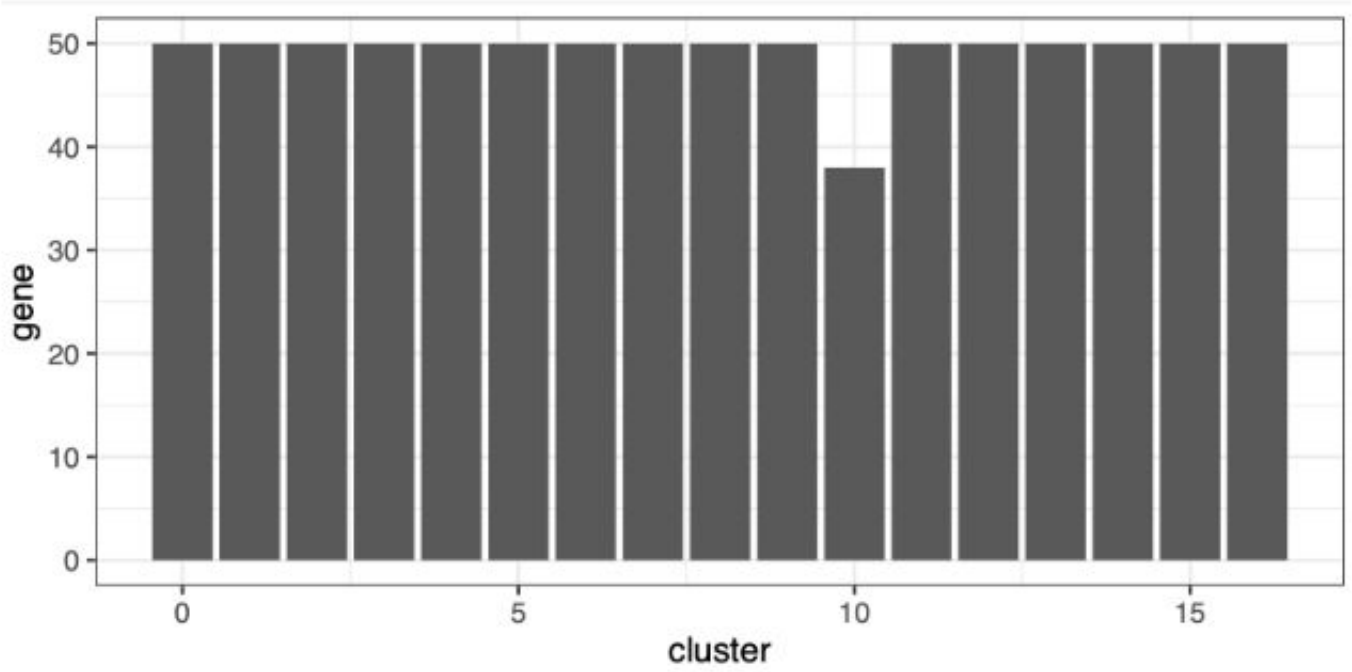


Markers: **7** vs all

What even is a marker?

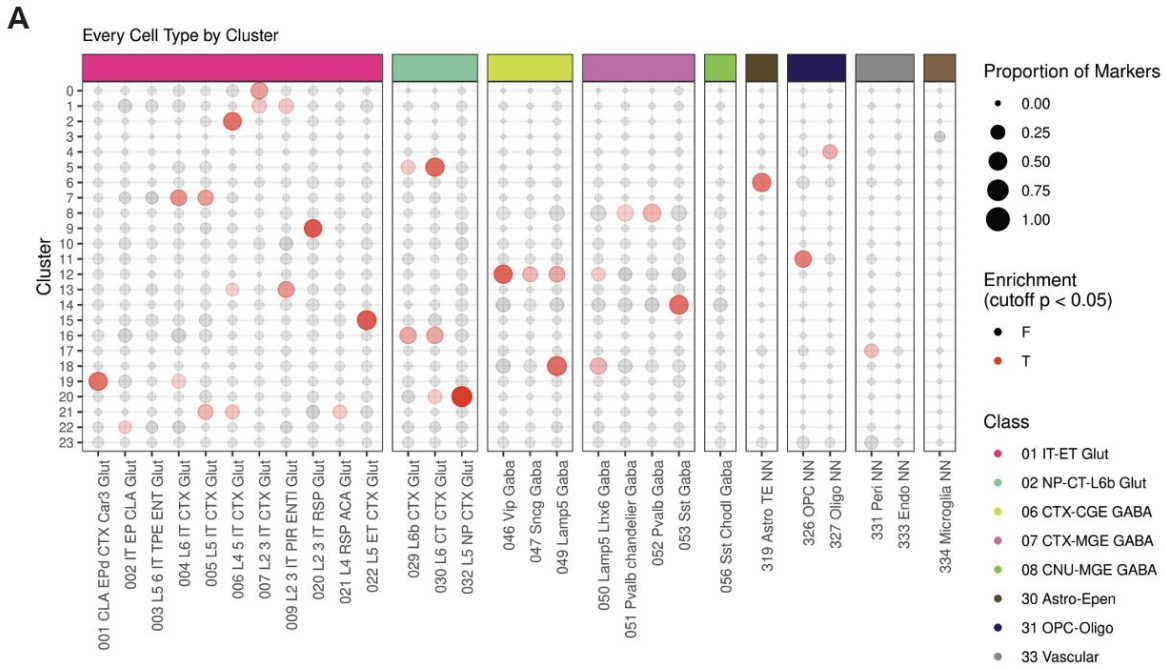


What even is a marker?



For more:
TuneMarkers() function in SAHA

Marker-based



Acri et al., *Preprint*
bioRxiv 10.64898/2026.07.30.741795

What is average expression?

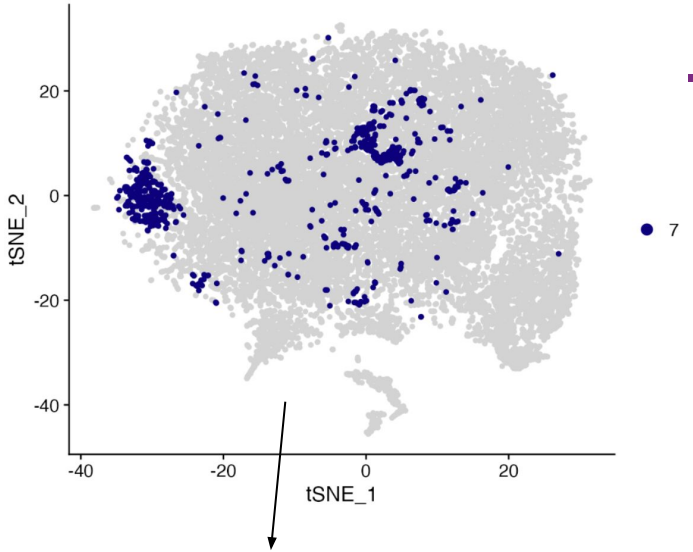
Query Input: Markers & Average Expression

Marker Genes:

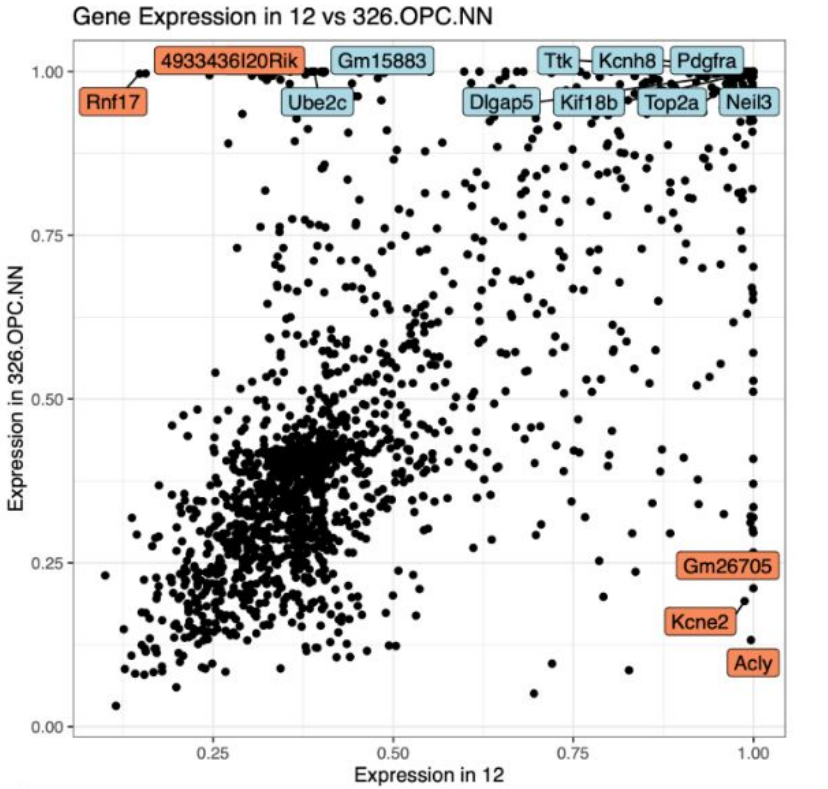
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Average Expression:

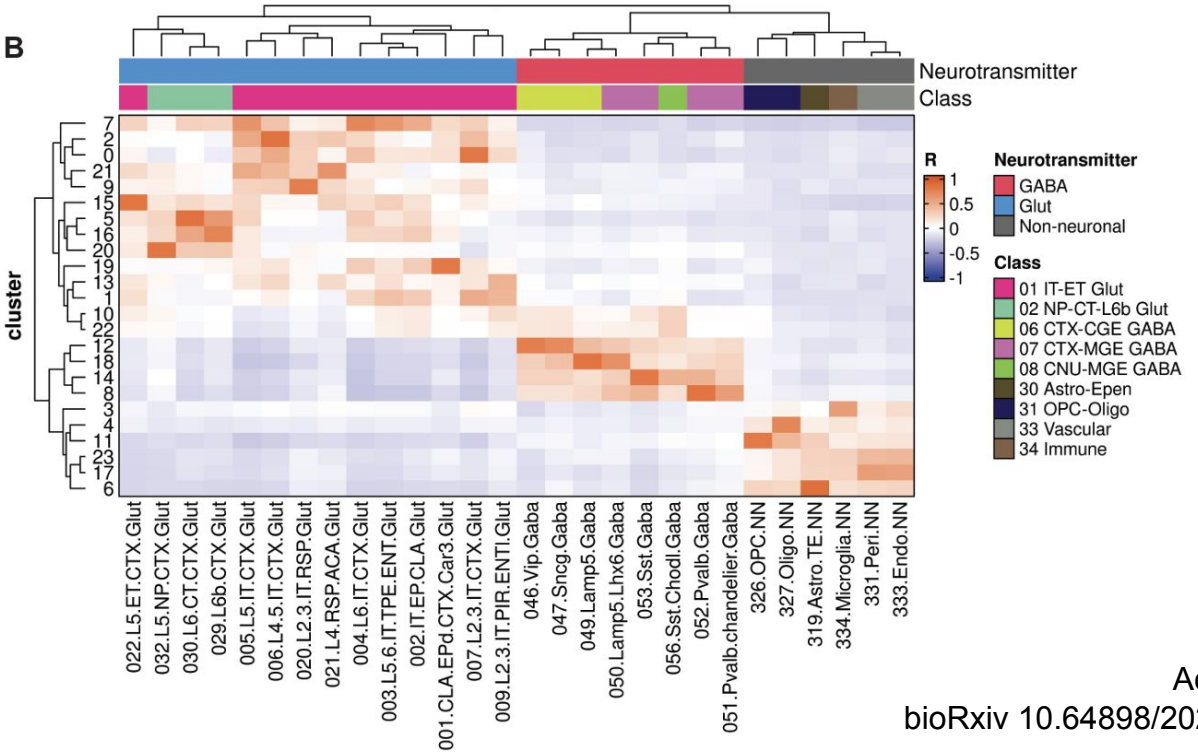
Exponentiated, log-normalized counts averaged together across a computationally-defined cluster, normalized on a 0-to-1 scale so that "Cluster 0" has top expressed genes (1), median expressed genes (0.5) and so on.



What is average expression?



Marker-based



Acri et al., *Preprint*
bioRxiv 10.64898/2026.07.30.741795

HTML report (and automated version → with caution)

SAHA Report

2026-04-14

Summary

```
## SAHA
##   Data Type: AvgExp & Markers
##   Description: 31140 avgexp query (13 clusters)
##                36601 avgexp db (24 possible annotations)
##                9238 query markers (13 clusters)
##                36601 database markers (24 possible annotations)
##   Analysis: self-similarity, marker-based, marker-free
```

Auto-annotation

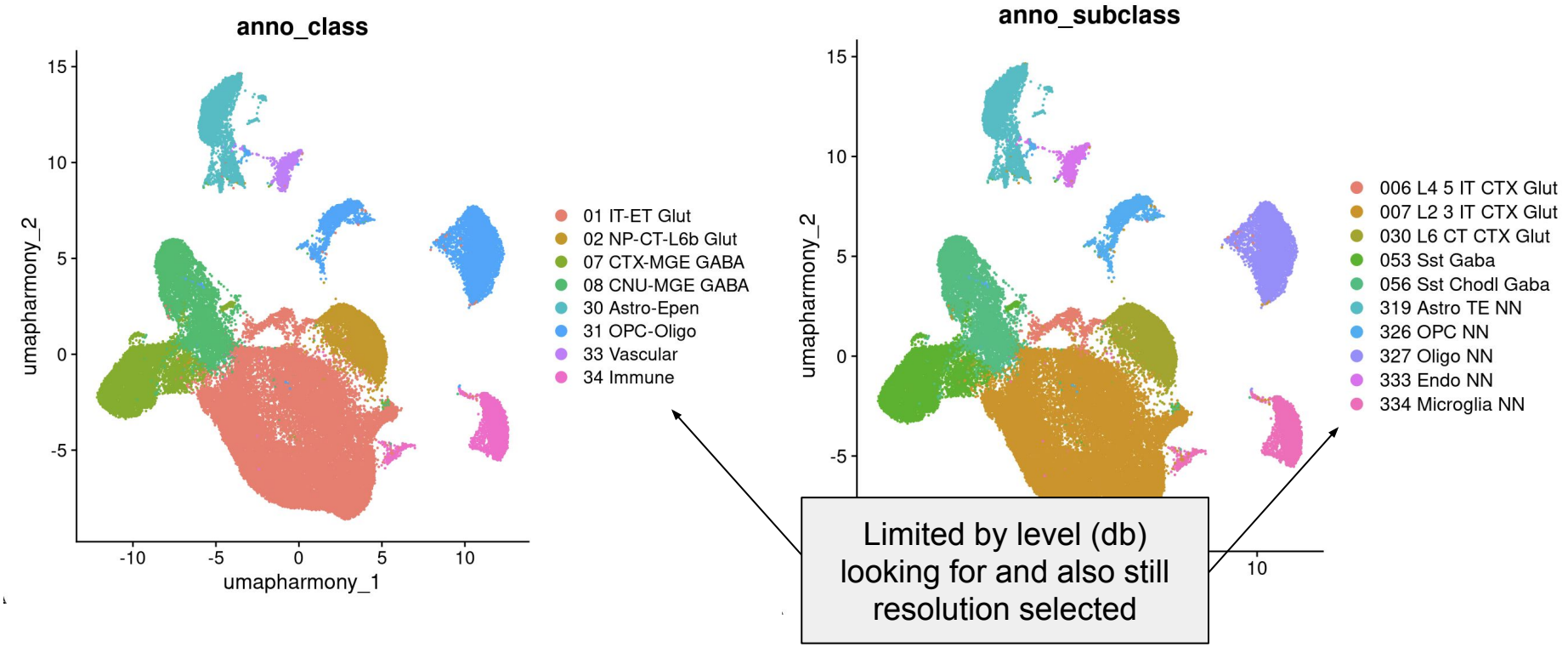
You performed auto-annotation on this data, using both Marker-Based and Marker-Free analyses.

This dataframe shows the agreement between the Marker-Based and Marker-Free analyses.

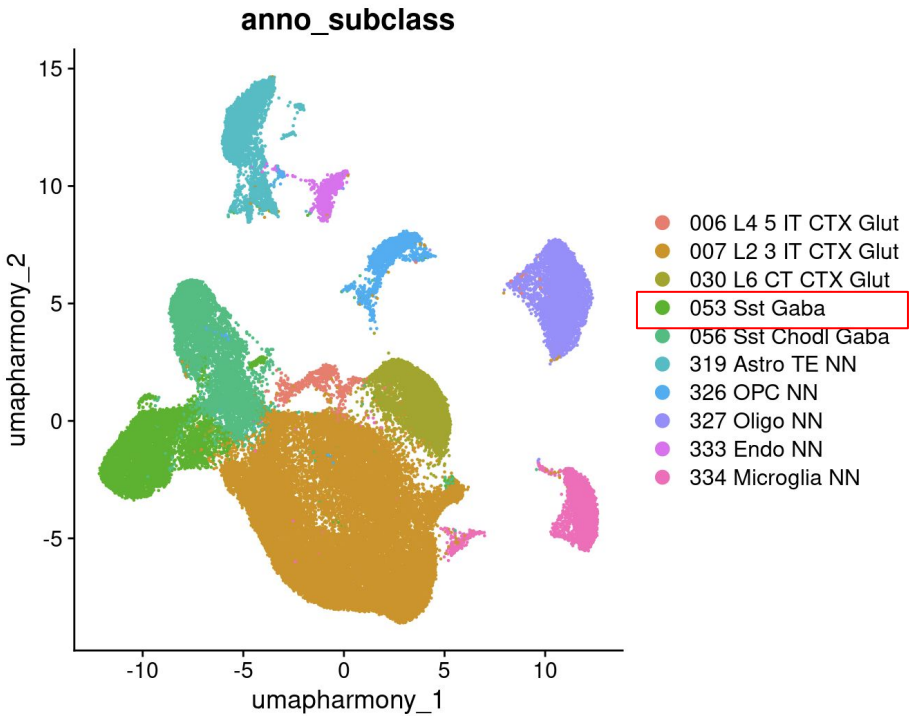
```
##    cluster    marker_based     marker_free    consensus      best_match
## 1       g0    INCONCLUSIVE         L2 3 IT INCONCLUSIVE    INCONCLUSIVE
## 2       g1           L4 IT           L5 IT DISAGREEMENT    INCONCLUSIVE
## 3       g2         L2/3 IT         L2 3 IT DISAGREEMENT    INCONCLUSIVE
## 4       g3       Astrocyte       Astrocyte        MATCH       Astrocyte
## 5       g4           Pvalb           Pvalb        MATCH           Pvalb
## 6       g5           Lamp5             Vip DISAGREEMENT    INCONCLUSIVE
## 7       g6           L5 IT           L5 IT        MATCH           L5 IT
## 8       g6           L6 CT           L5 IT DISAGREEMENT    INCONCLUSIVE
## 9       g7             OPC             OPC        MATCH             OPC
## 10      g8       Astrocyte         L2 3 IT DISAGREEMENT    INCONCLUSIVE
## 11      g9    INCONCLUSIVE           L4 IT INCONCLUSIVE    INCONCLUSIVE
## 12     g10 Oligodendrocyte Oligodendrocyte        MATCH Oligodendrocyte
## 13     g11   Microglia PVM   Microglia PVM        MATCH   Microglia PVM
## 14     g12     Endothelial            VLMC DISAGREEMENT    INCONCLUSIVE
```

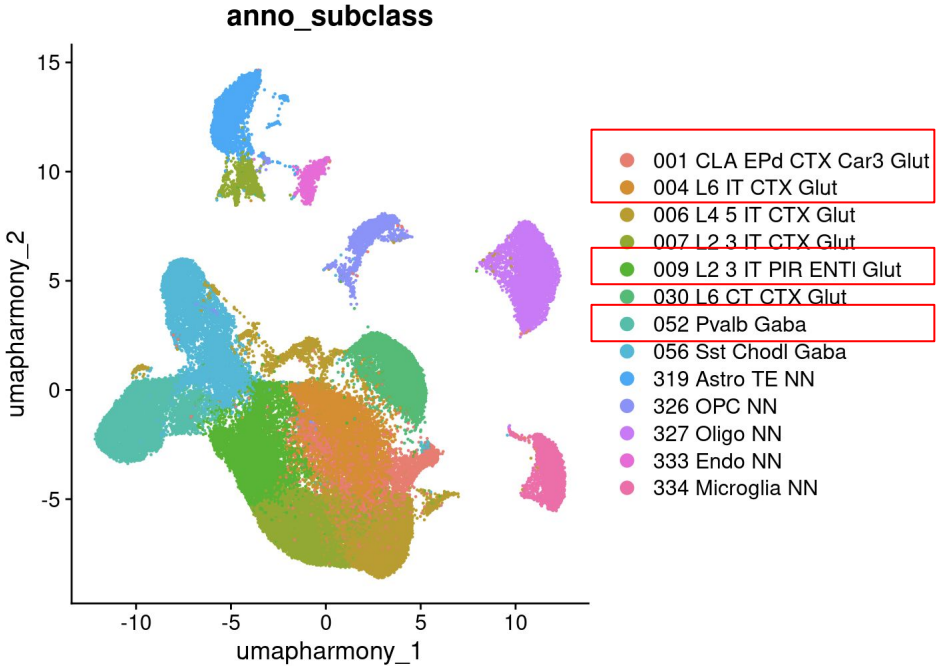
Acri et al., *Preprint*
bioRxiv 10.64898/2026.07.30.741795

What this enables



What this enables





Redone at res = 0.5 instead of 0.1

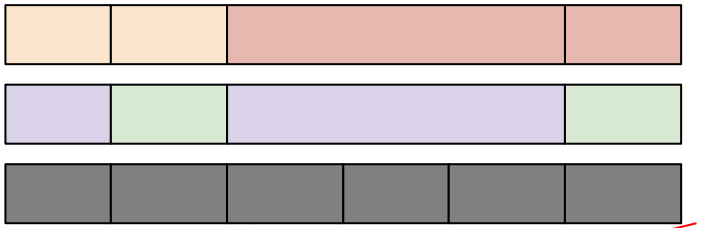
Part 3: *Differential Expression*

Principles of differential expression

	Control 1	Control 2	Control 3	Sample 1	Sample 2	Sample 3
Gene A	6	5	7	17	10	11
Gene B	11	11	10	3	4	2
Gene C	200	150	355	50	1	3
Gene D	0	1	0	2	0	1

1. Annotating the list of genes from the count matrix
2. Filtering out lowly expressed genes
3. Normalizing count data
4. More quality control
5. The actual testing!

Principles of sc differential expression



	Control 1	Control 2	Control 3	Sample 1	Sample 2	Sample 3
Gene A	6	5	7	17	10	11
Gene B	11	11	10	3	4	2
Gene C	200	150	355	50	1	3
Gene D	0	1	0	2	0	1

With added dimensionality there is that many more DE experiments to perform:

Testing condition1 vs condition2
10 celltypes = 10 DE experiments

Options for differential expression (generalized)

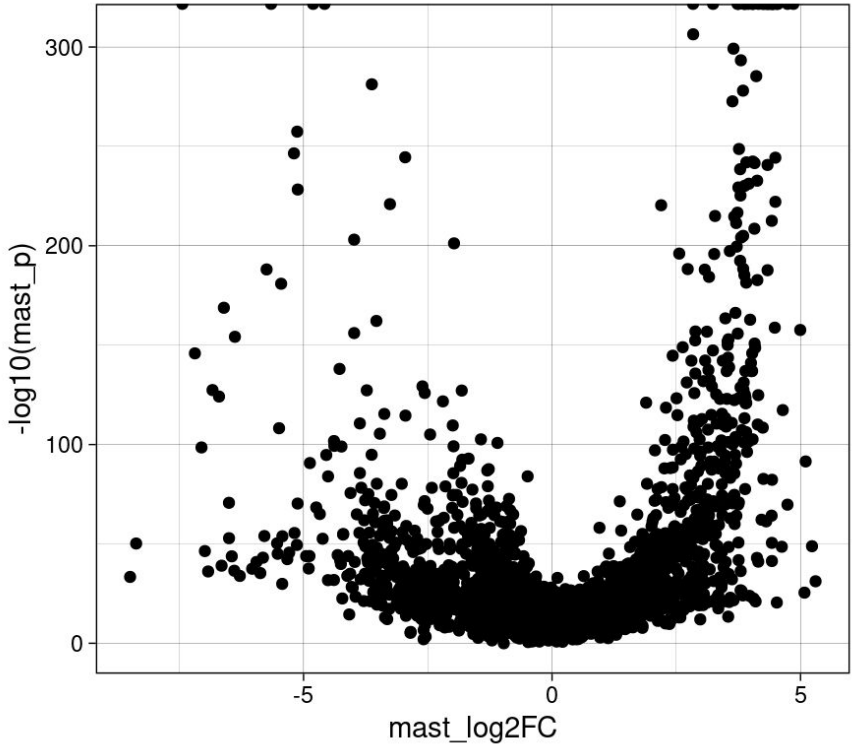
SC differential expression

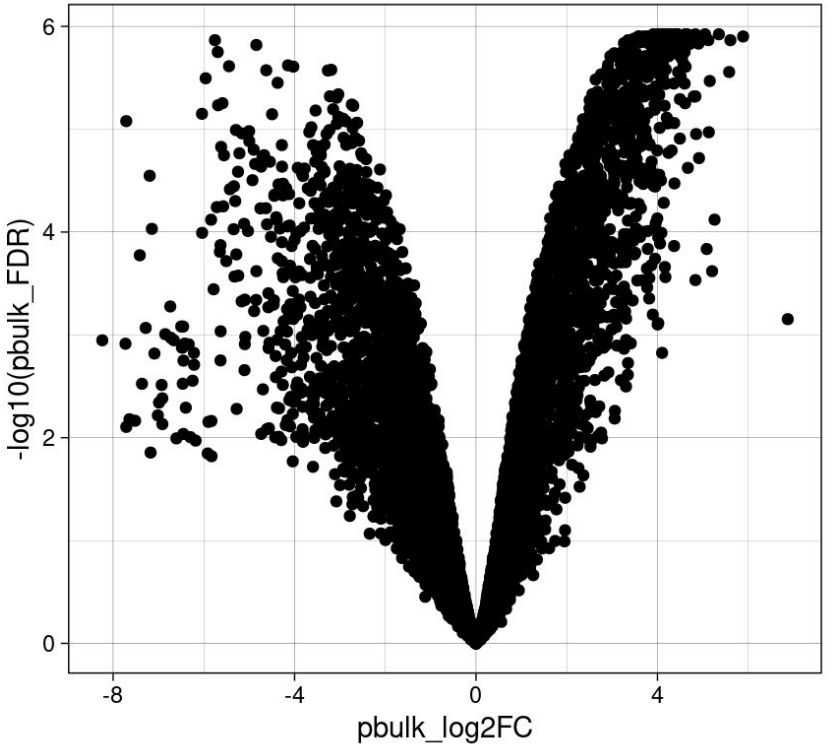
1. Load in single cell object
2. Subset to a cell type
3. Specify comparisons to be made (i.e. FindMarkers = condition instead of cluster)
4. Perform for every cluster

Pseudobulk

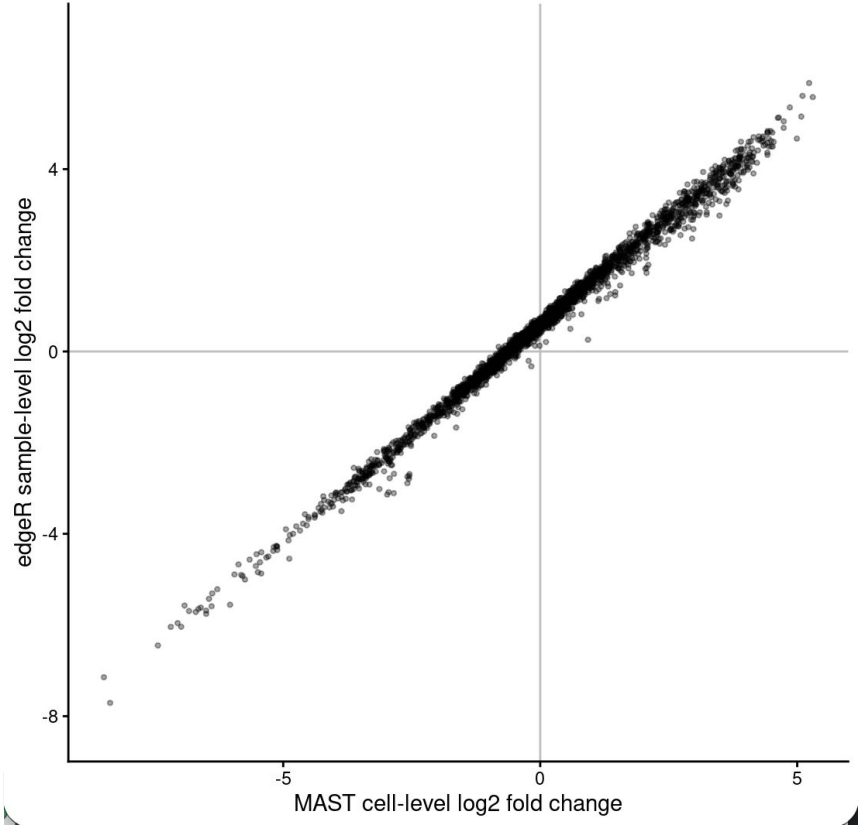
1. Load in single cell object
2. Sum reads per sample x per celltype (i.e. S1_microglia)
3. Use sample meta data similar to differential expression (DESeq2, limma, dream, etc)
4. Perform for every cluster

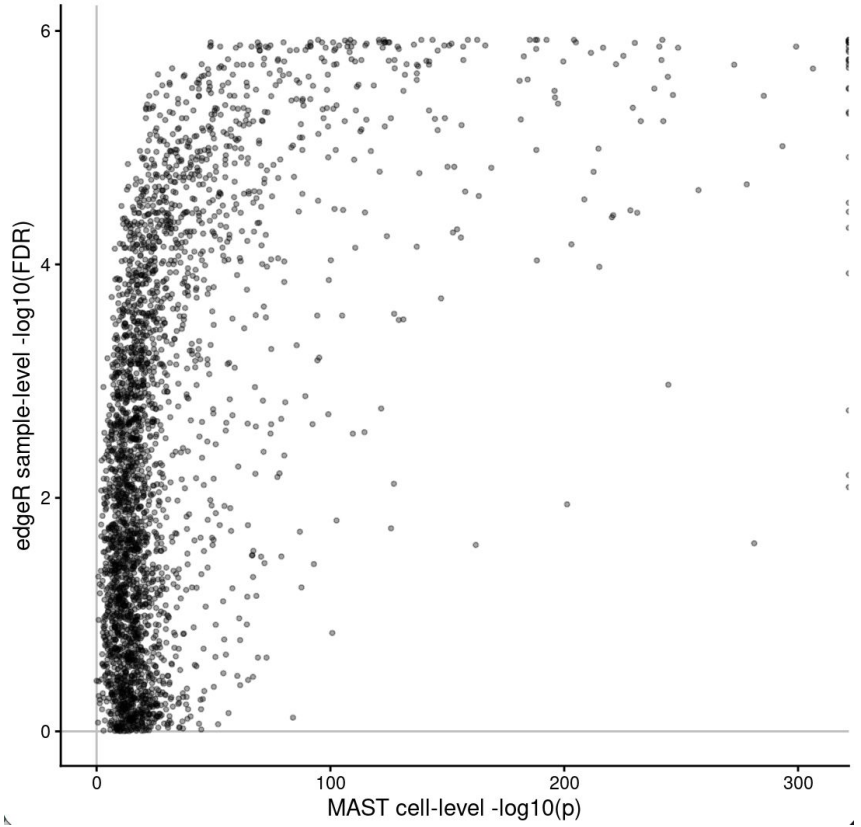
Side-by-side comparisons



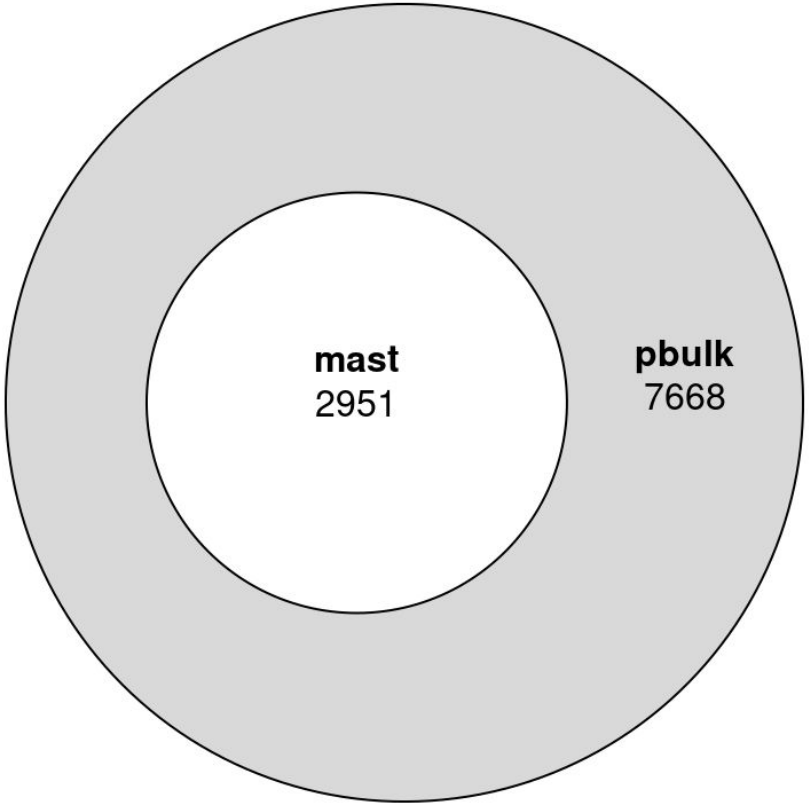


Side-by-side comparisons





Side-by-side comparisons



Part 4: *Advanced Analyses*

Cell-cell communication

nature protocols

https://doi.org/10.1038/s41596-024-01045-4

Protocol

Check for updates

CellChat for systematic analysis of cell–cell communication from single-cell transcriptomics

Suoqin Jin [1,2] ✉, Maksim V. Plikus [3,4] & Qing Nie [3,4,5] ✉

Inference and analysis of cell-cell communication using CellChat

Suoqin Jin

14 February, 2024

- Load the required libraries
- Part I: Data input & processing and initialization of CellChat object
  - Prepare required input data for CellChat analysis
    - (A) Starting from a count data matrix
    - (B) Starting from a Seurat object
    - (C) Starting from a SingleCellExperiment object
    - (D) Starting from an Anndata object
  - Create a CellChat object
    - (A) Starting from the digital gene expression matrix and cell label information
    - (B) Starting from a Seurat object
    - (C) Starting from a SingleCellExperiment object
    - (D) Starting from an AnnData object
  - Set the ligand-receptor interaction database
  - Preprocessing the expression data for cell-cell communication analysis
- Part II: Inference of cell-cell communication network
  - Compute the communication probability and infer cellular communication network
  - Extract the inferred cellular communication network as a data frame
  - Infer the cell-cell communication at a signaling pathway level
  - Calculate the aggregated cell-cell communication network
- Part III: Visualization of cell-cell communication network
  - Visualize each signaling pathway using Hierarchy plot, Circle plot or Chord diagram
    - Compute the contribution of each ligand-receptor pair to the overall signaling pathway and visualize cell-cell communication mediated by a single ligand-receptor pair
    - Automatically save the plots of the all inferred network for quick exploration
  - Visualize cell-cell communication mediated by multiple ligand-receptors or signaling pathways
    - (A) Bubble plot
    - (B) Chord diagram
  - Plot the signaling gene expression distribution using violin/dot plot

Cell-cell communication

Are you ready?

- Yes, Seurat object can be used for direct input to CellChat vignette
- All in R
- Can add another script to this folder

What to do next:

- *Recommend*: looking for signalling at the anno_class level
- *Advanced*: do differential between reference and 5xFAD

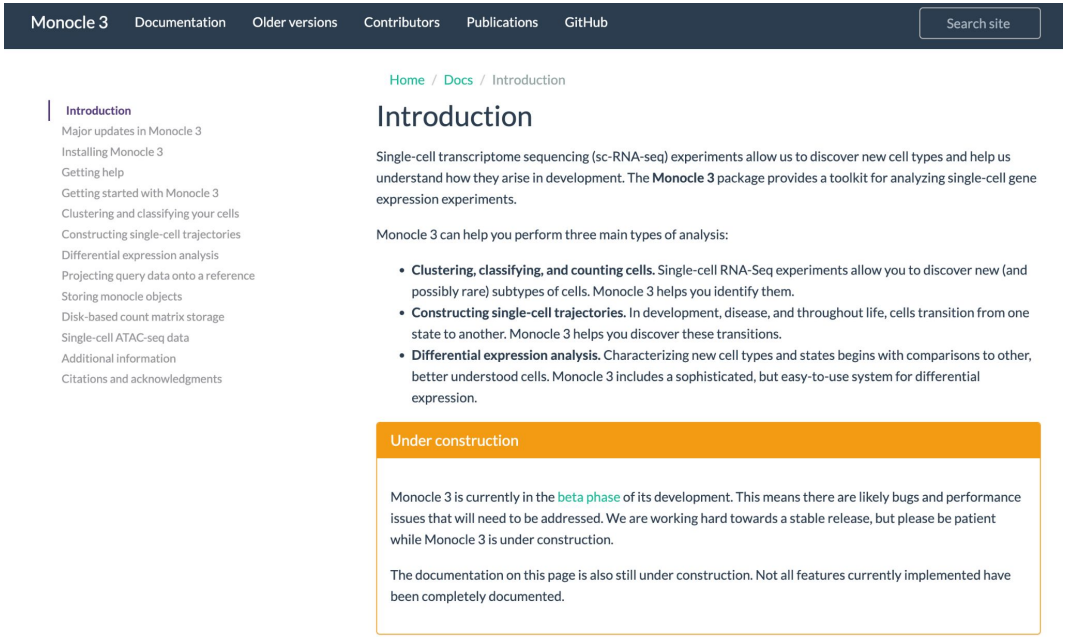
Pseudotime

nature biotechnology

LETTERS

The dynamics and regulators of cell fate decisions are revealed by pseudotemporal ordering of single cells

Cole Trapnell[1,2,6], Davide Cacchiarelli[1–3,6], Jonna Grimsby[2], Prapti Pokharel[2], Shuqiang Li[4], Michael Morse[1,2], Niall J Lennon[2], Kenneth J Livak[4], Tarjei S Mikkelsen[1–3] & John L Rinn[1,2,5]



Pseudotime

Are you ready?

- From a resource perspective, yes
- Vignette includes reclustering
- Hypothesis generation ok, however several hyperparamater tuning may be necessary

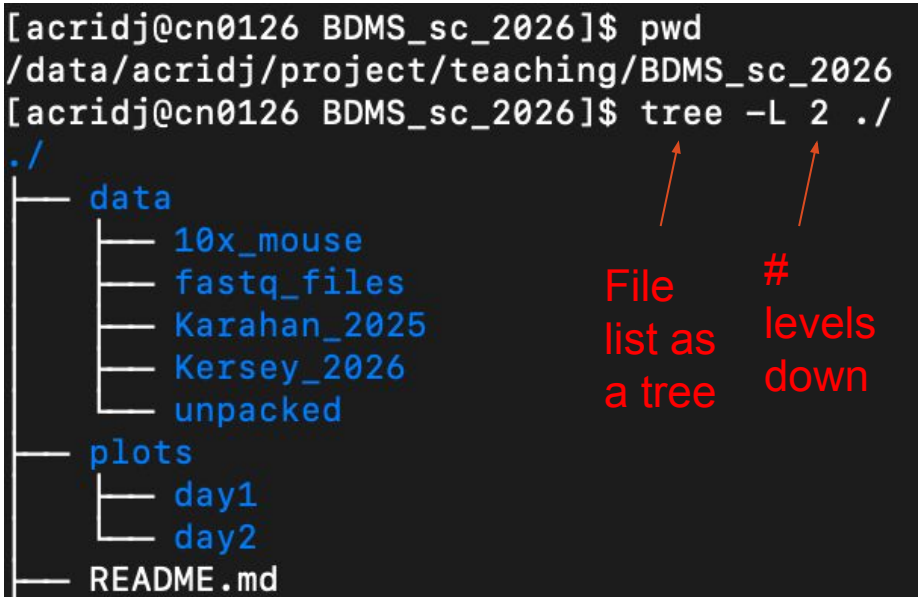
What to do next:

- *Recommend 1*: run the vignette with their demo data
- *Recommend 2*: use an LLM informed to this project to adapt a script **(next page)**
- *Advanced*: subcluster microglia and try to identify a ptime trajectory similar to those published (PMID28602351)

Using an LLM

What I do:

1. Train your model to your file system
2. Write our plan and end prompt with “confirm and hold”
3. Read the plan before “vibe coding” the answer
4. Run code line-by-line instead of sourcing scripts



Cell velocity

LETTER

https://doi.org/10.1038/s41586-018-0414-6

RNA velocity of single cells

Gioele La Manno[1,2], Ruslan Soldatov[3], Amit Zeisel[1,2], Emelie Braun[1,2], Hannah Hochgerner[1,2], Viktor Petukhov[3,4], Katja Lidschreiber[5], Maria E. Kastriti[6], Peter Lönnerberg[1,2], Alessandro Furlan[1], Jean Fan[3], Lars E. Borm[1,2], Zehua Liu[3], David van Bruggen[1], Jimin Guo[3], Xiaoling He[7], Roger Barker[7], Erik Sundström[8], Gonçalo Castelo-Branco[1], Patrick Cramer[5,9], Igor Adameyko[6], Sten Linnarsson[1,2]* & Peter V. Kharchenko[3,10]*



Welcome to the velocyto homepage!

`velocyto` (velox + κύτος, quick cell) is a package for the analysis of expression dynamics in single cell RNA seq data. In particular, it enables estimations of RNA velocities of single cells by distinguishing unspliced and spliced mRNAs in standard single-cell RNA sequencing protocols (see pre-print below for more information).

Implementations

`velocyto` currently has two different implementations:

Cell velocity

Are you ready?

- Not necessarily
- Splicing rates (assuming mature and immature RNA) is necessary, so sn (mainly pre-mRNA) breaks this assumption
- Specific use-cases for single-cell RNAseq

What to do next:

- *Recommend*: Trying vignette with their data
- *Advanced*: Use in a system with rapid change and whole cell recovery (stimulated cells in culture?)

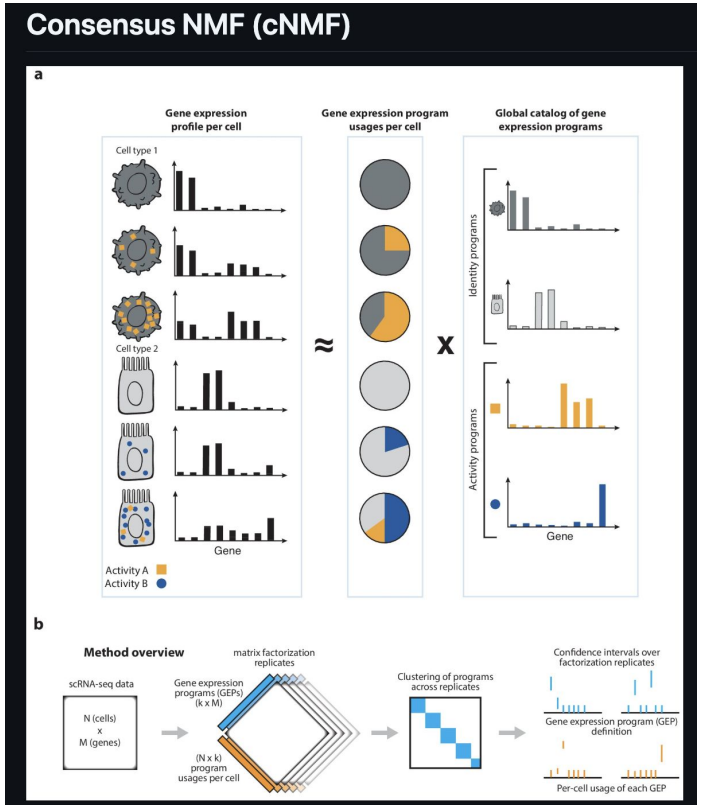
Latent factor analyses

eLIFE elifesciences.org TOOLS AND RESOURCES

Identifying gene expression programs of cell-type identity and cellular activity with single-cell RNA-Seq

Dylan Kotliar[1,2,3†]*, Adrian Veres[1,3,4†], M Aurel Nagy[3,5], Shervin Tabrizi[2], Eran Hodis[3,6], Douglas A Melton[4,7], Pardis C Sabeti[1,2,7]

[1]Department of Systems Biology, Harvard Medical School, Boston, United States; [2]Broad Institute of MIT and Harvard, Cambridge, United States; [3]Harvard-MIT Division of Health Sciences and Technology, Massachusetts Institute of Technology, Cambridge, United States; [4]Harvard Stem Cell Institute, Harvard University, Cambridge, United States; [5]Department of Neurobiology, Harvard Medical School, Boston, United States; [6]Biophysics Program, Harvard University, Cambridge, United States; [7]Howard Hughes Medical Institute, Chevy Chase, United States



Latent factor analyses

Are you ready?

- Maybe
- Latent = “hidden things”
- Latent variables = descriptor of things that are unreported
- Useful for cross-cell type discovery (i.e. things shared in 2 neuron types but no others)

What to do next:

- *Recommend*: Running vignette
- *Advanced*: Use an LLM to write a script for finding a latent variable associated with study (hidden batch effect?)

Transcription factor analyses

Resource

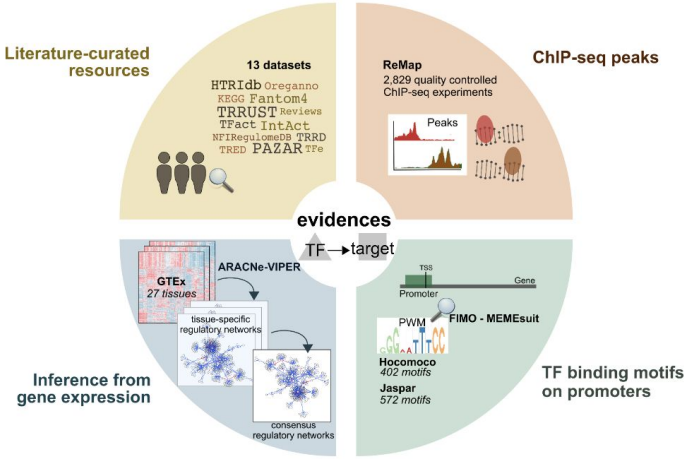
Benchmark and integration of resources for the estimation of human transcription factor activities

Luz Garcia-Alonso,[1,2] Christian H. Holland,[3,4] Mahmoud M. Ibrahim,[3,5] Denes Turei,[3,4] and Julio Saez-Rodriguez[1,2,3,4]

DoRothEA: collection of human and mouse regulons

Overview

DoRothEA is a gene regulatory network (GRN) containing signed transcription factor (TF) - target gene interactions. DoRothEA regulons, the collection of a TF and its transcriptional targets, were curated and collected from different types of evidence for both human and mouse. A confidence level was assigned to each TF-target interaction based on the number of supporting evidence.



These regulons, coupled with any statistical method, can be used to infer TF activities from bulk or single-cell transcriptomics.

This is an R package for storing the regulons. To infer TF activities, please check out decoupleR, available in R or python.

Transcription factor analyses

Are you ready?

- Yes
- Regulon: a group of genes controlled by a transcription factor
- Can directly use either microglia pseudobulk inputs or follow their single cell vignette

What to do next:

- *Recommend*: Starting with vignette to understand all the different tools that are used
- *Advanced*: Run directly using the objects from this workshop

Multi-omics

nature reviews molecular cell biology

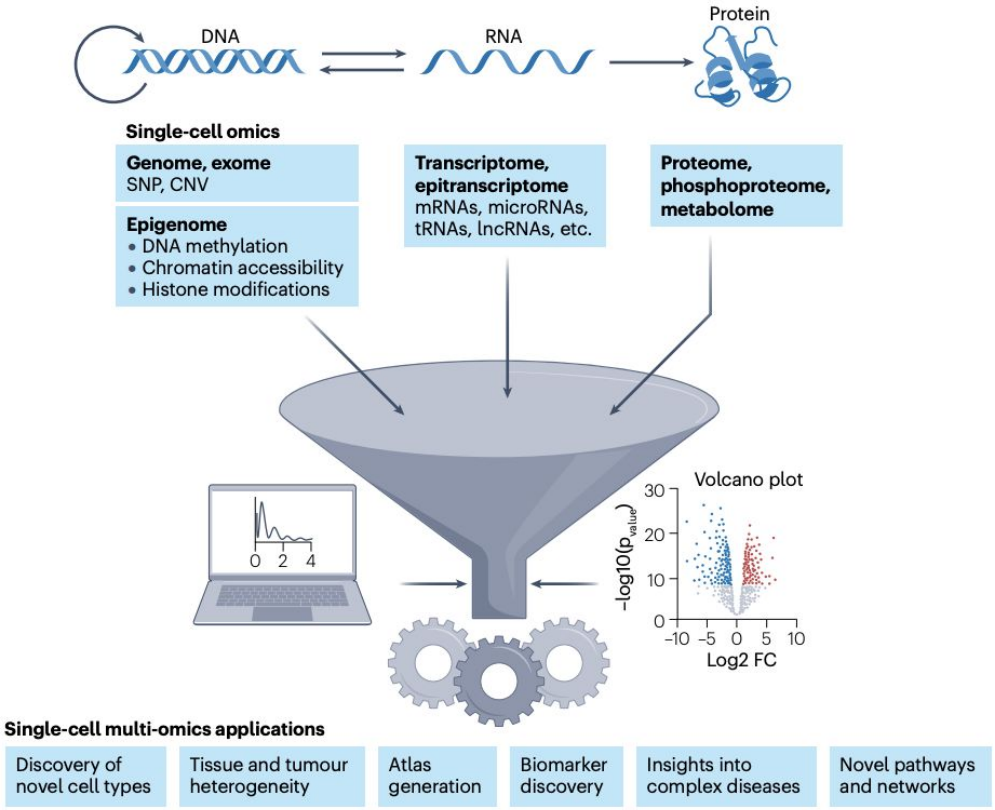
https://doi.org/10.1038/s41580-023-00615-w

Review article

Check for updates

The technological landscape and applications of single-cell multi-omics

Alev Baysoy[1,6], Zhiliang Bai[1,6], Rahul Satija[2,3] & Rong Fan[1,4,5]



Multi-omics

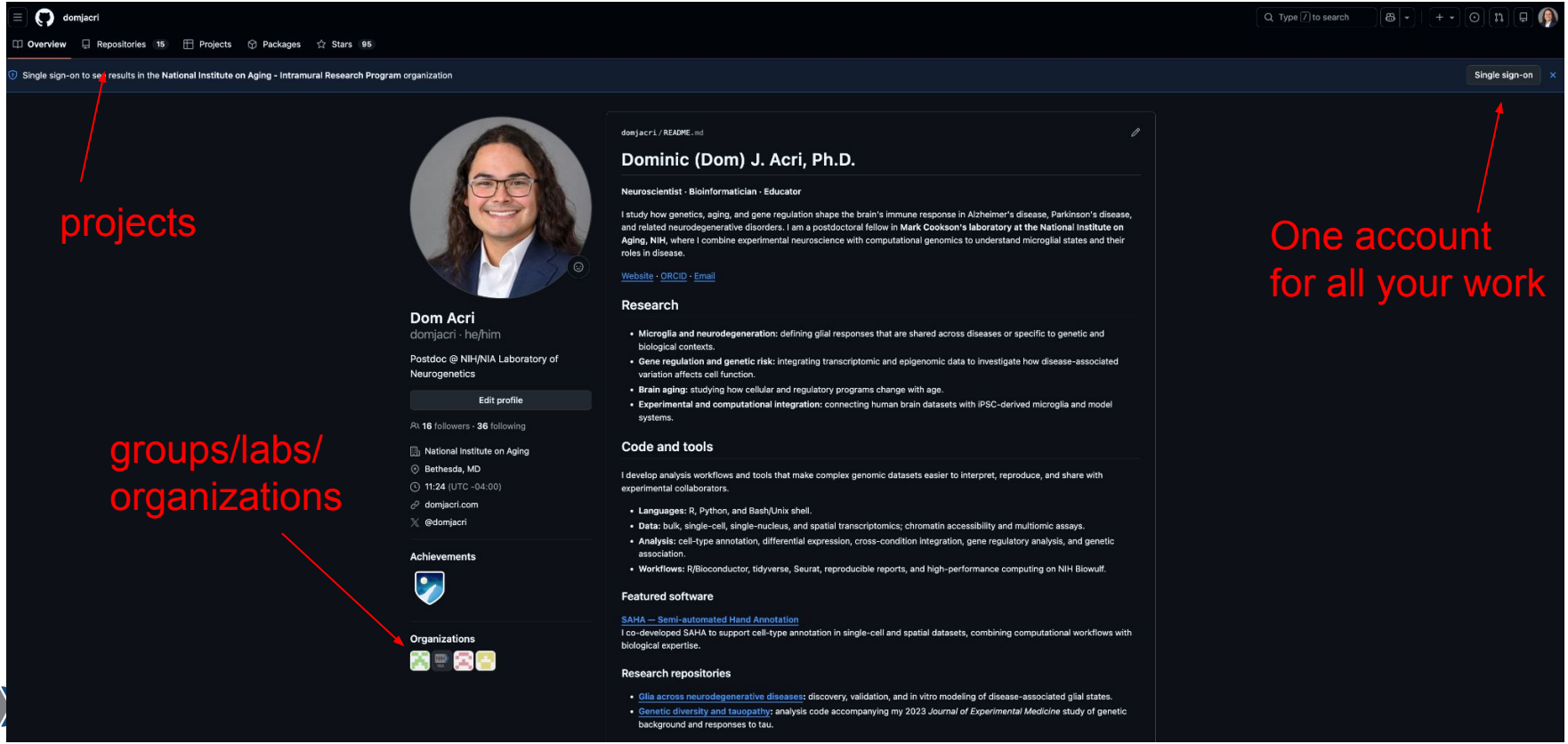
Are you ready?

- Maybe
- Everything for RNA can be done again for another omic
- But what is a “cell type marker” for an ATAC peak?

What to do next:

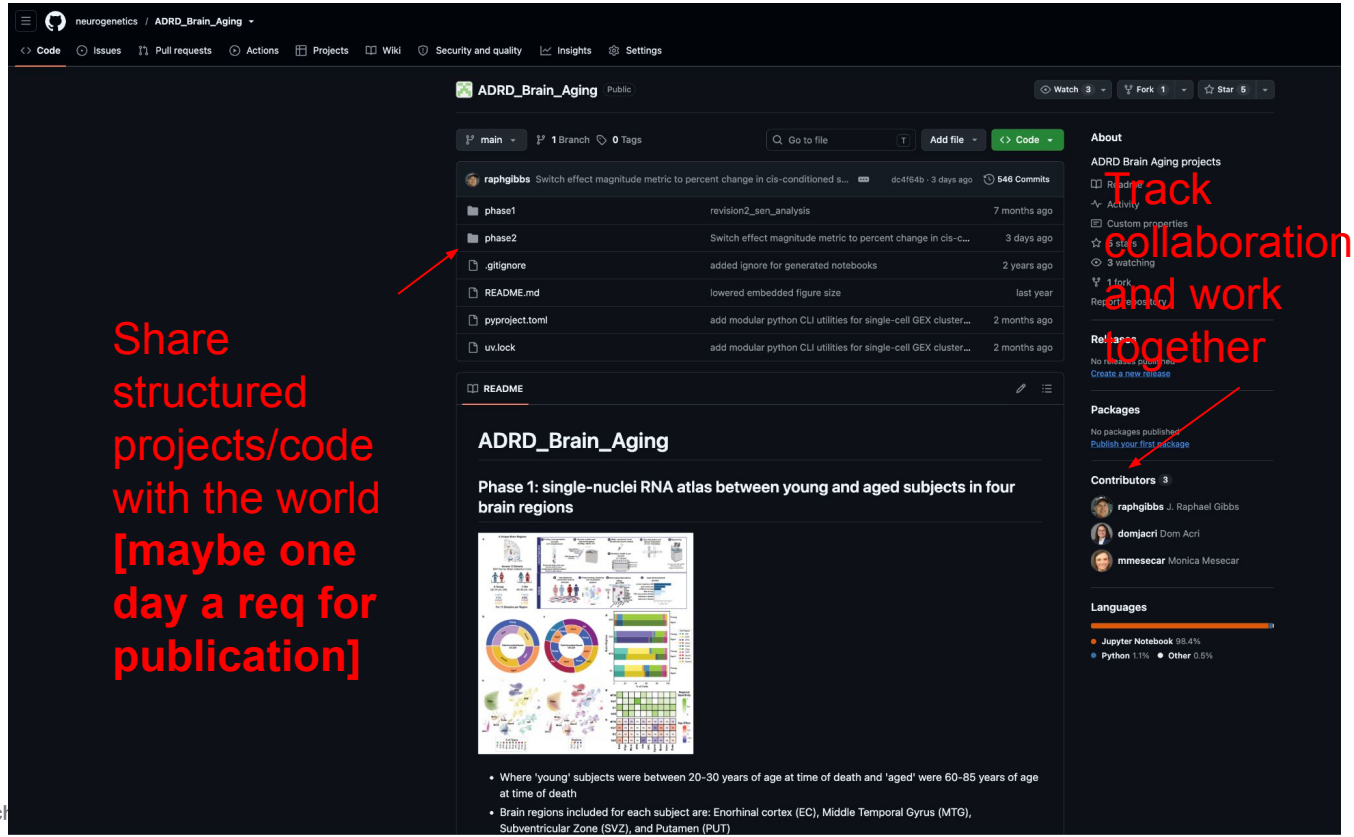
- *Recommend*: Taking a class on epigenetic single cell assays, or another multiomic

A word about GitHub



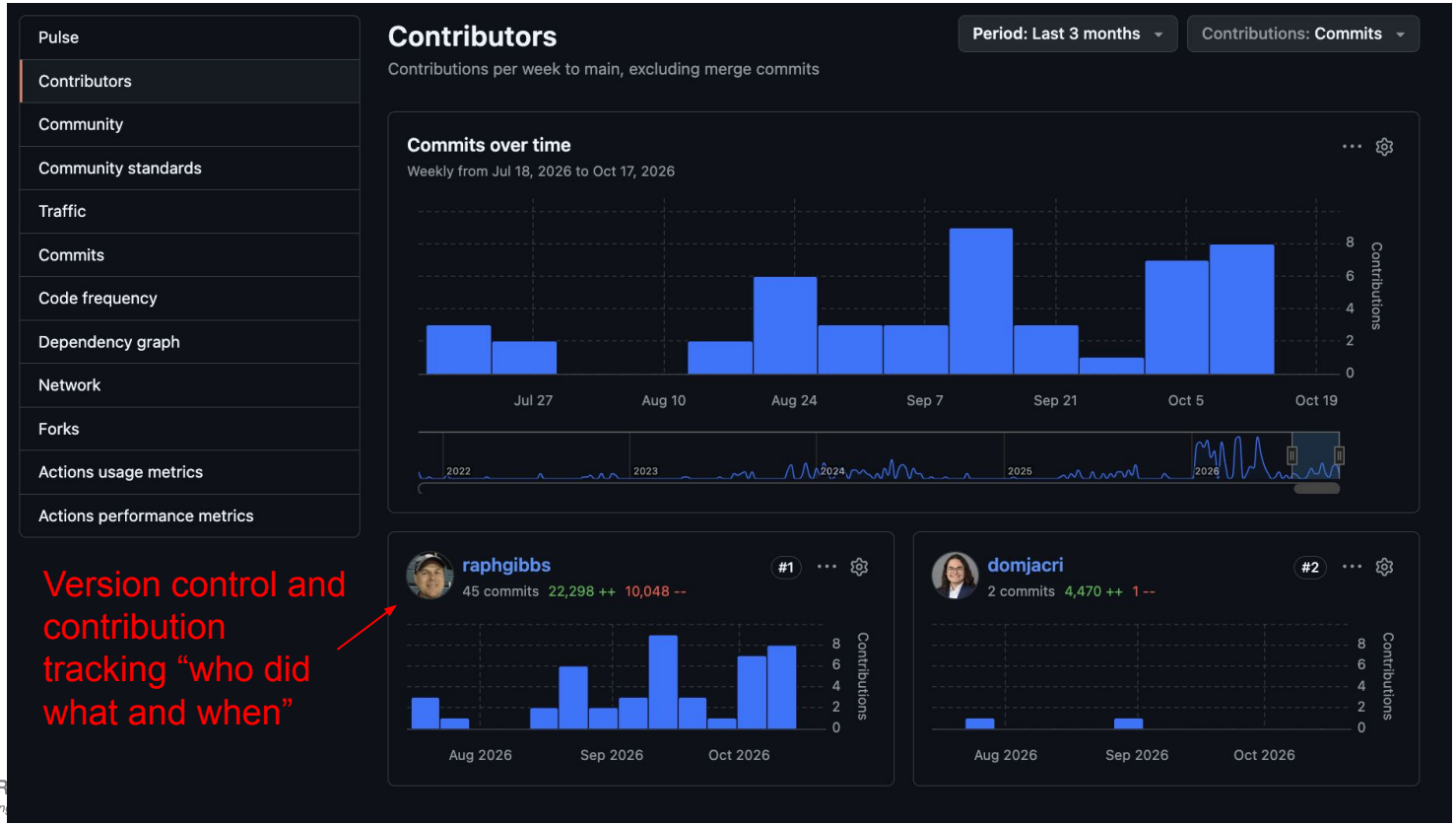
A word about GitHub

Share structured projects/code with the world **[maybe one day a req for publication]**



Track collaborations and work together

A word about GitHub



Questions to:

Dom Acri
(dominic.acri@nih.gov)

Resources available

This series (and the instructors)

Biomedical Data Science Discussion - monthly

LNG/Data Tecnica - biweekly roundtables

Many OITE/FAES workshops and courses

Spring 2027 paired FAES courses:

BIOF521 - Bioinformatic analysis of NGS

BIOF544 - From read counts to cell counts

Other NIH resources - i.e. NCI BTEP

The best type of learning…

via collaboration

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MANY PEOPLE,
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irp.nih.gov